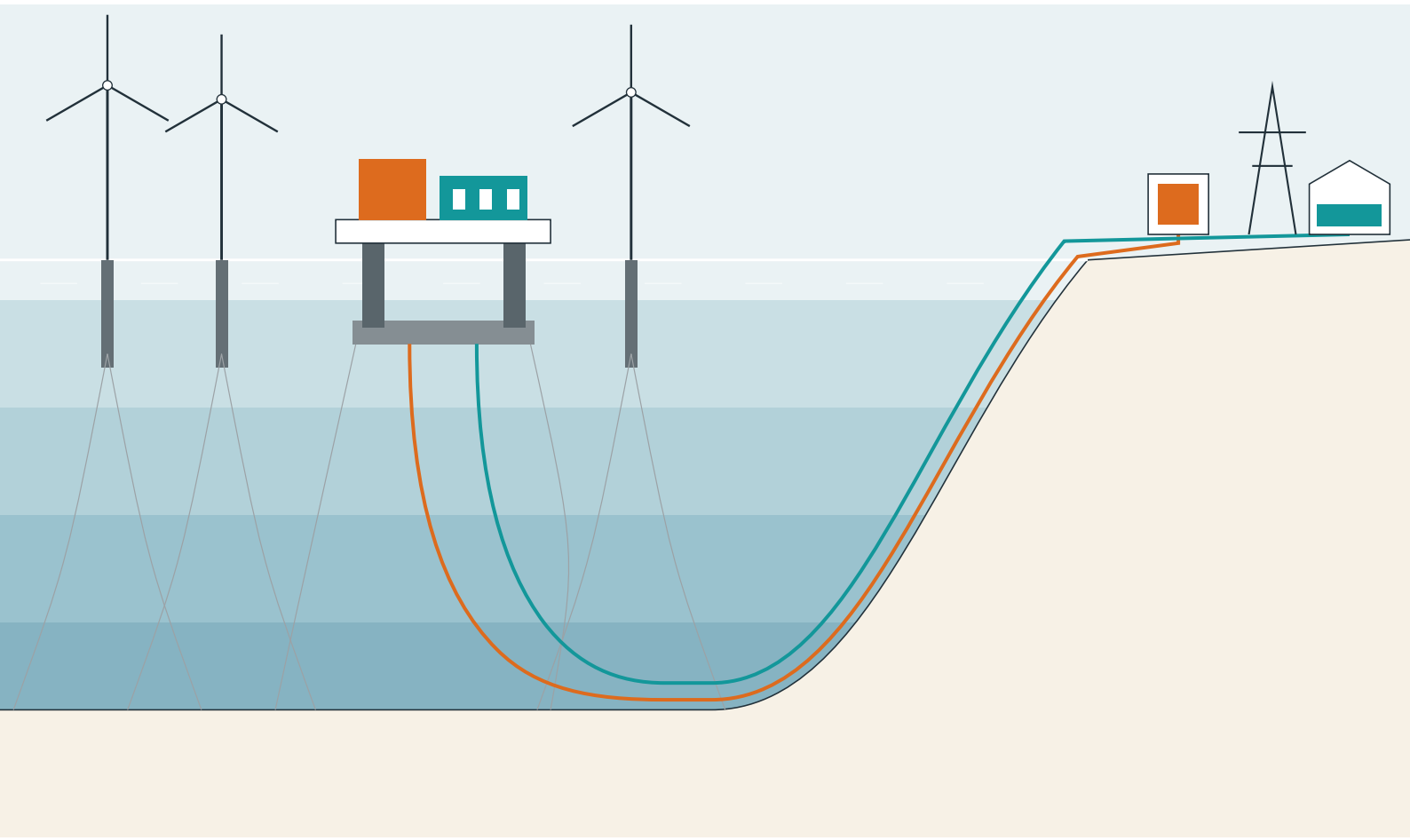


EVOLVE · Sustainable Blue Economy Partnership

# EVOLVE Planner

## Model and user handbook

Optimal planning of floating energy islands and their onshore plant, on one year of hours reduced with topological data analysis

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Software version 0.10.0 · October 2026

# About this handbook

The EVOLVE Planner is a web application that finds the lowest-cost design of an offshore energy island and the onshore plant that serves it. This handbook documents the planner in two ways. For modellers it gives the complete mathematical model: every variable, parameter and constraint, the objective function and the time-series reduction based on topological data analysis (TDA). For users it explains how to run the app, how to prepare and import data, and how to read the results, ending with a worked example that can be reproduced from the files shipped with the software.

**How to read it.** Chapter 1 sets out what the planner does and where its limits are. Chapter 2 describes the physical system and the choices a user makes. Chapter 3 is the optimisation model, written so that each equation can be traced to the source code; Chapter 4 gives every cost formula, its value and its source, and compares the defaults with the ACER 2026 unit investment cost indicators; Chapter 5 is the time-series reduction and the full-year check. Readers who only want to use the app can go from Chapter 1 straight to Chapter 6 (the app, step by step), Chapter 7 (data) and Chapter 8 (example). Appendix A lists all 203 inputs with their keys, units, default values and sources.

**Conventions.** Inputs are named in the text by their label in the app, in **bold sans serif**, and by their key in the inputs workbook, in `typewriter`. Power is in MW, energy in MWh, hydrogen in kg and kg/h, money in EUR of the base year and time in hours. Hydrogen energy uses the lower heating value (LHV). Offshore quantities carry the superscript off (the island) and onshore quantities the superscript on.

> **Where the costs come from**
>
> Export cables, HVDC converters, AC transformers, hydrogen pipelines and compressors follow the ACER 2026 unit investment cost indicators (Chapter 4). ACER publishes no indicator for wind, PV, electrolysers, fuel cells, batteries, tanks or the island platform. Most of these keep the values of the project's v9 model; the island platform and floating wind follow the published sources reviewed in Section 4.7. Values without a referenced source are marked with $^{\bullet}$ in Appendix A and flagged in the app until Step 3 of EVOLVE replaces them. Results illustrate the method; they are not cost estimates for a project.

**Software version.** This handbook describes version 0.10.0 of the planner (repository `EVOLVE-Floating-Shore`). The equations were checked against the source files `evolve_planner/model.py`, `params.py`, `tda.py`, `workflow.py` and `levelized.py` of that version. Version 0.8.0 introduced the cost model of Chapter 4, with nine pipeline sizes with capacities from gas flow and ACER 2026 values. Version 0.9.0 replaced the cable classes and the 60 km HVAC limit by a catalogue of 32 AC options with their physical limits and reactive compensation and an HVDC cost function fitted to realised projects, chosen by an exact branch and bound. Version 0.9.1 charges the TSO grid fee only on the capacity injected into the existing grid, so power that only feeds

onshore electrolysers pays no fee, and corrects the power-related battery cost to 100 kEUR/MW. Version 0.10.0 rebuilds the island platform cost (Section 4.7): the DNV GL coefficients are kept as structure costs with a documented installation, engineering and contingency factor; the floating factor of 2 is replaced by factors built from the jacket share (about 1.6 and 1.5 for a floating central platform, depth-dependent for a jacket); a floating platform carries a dynamic export-cable section and a warning above 66 kV; concept C has its own costs (turbine integration, electrolyser scale penalty, in-field lines, array-cable credit, yield gain); and floating wind costs 1.35 times bottom-fixed.

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# 1

# Introduction

## 1.1 Why an island planner

Offshore wind is moving to deeper water and further from shore, where floating foundations replace fixed ones and the export link becomes a large share of the cost. An energy island concentrates the conversion equipment of a wind farm on one platform. On it, electricity can be converted to hydrogen, stored, or sent to shore, and the onshore end can host more electrolysers, storage and a grid connection. Choosing how much of each to build, and where, is a question of balancing three things: the variable weather, the contracts the project must honour, and the cost of every component at sea and on land.

The EVOLVE Planner answers that question for one site at a time. Given the coordinates of the island and of its onshore connection points, one year of hourly weather and prices, the contracts to serve and the cost of each technology, it returns the design with the lowest net annual cost: capacities, the operation in every hour, annual costs and revenues, the levelized cost of electricity (LCOE) and of hydrogen (LCOH), and drawings of the result.

## 1.2 What the planner does

**Sizes the whole chain.** Offshore wind, floating PV, electrolysers (on the island and onshore), water treatment, compressors, fuel cells, batteries and hydrogen tanks (on both sides), the export link (AC in whole circuits, with the voltage, conductor and reactive compensation chosen, or HVDC) and a hydrogen pipeline chosen from nine sizes.

**Serves contracts.** Electricity under a power purchase agreement (PPA), sold at the day-ahead price (merchant) or under a two-sided contract for difference (CfD); hydrogen at a fixed price. Each contract can be an annual volume, a constant hourly band or an hourly profile.

**Respects space.** Wind and PV share the usable sea area; equipment on the island must fit on the deck; onshore equipment must fit on the plot.

**Compares connections.** The island can export through its own cable or through a nearby offshore substation with spare capacity, and hydrogen can leave through its own pipeline or a tie-in to an existing one. The planner can size every combination and keep the cheapest.

**Keeps the chronology.** One year of hours is reduced to about 1 460 variable-length blocks in calendar order with a method based on topological data analysis (Chapter 5). Storage is tracked across the whole year, so seasonal shifting is represented. The design is then checked, and repaired if needed, on all 8 760 hours.

## 1.3 What it does not do

The planner is a screening and design tool. Some limits follow from its scope:

- It is deterministic: it optimises against one weather and price year. Robustness across years is checked by running other years (the Belle-Île example ships 2019 and 2024).
- It is linear apart from three discrete choices: the export-link option, its whole circuits and the pipeline class. Other costs are proportional to capacity, except the fixed tie-in cost. Part-load efficiency curves, minimum loads, ramp limits and start-up costs of electrolysers are not modelled.
- The electrical network is a pair of copper plates (island and onshore) joined by a lossy link; no AC power flow or N-1 security is modelled. The physics of the AC cable (charging current, the derating it causes, reactive compensation, load and no-load losses) enters through the ratings, costs and losses of the options (Section 4.4); voltages and reactive flows in the grid are not modelled.
- Route lengths come from coordinates and a 1 km coastline mask; bathymetry, seabed conditions, protected areas and existing seabed users are not modelled.
- No database of substations or pipelines is searched: the user supplies the coordinates of the grid point, the hydrogen offtake, a hub or a tie-in (Section 2.7 lists where to find them).

## 1.4 How a run works

Figure 1.1 shows the workflow. The user describes the project in six steps. A run then (i) reduces the year with TDA, (ii) sizes every component on the reduced year, and (iii) operates the design on the full year, scaling groups of capacities up where some hours need it. The reduced year makes the sizing problem roughly six times smaller than the full year while keeping every persistent surplus and deficit episode in place; the full-year check makes sure the final design is feasible hour by hour.

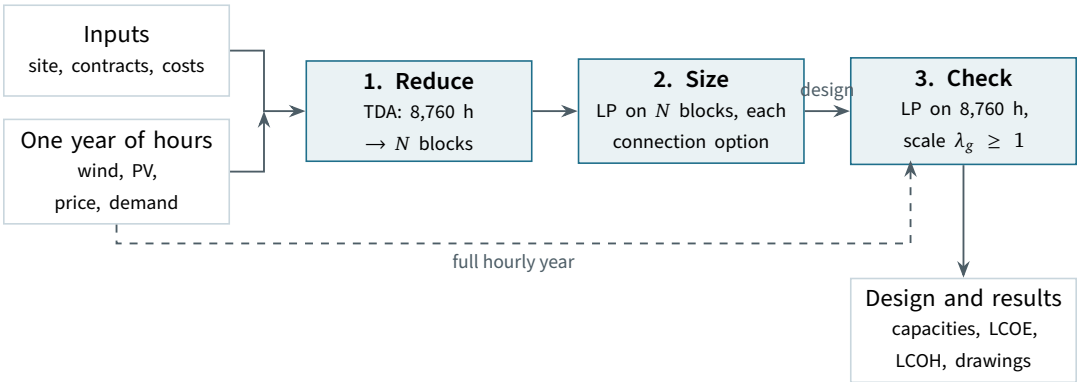



**Figure 1.1.** Workflow of a run. The model of Chapter 3 is solved twice: on the reduced year to size the system, then on the full year with the capacities fixed up to group scale factors.

## 1.5 Software

The planner is written in Python. The optimisation model is built with Pyomo [6] and solved with HiGHS [30]; GLPK can be used instead. Persistence diagrams are compared with GUDHI [45]. Weather data can be downloaded from ERA5 [27], converted with the IEA 15 MW reference turbine [24] and pvlib [29]. The interface is a Streamlit app that partners open in a browser; no installation is needed to use the hosted version (Appendix C).

## 1.6 Structure of the handbook

Chapter 2 describes the system and the user's choices in words. Chapter 3 gives the optimisation model and Chapter 5 the time reduction and the full-year check. Chapter 6 walks through the app, Chapter 7 explains the data and how to import it, and Chapter 8 applies everything to the Belle-Île case of the EVOLVE project. The appendices list all inputs, the output files, and the steps to host your own copy.

# 2

# The system and the choices it offers

This chapter describes the system the planner designs and the decisions left to the user. The equations follow in Chapter 3; the symbols in Figure 2.1 are the ones used there.

## 2.1 Two nodes and two carriers

The planner sees the project as two places joined by up to two links (Figure 2.1). The *island* collects the output of the wind farm and of any floating PV, and may host electrolysers, compressors, fuel cells, batteries and hydrogen tanks. The *onshore* site is where the export cable lands and where the hydrogen offtaker is reached; it may host its own electrolysers, compressors, fuel cells, batteries and tanks. The two places exchange electricity through the export cable, in either direction, and hydrogen through the pipeline, from the island to the shore.

Each place has one electricity balance and one hydrogen balance per time step. Electricity reaches the grid only from the onshore node; hydrogen reaches the offtaker only from the onshore node. Hydrogen made on the island must therefore be piped to shore, and electricity made into hydrogen onshore must first cross the cable.

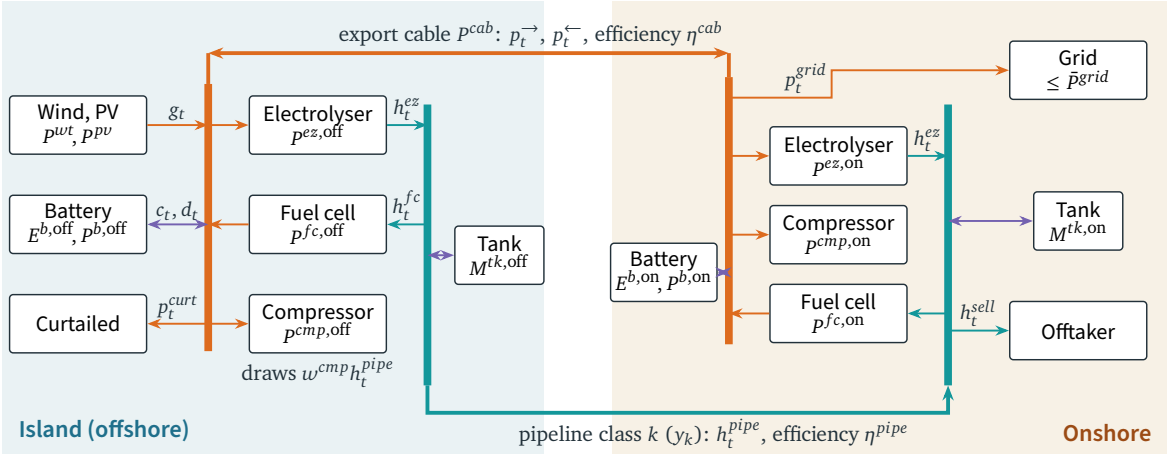



**Figure 2.1.** The two-node system. Orange: electricity; teal: hydrogen; purple: storage, which charges and discharges. Thick bars are the electricity and hydrogen balances of each place. Capacities to be sized are written inside the units, hourly flows on the arrows. Island compressors serve the pipeline; onshore compressors serve hydrogen produced onshore.

## 2.2 Island concepts

The water depth at the island decides the foundation, and one further switch decides whether conversion is centralised. Three concepts follow, as in the EVOLVE work package 2 model (Table 2.1). The cost of the platform in each concept is derived in Section 4.7.

**Table 2.1.** Island concepts and what changes in the model.

| Concept | Selected when | Effect in the model |
|---|---|---|
| A. Bottom-fixed | depth ≤ **Fixed/floating threshold depth** (60 m) | Bottom-fixed wind CAPEX and O&M; central platform on a jacket whose cost grows with depth (factors 1.23 and 1.18 at 55 m); platform O&M 2 %; no HVDC offshore-station premium. |
| B. Floating, centralised | deeper, and turbine-level electrolysis off | Floating wind CAPEX (3.43 MEUR/MW) and O&M; central platform on a floating hull (factors 1.63 for the process part and 1.52 for the electrical part, from the jacket share, hull ratio and topside uplift); platform O&M 3 %; a dynamic section of 0.5 km on each export circuit at +75 %, with a warning above 66 kV; HVDC offshore converter station × **Floating offshore-station premium** (1.5). |
| C. Floating, decentralised (Dolphyn type) | deeper, and **Turbine-level electrolysis** on | No export cable and no central platform; an electrolyser in each turbine with an integration cost, a 1.3 scale penalty on electrolyser cost, 1 km of in-field hydrogen line per turbine and a collection manifold; array cables credited (145 kEUR/MW); 0.5 % more hydrogen per MWh; island electrolysers limited to the installed generation, $P^{ez,\text{off}} \leq P^{wt} + P^{pv}$; energy leaves as hydrogen. |

## 2.3 What the island delivers

**Use case.** **Power-led**: electricity to the grid, with hydrogen only as internal storage (or sold as a by-product if **Sell the other carrier as a by-product** is on). **Hydrogen-led**: hydrogen to an offtaker; electricity is sold only if the by-product switch is on and the revenue is merchant or CfD. **Hybrid**: a contract for each.

**Delivery profile.** Each contract is one of three forms: *annual* (only the yearly volume counts, in any hours), *firm* (the same power or flow in every hour, equal to the annual volume divided by 8 760), or *profile* (an hourly series uploaded with the data, `elec_demand_mw` or `h2_demand_kgh`).

**Electricity revenue.** *PPA*: the contracted volume is delivered and paid at the PPA price. *Merchant*: every MWh delivered earns the hourly day-ahead price and the volume is free. *CfD*: a two-sided contract for difference in the style of the French offshore tenders: the strike price is paid in hours with a non-negative day-ahead price and the market price otherwise; the volume is free. With merchant and CfD revenue no electricity volume is imposed: the optimiser sells what pays.

## 2.4 Technologies

Table 2.2 lists what can be sized. On the **Technologies** step, PV, the electrolysers, fuel cells, batteries and tanks at each location, the export cable and the pipeline can be switched off; a switched-off technology has its capacity fixed at zero. The other components follow: the island compressor exists only with a pipeline, the onshore compressor only with onshore electrolysers, water treatment with any electrolyser, and the platform cost with the cable and island equipment (in concept C, the turbine-level integration and the in-field lines). Offshore wind is always available.

**Table 2.2.** Technologies, where they can be placed and how they are sized.

| Technology | Where | Sized as | Role |
|---|---|---|---|
| Offshore wind | sea area | MW | Main source; floating or bottom-fixed by depth. |
| Floating PV | sea area | $MW_p$ | Second source sharing the sea area. |
| Electrolyser | island, onshore | MW (electric input) | Power to hydrogen at a fixed LHV efficiency. |
| Water treatment | with electrolysers | MW of electrolysis | Desalination and purification, costed per MW of electrolysis. |
| Compressor | island, onshore | MW (electric) | Island: lifts pipeline hydrogen to pipeline pressure. Onshore: compresses onshore production. |
| Fuel cell | island, onshore | MW (electric output) | Hydrogen back to power. |
| Battery | island, onshore | MWh and MW | Short-term electricity storage, energy and power sized separately. |
| Hydrogen tank | island, onshore | kg | Hydrogen storage. |
| Export link | link | option, whole circuits and MW | One of 32 AC options (voltage, conductor, compensation) in whole circuits, or an HVDC link sized in MW (in 2 GW units); carries power both ways. |
| Hydrogen pipeline | link | one of nine classes | New 4 to 36-inch or repurposed 36-inch, each with a capacity in kg/h from gas flow over the route. |
| Island platform | island | from the equipment | Structure cost grows with the AC export capacity and the MW of island equipment, by concept and depth (Section 4.7). |

## 2.5 The export link

The export link is chosen from a catalogue (Section 4.4). An AC link is one of four voltages (66, 132, 220 or 275 kV), one of four conductor sizes (500 to 1 000 $mm^2$) and one of two compensation layouts (shunt reactors at both ends, or also halfway on a small platform), built in whole circuits. Each AC option has a usable rating per circuit that falls with the route length $L$, because the cable's charging current takes up part of the conductor's thermal rating; a cost per circuit (cable, transformers, reactors); losses that grow with the square of the load; and no-load losses. The HVDC option is a VSC link whose cable and converter stations are sized

to the capacity, with a fixed cost per 2 GW unit and fixed converter losses. No distance rule separates AC and DC.

With **Export link** on **auto**, the optimiser chooses the option, its number of circuits and the capacity used, together with the rest of the design, and the choice is exact (Section 3.8). A user can restrict it to one AC voltage or to HVDC. Before a run the app previews the option that is cheapest at the grid injection limit, before load losses; after the run it shows the link built, its circuits, its losses and its reactive compensation.

A transmission fee per MW of capacity is charged every year. The **TSO grid fee** (`tso_fee_eur_mw_yr`) applies when the island connects directly to the existing grid through its own link, AC or HVDC, and is charged per MW of capacity injected into the grid, not per MW of cable: power that crosses the cable only to feed onshore electrolysers stays behind the meter and pays no fee. With a hub connection the hub access fee is charged per MW of the cable to the hub instead. **Cable cost model** = **v9 linear** restores the per-MW cable cost of the project's v9 model, with its 60 km HVAC limit, for comparison (Section 4.4.8).

## 2.6 Connections to existing infrastructure

A new island may be close to an offshore substation or an export cable with spare capacity, or to a gas or hydrogen pipeline. The planner offers both options as alternatives to building its own link to shore.

**Hub (electricity).** A short cable from the island to a nearby offshore substation or cable. The hub owner carries the power to shore. The cable length is the route from the island to the hub; the export is capped by the hub's spare capacity, $\bar{P}^{grid} = \min(\text{grid injection limit}, \text{hub spare MW})$; the cable is AC at the catalogue voltage nearest to the hub voltage (66, 132, 220 or 275 kV), with reactors at both ends and the conductor and number of circuits chosen by the optimiser; the hub access fee replaces the TSO fee.

**Tie-in (hydrogen).** A short spur pipeline to a nearby pipeline. The spur length is the route to the tie-in point; the flow is capped by the pipeline's spare capacity; the compressors deliver at the pipeline's operating pressure, which changes the compression work and the pipe capacities; the tie-in cost (hot tap, metering) is added to the investment of whichever pipeline class is built.

**Let the model choose.** Every combination of the electricity options (direct, hub) and hydrogen options (direct, tie-in) is sized on the reduced year (the hub only if the export cable is allowed, the tie-in only if the pipeline is allowed); the cheapest is checked on the full year and the others are reported with their cost difference.

## 2.7 Route lengths and connection points

All end points are coordinates entered by the user: the island, the grid connection point, the hydrogen offtake, and, if used, the hub and the tie-in. The planner does not search a database of substations or pipelines. From the coordinates it measures each route with the first of these that applies; the user chooses between the second and the third with **How route lengths are found** (`route_method`):

1. **Entered length.** **Cable length, if known** and **Pipeline length, if known** (when above zero) replace the computed length of the island's own cable and pipeline. A hub cable and a tie-in spur are always measured.
2. **Sea route** (default). The shortest path through the sea on a 1 km land mask (the GLOBE data set [26], shipped with the app), around islands and headlands, from the island to the coast nearest the onshore point, followed by a straight onshore leg to that point. Cells within 1.5 km of land cost up to four times more to cross, so the route keeps off the coast until the landfall. The length used is the length of the smoothed path, and never less than the great-circle distance. Large areas are coarsened so that a route across a whole sea is found in about a second. If the coastline file is missing, the route cannot be found, or the two ends are within 0.5 km of each other, the straight-line method is used instead.
3. **Straight line × routing factor.** The great-circle distance multiplied by the **Routing factor** (1.1 to 1.3 is typical).

The length used, and how it was found, is shown on the **Site** step and stored with the results. For a new site with no known substation, the **Site** step also gives the nearest coast to the island and can place the grid point or the offtake there as a first guess, and it links to three public maps for finding real connection points: OpenInfraMap (substations, power lines, pipelines, from OpenStreetMap), EMODnet Human Activities (offshore cables, pipelines, wind farms) and the ENTSO-E grid map.

## 2.8 Space

Three areas limit the design. The **Usable sea area** ($km^2$) is shared by wind and PV through their power densities ($MW/km^2$). The **Island deck area** limits island electrolysers, compressors, fuel cells, batteries and tanks through their footprints per MW, MWh or kg. The **Onshore plot area** does the same onshore.

# 3

# Optimisation model

The planner solves one optimisation model, built in `evolve_planner/model.py`. It is a linear programme except for three discrete choices: the export-link option (AC voltage, conductor and compensation, or HVDC), its whole number of circuits or converter units, and the pipeline class. The same model is solved on the reduced year to size the system and on the full year to check the design (Chapter 5); only the time axis and a few added variables differ. This chapter states the model in full: sets, parameters, variables, the objective function and every constraint, grouped by technology.

## 3.1 Time axis

The year is a sequence of time steps $t \in \mathcal{T} = \{1, \ldots, N\}$ in calendar order. Step $t$ lasts $\Delta_t$ hours, with $\sum_t \Delta_t = 8760$. On the full year $N = 8\,760$ and $\Delta_t = 1$. On the reduced year $N$ is the number of TDA blocks (1 460 by default) and $\Delta_t$ is the length of block $t$, between one hour and a few days (Chapter 5). Every time-dependent input (capacity factors, price, demand profiles) enters as its mean over the step. Power and flow variables are constant within a step, and the energy in a step is the power times $\Delta_t$. Writing every annual sum as $\sum_t \Delta_t(\cdot)$ makes the same equations valid for both time axes.

## 3.2 Sets and indices

**Table 3.1.** Sets and indices.

| Symbol | Meaning |
|---|---|
| $t \in \mathcal{T}$ | time steps, $\mathcal{T} = \{1, \ldots, N\}$, durations $\Delta_t$ [h] |
| $\ell \in \{\text{off}, \text{on}\}$ | location: the island (off) or the onshore site (on) |
| $k \in \mathcal{K}$ | pipeline classes allowed: new 4, 6, 8, 12, 16, 20, 24 and 36-inch, and repurposed 36-inch if an existing pipeline can be repurposed; $\mathcal{K} = \emptyset$ if the pipeline is switched off (Section 4.5) |
| $v \in \mathcal{V}$ | export-link options offered: 32 AC options (66, 132, 220 or 275 kV; 500, 630, 800 or 1000 mm$^2$; reactors at both ends, or also mid-route) that can carry power over the route, and one HVDC option; only the AC options of one voltage, or only HVDC, if the user fixes the export voltage; for a hub cable the AC options at the catalogue voltage nearest the hub voltage, with reactors at both ends; empty if no cable may be built (Section 4.4) |
| $j \in \{1, 2, 3\}$ | segments of the piecewise-linear AC loss function |
| $s \in \mathcal{S}$ | storage units: battery and hydrogen tank at each location, $\mathcal{S} = \{b^{\text{off}}, b^{\text{on}}, tk^{\text{off}}, tk^{\text{on}}\}$ |
| $c \in \mathcal{C}$ | cost components (Table 3.5) |
| $g \in \mathcal{G}$ | capacity groups of the full-year check: generation, conversion and export, storage |

## 3.3 Decision variables

All variables are non-negative unless stated otherwise. Table 3.2 lists the capacities, which are the design; Table 3.3 lists the operation in each step.

**Table 3.2.** Capacity variables (the design).

| Symbol | Unit | Meaning | Code name |
|---|---|---|---|
| $P^{wt}$ | MW | offshore wind capacity | `C_wt` |
| $P^{pv}$ | MW | floating PV capacity | `C_pv` |
| $P^{ez,\ell}$ | MW | electrolyser capacity (electric input) at $\ell$ | `C_ez_off`, `C_ez_on` |
| $P^{cmp,\ell}$ | MW | compressor capacity (electric) at $\ell$ | `C_cmp_off`, `C_cmp_on` |
| $P^{fc,\ell}$ | MW | fuel cell capacity (electric output) at $\ell$ | `C_fc_off`, `C_fc_on` |
| $P^{cab}$ | MW | export cable capacity used | `C_cab` |
| $P^{inj}$ | MW | grid-injection capacity on which the TSO fee is paid | `C_grid` |
| $n^{cab}$ | integer | number of AC circuits, or of 2 GW HVDC units | `n_cab` |
| $v$ | option | export-link option built (one of $\mathcal{V}$) | set by `set_cable_option` |
| $E^{b,\ell}$, $P^{b,\ell}$ | MWh, MW | battery energy and power capacity at $\ell$ | `E_bat_*`, `P_bat_*` |
| $M^{tk,\ell}$ | kg | hydrogen tank capacity at $\ell$ | `E_tank_off`, `E_tank_on` |
| $y_k$ | $\{0, 1\}$ | 1 if pipeline class $k$ is built | `y_pipe[k]` |

**Table 3.3.** Operation variables, one value per time step $t$.

| Symbol | Unit | Meaning | Code name |
|---|---|---|---|
| $p_t^{ez,\ell}$ | MW | power fed to the electrolysers at $\ell$ | `p_we`, `p_pe` |
| $p_t^{\rightarrow}$ | MW | power sent from the island into the cable | `p_cab_f` |
| $p_{j,t}^{\rightarrow}$ | MW | part of $p_t^{\rightarrow}$ in loss segment $j$ | `pf_seg` |
| $p_t^{\leftarrow}$ | MW | power sent from shore to the island | `p_cab_r` |
| $p_t^{curt}$ | MW | curtailed generation | `p_curt_off` |
| $p_t^{grid}$ | MW | power delivered to the grid | `p_grid` |
| $c_t^{b,\ell}$, $d_t^{b,\ell}$ | MW | battery charge and discharge at $\ell$ | `bat_*_ch`, `bat_*_dis` |
| $h_t^{ez,\ell}$ | kg/h | hydrogen produced at $\ell$ | `h2_off`, `h2_on` |
| $h_t^{pipe}$ | kg/h | hydrogen sent into the pipeline | `h2_pipe` |
| $h_t^{fc,\ell}$ | kg/h | hydrogen fed to the fuel cells at $\ell$ | `h2fc_off`, `h2fc_on` |
| $h_t^{sell}$ | kg/h | hydrogen delivered to the offtaker | `h2_sell` |
| $h_t^{in,\ell}$, $h_t^{out,\ell}$ | kg/h | hydrogen into and out of the tank at $\ell$ | `tank_*_in`, `tank_*_out` |
| $e_0^s$ | MWh, kg | storage level at the start of the year | `*_start` |
| $\delta_t^s$, $\underline{\delta}^s$, $\overline{\delta}^s$ | MWh, kg | change of level since the start of the year, and its minimum and maximum (free sign) | `*_dev`, `*_dmin`, `*_dmax` |

## 3.4 Parameters

Table 3.4 lists the parameters that appear in the equations, with the input keys they come from. Quantities computed from several inputs (efficiencies, cable cost, compression work) are derived in Section 3.7. The full list of inputs with default values is in Appendix A.

**Table 3.4.** Parameters of the model.

| Symbol | Unit | Meaning | Input key(s) |
|---|---|---|---|
| **Time series (mean over step $t$)** | | | |
| $f_t^{wt}$, $f_t^{pv}$ | – | wind and PV capacity factors | `wind_cf`, `pv_cf` |
| $\pi_t$ | EUR/MWh | day-ahead price | `price_eur_mwh` |
| $\hat{d}_t^e$, $\hat{d}_t^h$ | MW, kg/h | delivery profiles | `elec_demand_mw`, `h2_demand_kgh` |
| **Contracts and revenue** | | | |
| $D^e$, $D^h$ | MWh/yr, kg/yr | electricity and hydrogen to deliver per year | `annual_demand_mwh`, `annual_h2_demand_kg` |
| $\pi^{ppa}$, $\pi^{str}$ | EUR/MWh | PPA price, CfD strike price | `ppa_price_eur_mwh`, `cfd_strike_eur_mwh` |
| $\pi^h$ | EUR/kg | hydrogen price | `h2_price_eur_kg` |
| $\bar{P}^{grid}$ | MW | grid injection limit (hub: also $\leq$ spare capacity) | `grid_injection_cap_mw`, `hub_spare_mw` |
| $\pi^{pen,e}$, $\pi^{pen,h}$ | EUR/MWh, EUR/kg | penalties for undelivered energy (full-year check only) | `shortfall_penalty_*` |
| **Conversion** | | | |
| $\kappa^{ez}$ | kg/MWh | hydrogen produced per MWh of electrolysis, Equation (3.34) | `electrolyzer_efficiency`, `lhv_h2_kwh_kg` |

Table 3.4 continued

| Symbol | Unit | Meaning | Input key(s) |
|---|---|---|---|
| $\varphi^{fc}$ | MWh/kg | electricity from one kg in a fuel cell, Equation (3.34) | `fc_efficiency_lhv` |
| $w^{cmp}$ | MWh/kg | compression work, Equation (3.35) | `comp_*` |
| $\eta_v$ | – | share of the power left after fixed losses: 1 for AC, 0.98 for HVDC (converters) | `hvdc_converter_loss` |
| $\rho_v$ | – | load loss of option $v$ at full use, Equation (4.4); for HVDC the cable loss | `cable_loss_factor`, `hvdc_cable_loss_per_100km` |
| $P_v^0$ | MW | no-load losses of one AC circuit, Equation (4.5) | `cable_tan_delta` |
| $\eta^{pipe}$ | – | pipeline delivery efficiency | `pipeline_efficiency` |
| $\eta^c, \eta^d$ | – | battery charge and discharge efficiency (each way) | `batt_eta` |
| **Limits** | | | |
| $\bar{H}_k$ | kg/h | capacity of pipeline class $k$ on the route, Equations (4.10) and (4.11) | `pipe_id_*in`, `pipe_*` |
| $R_v$ | MW | usable rating of one circuit of option $v$ on the route, Equation (4.3); 2 000 MW per HVDC unit | `hvdc_unit_mw` |
| $\bar{H}^{spare}$ | kg/h | spare capacity at the tie-in (tie-in only) | `tiein_spare_kgh` |
| $A^{sea}, \rho^{wt}, \rho^{pv}$ | km$^2$, MW/km$^2$ | usable sea area, wind and PV power densities | `sea_area_km2`, `*_density_mw_km2` |
| $A^{deck}, A^{plot}$ | m$^2$ | island deck area, onshore plot area | `deck_area_max_m2`, `area_onshore_max_m2` |
| $a^x$ | m$^2$/MW, /MWh or /kg | footprint of technology $x \in \{ez, cmp, fc, b, tk\}$ | `area_*` |
| $\bar{P}, \bar{P}^{ren}$ | MW | bounds on converter and cable capacities, and on wind and PV | `cap_ub_mw`, `cap_renewable_ub_mw` |
| $\bar{E}^b, \bar{P}^b, \bar{M}^{tk}$ | MWh, MW, kg | largest battery and tank at each location | `batt_max_*`, `tank_max_kg` |
| **Costs** | | | |
| $r$ | – | discount rate (real WACC) | `discount_rate` |
| $n_c, \phi_c$ | yr, 1/yr | lifetime and fixed O&M (share of CAPEX per year) of component $c$ | `*_life`, `*_fom` |
| $I_c$ | EUR | investment of component $c$, linear in the capacities (Table 3.5) | `*_capex*` |
| $\mu$ | – | offshore cost premium on island equipment | `offshore_premium` |
| $I_v^{circ}, I_v^{fix}, c_v^{MW}$ | EUR, EUR, EUR/MW | investment of option $v$ per circuit or unit, once per link (mid-route platform) and per MW of capacity (HVDC), Equations (4.6) and (4.8) | `ac_*`, `reactor_*`, `vh_*` |
| $\pi^{loss}$ | EUR/MWh | value of no-load losses | `loss_value_eur_mwh` |
| $F^{tso}, F^{hub}$ | EUR/MW/yr | TSO fee per MW injected; hub access fee per MW of cable, Equation (3.36) | `tso_fee*`, `hub_access_fee_eur_mw_yr` |
| $v^b, v^{tk}$ | EUR/MWh, EUR/kg | variable O&M of batteries and tanks | `batt_vom`, `tank_vom` |

## 3.5 Objective function

The model minimises the net annual cost: annualised investment plus fixed O&M of every component, the grid fee, variable O&M, minus the revenue from electricity and hydrogen,

$$\min\ Z\ =\ \sum_{c\in C}\left(\mathrm{CRF}(r,n_c)+\phi_c\right)I_c + F^{tso}P^{inj} + F^{hub}P^{cab} + 8760\,\pi^{loss}P_v^0\,n^{cab} + C^{vom} - R^e - R^h\ \left[+\,\Pi\right], \tag{3.1}$$

where the third term values the no-load losses of the AC circuits, which flow all year and are therefore not in the energy balances, and the capital recovery factor annualises an investment over its lifetime,

$$\mathrm{CRF}(r,n) = \frac{r(1+r)^n}{(1+r)^n-1}\quad (r>0), \qquad \mathrm{CRF}(0,n) = \frac{1}{n}. \tag{3.2}$$

The penalty $\Pi$ in brackets appears only in the full-year check (Section 5.6). A negative $Z$ is an annual surplus.

**Investment of each component.** Table 3.5 gives $I_c$. Costs in EUR/kW are multiplied by 1000; island electrolysers, compressors, fuel cells, batteries and tanks carry the offshore premium $\mu$. Wind uses the floating CAPEX and O&M when the island floats (Section 2.2).

**Table 3.5.** Investment $I_c$ of each cost component, its lifetime and fixed O&M.

| Component $c$ | Investment $I_c$ [EUR] | $n_c, \phi_c$ from |
|---|---|---|
| Offshore wind | $\left(c^{wt} - c^{arr}\mathbb{1}^C\right)P^{wt}$ (floating or bottom-fixed CAPEX) | `wind_life`, `wind_fom`* |
| Floating PV | $1000\,c^{pv}P^{pv}$ | `pv_life`, `pv_fom` |
| Electrolyser, island | $1000\,c^{ez}\mu\,k^{scale}P^{ez,\mathrm{off}}$ ($k^{scale}=1$ in A and B) | `elec_life`, `elec_fom` |
| Electrolyser, onshore | $1000\,c^{ez}P^{ez,\mathrm{on}}$ | `elec_life`, `elec_fom` |
| Water treatment | $c^{wtr}\left(P^{ez,\mathrm{off}}+P^{ez,\mathrm{on}}\right)$ | `wt_life`, `wt_fom` |
| Compressors | $c^{cmp}\left(\mu P^{cmp,\mathrm{off}}+P^{cmp,\mathrm{on}}\right)$ | `comp_life`, `comp_fom` |
| Fuel cells | $1000\,c^{fc}\left(\mu P^{fc,\mathrm{off}}+P^{fc,\mathrm{on}}\right)$ | `fc_life`, `fc_fom` |
| Export link | $I_v^{circ}n^{cab}+I_v^{fix}+c_v^{MW}P^{cab}$ (Equations (4.6) and (4.8)) | `cable_life`, `cable_fom_*` |
| Hydrogen pipeline | $\sum_{k\in\mathcal{K}}\left(c_k^{pipe}L^{pipe}+C^{tie}\right)y_k$ | `pipeline_life`, `pipeline_fom` |
| Island platform (A, B) | $1000\,k^{esc}k^{inst}\left(f_E\,c_E\,\mathbb{1}_v^{AC}P^{cab}+f_H\,c_H\,S^H\right)+C_0$ (Equation (4.12)) | `platform_life`, `platform_fom`* |
| Turbine-level integration (C) | $1000\,k^{esc}k^{inst}f_C\,c_H\,S^H$ (Equation (4.16)) | `platform_life`, `platform_fom` |
| In-field hydrogen lines (C) | $c^{inf}\ell^{inf}P^{wt}/P^{wt}_{unit}$ | `pipeline_life`, `infield_pipe_fom` |
| Batteries | $\mu\left(c^E E^{b,\mathrm{off}}+c^P P^{b,\mathrm{off}}\right)+c^E E^{b,\mathrm{on}}+c^P P^{b,\mathrm{on}}$ | `batt_life`, `batt_fom`† |
| Hydrogen tanks | $c^{tk}\left(\mu M^{tk,\mathrm{off}}+M^{tk,\mathrm{on}}\right)$ | `tank_life`, `tank_fom`† |

Here $c^{wt}$ [EUR/MW], $c^{pv}$, $c^{ez}$, $c^{fc}$ [EUR/kW], $c^{wtr}$, $c^{cmp}$ [EUR/MW], $c^E$ [EUR/MWh], $c^P$ [EUR/MW] and $c^{tk}$ [EUR/kg] are the unit costs; $c_k^{pipe}$ [EUR/km] is the cost of class $k$, $L^{pipe}$ the pipeline route length and $C^{tie}$ the tie-in cost (zero without a tie-in); $L$ is the cable route length.

The island platform is described in Section 4.7: $c_E$ and $c_H$ [kEUR/MW] are the electrical and process platform coefficients, $S^H = P^{ez,\text{off}} + w_c P^{cmp,\text{off}} + w_f P^{fc,\text{off}} + w_b P^{b,\text{off}}$ the weighted MW of island equipment, $k^{inst}$ the installation, engineering and contingency factor, $k^{esc}$ a price escalation and $C_0$ an optional fixed cost; the configuration factors $f_E$ and $f_H$ depend on concept and depth (Equations (4.14) and (4.15)). The cable part applies to AC links only, $\mathbb{1}_v^{AC}$ (an HVDC link has its own converter platform in $I_v^{circ}$ and $c_v^{MW}$; with the **v9 linear** cable cost it applies to both). The platform's fixed O&M is `platform_fom_floating` (3 %) for a floating central platform and `platform_fom` (2 %) otherwise. In concept C ($\mathbb{1}^C = 1$) there is no central platform ($f_E = f_H = 0$); instead each turbine pays an integration cost with factor $f_C$, $k^{scale}$ is the cost penalty of small electrolysers, $c^{inf}$ [EUR/km] and $\ell^{inf}$ [km per turbine] price the in-field hydrogen lines, $P_{unit}^{wt}$ is the turbine rating, and $c^{arr}$ [EUR/MW] is the array cable that is no longer needed. The pipeline's fixed cost $C^{tie}$ then also includes a collection manifold of `infield_manifold_eur`. †On the island, batteries and tanks carry an extra fixed O&M of `offshore_fom_adder` (0.5 % a year, so 2.5 % instead of 2.0 % as in the v9 model); their two parts are annualised separately.

**Variable O&M.** Battery O&M is charged per MWh discharged and tank O&M per kg stored:

$$C^{vom} = v^b \sum_t \Delta_t \left(d_t^{b,\text{off}} + d_t^{b,\text{on}}\right) + v^{tk} \sum_t \Delta_t \left(h_t^{in,\text{off}} + h_t^{in,\text{on}}\right). \tag{3.3}$$

**Revenue.** Hydrogen earns $R^h = \pi^h \sum_t \Delta_t \, h_t^{sell}$. Electricity earns, by contract,

$$R^e = \begin{cases} \pi^{ppa} \sum_t \Delta_t \, p_t^{grid} & \text{PPA,} \\ \sum_t \Delta_t \, \pi_t \, p_t^{grid} & \text{merchant,} \\ \sum_t \Delta_t \, \tilde{\pi}_t \, p_t^{grid}, \quad \tilde{\pi}_t = \begin{cases} \pi^{str} & \pi_t \geq 0 \\ \pi_t & \pi_t < 0 \end{cases} & \text{CfD,} \\ 0 & \text{no electricity sales.} \end{cases} \tag{3.4}$$

With a PPA and a contracted volume or profile, the delivered energy is fixed by Section 3.6.10, so $R^e$ is a constant; with a PPA and no volume, every MWh delivered earns the PPA price. Under the power-led use case the hydrogen price is set to zero unless the by-product switch is on; under the hydrogen-led use case electricity earns nothing unless the switch is on and the revenue is merchant or CfD.

## 3.6 Constraints

All constraints hold for every $t \in \mathcal{T}$ unless a sum over $t$ appears.

### 3.6.1 Generation and curtailment

Generation is the installed capacity times the capacity factor of the step,

$$g_t = P^{wt} f_t^{wt} + P^{pv} f_t^{pv}, \tag{3.5}$$

and only generated power can be curtailed,

$$p_t^{curt} \leq g_t. \tag{3.6}$$

Wind and PV share the usable sea area,

$$\frac{P^{wt}}{\rho^{wt}} + \frac{P^{pv}}{\rho^{pv}} \le A^{sea}. \tag{3.7}$$

### 3.6.2 Electricity balances

On the island, generation, power arriving through the cable, fuel cell output and battery discharge equal the power used by electrolysers and compressors, sent to shore, curtailed and stored:

$$g_t + (\eta_v - \rho_v)\, p_t^{\leftarrow} + \varphi^{fc} h_t^{fc,\mathrm{off}} + d_t^{b,\mathrm{off}} = p_t^{ez,\mathrm{off}} + w^{cmp} h_t^{pipe} + p_t^{\rightarrow} + p_t^{curt} + c_t^{b,\mathrm{off}}. \tag{3.8}$$

Onshore, power arriving from the island, fuel cell output and battery discharge equal the power used by electrolysers and compressors, sent back to the island, delivered to the grid and stored:

$$\eta_v p_t^{\rightarrow} - \ell_t + \varphi^{fc} h_t^{fc,\mathrm{on}} + d_t^{b,\mathrm{on}} = p_t^{ez,\mathrm{on}} + w^{cmp} h_t^{ez,\mathrm{on}} + p_t^{\leftarrow} + p_t^{grid} + c_t^{b,\mathrm{on}}, \tag{3.9}$$

where $\ell_t$ is the load loss of the power sent to shore. For AC it is the quadratic loss $\rho_v (p_t^{\rightarrow})^2/(R_v n^{cab})$ written as chords over three equal segments of the installed rating $R_v n^{cab}$,

$$p_t^{\rightarrow} = \sum_{j=1}^{3} p_{j,t}^{\rightarrow}, \qquad 0 \le p_{j,t}^{\rightarrow} \le \tfrac{1}{3} R_v n^{cab}, \qquad \ell_t = \sum_{j=1}^{3} \frac{2j-1}{3}\, \rho_v\, p_{j,t}^{\rightarrow}, \tag{3.10}$$

which is exact at 0, $\frac{1}{3}$, $\frac{2}{3}$ and the full rating and slightly above the quadratic in between (by at most $\rho_v/36$ of the rating). Because the slopes increase, the segments fill in order without binaries. For HVDC, whose cable losses are small, $\ell_t = \rho_v p_t^{\rightarrow}$. Power sent from shore to the island, which is small, is charged the full-load share in both cases. Island compressors lift the hydrogen that enters the pipeline; onshore compressors lift the hydrogen produced onshore.

### 3.6.3 Hydrogen balances

$$h_t^{ez,\mathrm{off}} + h_t^{out,\mathrm{off}} = h_t^{pipe} + h_t^{fc,\mathrm{off}} + h_t^{in,\mathrm{off}}, \tag{3.11}$$

$$h_t^{ez,\mathrm{on}} + \eta^{pipe} h_t^{pipe} + h_t^{out,\mathrm{on}} = h_t^{sell} + h_t^{fc,\mathrm{on}} + h_t^{in,\mathrm{on}}. \tag{3.12}$$

### 3.6.4 Electrolysers

Electrolysers convert power to hydrogen at a constant efficiency and run up to their capacity:

$$h_t^{ez,\ell} = \kappa_\ell^{ez}\, p_t^{ez,\ell}, \qquad p_t^{ez,\ell} \le P^{ez,\ell}, \qquad \ell \in \{\mathrm{off}, \mathrm{on}\}. \tag{3.13}$$

with $\kappa_{\mathrm{on}}^{ez} = \kappa^{ez}$ (Equation (3.34)) and $\kappa_{\mathrm{off}}^{ez} = \kappa^{ez}(1 + g\,\mathbb{1}^C)$: in concept C the turbine feeds its electrolyser directly and avoids the array cable and transformer losses, worth $g$ = `turbine_ez_eff_gain` (0.5 %) of hydrogen output [34]. In concept C the island electrolysers cannot exceed the installed generation,

$$P^{ez,\mathrm{off}} \le P^{wt} + P^{pv}. \tag{3.14}$$

### 3.6.5 Compressors

The electric power drawn by compression may not exceed the compressor capacity:

$$w^{cmp} h_t^{pipe} \le P^{cmp,\mathrm{off}}, \qquad w^{cmp} h_t^{ez,\mathrm{on}} \le P^{cmp,\mathrm{on}}. \tag{3.15}$$

### 3.6.6 Fuel cells

$$\varphi^{fc} h_t^{fc,\ell} \le P^{fc,\ell}, \qquad \ell \in \{\mathrm{off}, \mathrm{on}\}. \tag{3.16}$$

### 3.6.7 Export cable and grid

The cable carries power in both directions up to its capacity, the capacity is limited by the circuits built, and the grid accepts at most its injection limit:

$$p_t^{\rightarrow} \le P^{cab}, \qquad p_t^{\leftarrow} \le P^{cab}, \qquad P^{cab} \le R_v\, n^{cab}, \qquad n^{cab} \in \{1, \dots, \bar{n}_v\}, \qquad p_t^{grid} \le P^{inj} \le \bar{P}^{grid}, \tag{3.17}$$

with at most $\bar{n}_v = 12$ AC circuits (`cable_max_circuits`) and as many HVDC units as $\bar{P}$ needs. Losses enter through Equations (3.8) to (3.10). $P^{cab}$, the capacity used, can be below the rating of the circuits built: the cost of the circuits is paid in full, and the hub access fee and the HVDC cost per MW follow $P^{cab}$. The TSO fee follows $P^{inj}$, the largest grid injection, which is zero for a hub connection and can be below $P^{cab}$ when part of the power feeds onshore electrolysers. In concept C, or when the cable is switched off, no option is built and $P^{cab} = 0$; the optimiser can also build no link ($n^{cab} = 0$) when no contract needs one (Section 3.8). With the **v9 linear** cable cost (Section 4.4.8) there are no circuits: $n^{cab} = 1$, $R_v$ is the bound $\bar{P}$ and the cost is per MW.

### 3.6.8 Hydrogen pipeline

At most one pipeline class is built, and the flow is limited by the class built and, with a tie-in, by the spare capacity of the existing pipeline:

$$\sum_{k \in \mathcal{K}} y_k \le 1, \qquad h_t^{pipe} \le \sum_{k \in \mathcal{K}} \bar{H}_k\, y_k, \qquad h_t^{pipe} \le \bar{H}^{spare} \ \text{(with a tie-in)}. \tag{3.18}$$

With $\mathcal{K} = \emptyset$ the right-hand side is zero and no hydrogen leaves the island by pipe.

### 3.6.9 Batteries and hydrogen tanks

For each storage unit $s$ let $q_t^s$ be the charging flow, $o_t^s$ the discharging flow, $S^s$ the capacity and $\eta^{s,c}, \eta^{s,d}$ the efficiencies:

| $s$ | $q_t^s$ | $o_t^s$ | $S^s$ | $\eta^{s,c},\ \eta^{s,d}$ |
|---|---|---|---|---|
| battery at $\ell$ | $c_t^{b,\ell}$ | $d_t^{b,\ell}$ | $E^{b,\ell}$ | $\eta^c,\ \eta^d$ |
| tank at $\ell$ | $h_t^{in,\ell}$ | $h_t^{out,\ell}$ | $M^{tk,\ell}$ | 1, 1 |

The level is written as the level at the start of the year plus a deviation that accumulates over the year, following the inter-period storage formulation of Kotzur et al. [32]:

$$\delta^s_0 = 0, \tag{3.19}$$

$$\delta^s_t = \delta^s_{t-1} + \Delta_t\left(\eta^{s,c} q^s_t - \frac{o^s_t}{\eta^{s,d}}\right), \qquad t = 1, \dots, N, \tag{3.20}$$

$$\delta^s_N = 0, \tag{3.21}$$

$$\underline{\delta}^s \le \delta^s_t \le \overline{\delta}^s, \qquad t = 0, \dots, N, \tag{3.22}$$

$$e^s_0 + \underline{\delta}^s \ge 0, \qquad e^s_0 + \overline{\delta}^s \le S^s. \tag{3.23}$$

The level at the end of step $t$ is $e^s_t = e^s_0 + \delta^s_t$, and Equations (3.22) and (3.23) are equivalent to $0 \le e^s_t \le S^s$ for every $t$. Equation (3.21) closes the year: storage ends where it started, so no energy is taken from outside the year. Battery charge and discharge are limited by the power capacity,

$$c^{b,\ell}_t \le P^{b,\ell}, \qquad d^{b,\ell}_t \le P^{b,\ell}. \tag{3.24}$$

No self-discharge is modelled. Tank flows are limited only by the hydrogen balances and the converters.

**Why level bounds at block ends are enough**

Within a step all flows are constant, so the level changes linearly between the ends of the step and its minimum and maximum over the step are reached at the ends. Bounding the level at block ends therefore bounds it everywhere, and the reduced model is an exact model of a year whose inputs are constant within each block. The code implements the general multi-period form of Kotzur et al. [32], which also supports representative periods; with the planner's single chronological period it reduces to Equations (3.19) to (3.23).

### 3.6.10 Deliveries

Electricity contracts apply when electricity revenue is a PPA (or absent); under merchant or CfD revenue the volume is free. The hydrogen contract applies whenever a hydrogen volume or profile is set. For the three delivery modes,

$$\text{annual:} \quad \sum_t \Delta_t\, p^{grid}_t = D^e, \qquad \sum_t \Delta_t\, h^{sell}_t = D^h, \tag{3.25}$$

$$\text{firm:} \quad p^{grid}_t = D^e/8760, \qquad h^{sell}_t = D^h/8760, \tag{3.26}$$

$$\text{profile:} \quad p^{grid}_t = \hat{d}^e_t, \qquad h^{sell}_t = \hat{d}^h_t. \tag{3.27}$$

In annual and firm mode a volume of zero imposes nothing; in profile mode the uploaded profile is imposed whenever its column is present, and its annual sum replaces the annual volume entered on the **Use case** step (it is what the app reports as the contract). Profile values are absolute: MW of electricity and kg/h of hydrogen in each hour, not capacity factors. The electricity profile must stay at or below the grid injection limit. Deliveries are equalities: the contract is met exactly, and any further energy is curtailed, stored or, if the price allows, converted and sold as the other carrier.

### 3.6.11 Space on the island and onshore

$$a^{ez} P^{ez,\mathrm{off}} + a^{cmp} P^{cmp,\mathrm{off}} + a^{fc} P^{fc,\mathrm{off}} + a^{b} E^{b,\mathrm{off}} + a^{tk} M^{tk,\mathrm{off}} \le A^{deck}, \tag{3.28}$$

$$a^{ez} P^{ez,\mathrm{on}} + a^{cmp} P^{cmp,\mathrm{on}} + a^{fc} P^{fc,\mathrm{on}} + a^{b} E^{b,\mathrm{on}} + a^{tk} M^{tk,\mathrm{on}} \le A^{plot}. \tag{3.29}$$

### 3.6.12 Bounds and technology switches

$$0 \le P^{wt} \le \bar{P}^{ren}, \qquad 0 \le P^{pv} \le \bar{P}^{ren}\,\mathbb{1}^{pv}, \tag{3.30}$$

$$0 \le P^{x} \le \bar{P}\,\mathbb{1}^{x} \quad x \in \{ez^{\mathrm{off}}, ez^{\mathrm{on}}, fc^{\mathrm{off}}, fc^{\mathrm{on}}, cab\}, \tag{3.31}$$

$$0 \le P^{cmp,\mathrm{off}} \le \bar{P}\,\mathbb{1}[\mathcal{K} \neq \emptyset], \qquad 0 \le P^{cmp,\mathrm{on}} \le \bar{P}\,\mathbb{1}^{ez^{\mathrm{on}}}, \tag{3.32}$$

$$0 \le E^{b,\ell} \le \bar{E}^{b}\mathbb{1}^{b^{\ell}}, \quad 0 \le P^{b,\ell} \le \bar{P}^{b}\mathbb{1}^{b^{\ell}}, \quad 0 \le M^{tk,\ell} \le \bar{M}^{tk}\mathbb{1}^{tk^{\ell}}, \tag{3.33}$$

where $\mathbb{1}^{x}$ is 1 if technology $x$ is switched on and 0 otherwise. The cable indicator is 0 in concept C.

## 3.7 Derived parameters

The following quantities are computed from the inputs before the model is built (`params.py`, function `derive`).

**Electrolysis and fuel cells.** With LHV in kWh/kg,

$$\kappa^{ez} = \frac{1000\,\eta^{ez}}{\mathrm{LHV}} \ \text{[kg/MWh]}, \qquad \varphi^{fc} = \frac{\eta^{fc}\,\mathrm{LHV}}{1000} \ \text{[MWh/kg]}. \tag{3.34}$$

The defaults $\eta^{ez} = 0.75$, $\eta^{fc} = 0.50$ and an LHV of 33.3 give 22.5 kg/MWh and 0.016 7 MWh/kg.

**Compression work.** Multi-stage isentropic compression of hydrogen from $p^{in}$ to $p^{out}$ in $n$ stages with intercooling to $T^{in}$, compressibility $Z$, heat capacity ratio $\gamma$, isentropic efficiency $\eta^{is}$ and molar mass $M_{\mathrm{H_2}} = 2.016 \times 10^{-3}$ kg/mol:

$$w^{cmp} = \frac{Z\,R\,T^{in}}{\eta^{is} M_{\mathrm{H_2}}} \cdot \frac{n\gamma}{\gamma - 1}\left[\left(\frac{p^{out}}{p^{in}}\right)^{\frac{\gamma-1}{n\gamma}} - 1\right] \cdot \frac{1}{3.6 \times 10^{9}} \quad \text{[MWh/kg]}, \tag{3.35}$$

with $R = 8.314$ J/(mol K). With a pipeline tie-in, $p^{out}$ is the tie-in operating pressure plus the spur's pressure-drop margin (`tiein_dp_bar`, 5 bar). The defaults (30 to 70 bar, two stages, 313 K, $Z = 1.05$, $\gamma = 1.41$, $\eta^{is} = 0.75$) give about 0.45 kWh/kg.

**Export-link options.** For each option $v$, `derive` (through `cables.py`) computes the usable rating $R_v$ on the route (Equation (4.3)), the investment terms $I_v^{circ}$, $I_v^{fix}$ and $c_v^{MW}$ (Equations (4.6) and (4.8)), the loss terms $\eta_v$, $\rho_v$ and $P_v^0$ (Equations (4.4) and (4.5)), the reactive power, the fixed O&M share and the fee. Chapter 4 gives every formula, value and source. An AC option is offered when its usable rating on the route is at least 5 % of its rating at zero length; there is no distance rule between AC and HVDC. A hub cable has the AC options at the catalogue voltage nearest the hub voltage, with reactors at both ends (Section 2.5).

**Pipeline classes.** For each class $k$, the cost per km $c_k^{pipe}$ follows Equation (4.9) and the capacity $\bar{H}_k$ the lower of the velocity and pressure-drop limits of Equations (4.10) and (4.11), evaluated on the pipeline route between the compressor outlet pressure and the delivery pressure (with a tie-in: between the tie-in pressure plus a margin and the tie-in pressure).

**Grid and hub fee.**

$$F^{tso} = \begin{cases} \texttt{tso_fee_eur_mw_yr} & \text{own link to the grid, AC or HVDC,} \\ \texttt{tso_fee_hvdc_eur_mw_yr} & \text{own HVDC link with the v9 linear cable cost,} \\ 0 & \text{connection through a hub,} \end{cases} \tag{3.36}$$

$$F^{hub} = \begin{cases} \texttt{hub_access_fee_eur_mw_yr} & \text{hub,} \\ 0 & \text{otherwise.} \end{cases} \tag{3.37}$$

The TSO fee is paid on the grid-injection capacity $P^{inj}$, the hub fee on the cable to the hub $P^{cab}$ (version 0.9.1; earlier versions charged both on $P^{cab}$).

**Route lengths.** $L$ is the cable route to the grid point or to the hub, and $L^{pipe}$ the pipeline route to the offtake or to the tie-in, measured as described in Section 2.7.

## 3.8 Solving the sizing problem

The discrete choices are the export-link option $v$, its circuits or units $n^{cab}$ and the pipeline class $k$ ($y_k$ with $\sum_k y_k \le 1$, or no pipe). For fixed $(k, v, n^{cab})$ the model is a linear programme, a *leaf*. The sizing problem is the cheapest leaf. With 33 options, up to 12 circuits and ten pipe choices there are a few thousand leaves, and a general mixed-integer solver needs many simplex solves of the large LP. The planner finds the cheapest leaf exactly with a branch and bound built on the structure of the problem (`model.py`, `solve_sizing`):

1. **Pipe bounds.** For each pipe choice $k$, solve the LP with an *ideal* link: lossless, without circuits, at $c_0$ per MW and year, the lowest annual cost per MW of any option used at its full rating (annuity, O&M, fee, island platform part and no-load losses). Every design with pipe $k$ costs at least this value $Z_k$. The smallest $Z_k$ is the lower bound reported for the whole problem.

2. **Link bounds.** Let $G_k(P)$ be the cost of everything except the link when the link is lossless and free with capacity $P$. It is the value of an LP as a function of a bound, so it is convex in $P$ and every LP solved at a fixed $P_i$, with the reduced cost $\sigma_i$ of $P^{cab}$, gives a tangent $G_k(P) \ge G_k(P_i) + \sigma_i(P - P_i)$. Up to eight such LPs at multiples of the capacity $P^*$ chosen in step 1, together with the tangent of slope $-c_0$ at $P^*$ itself, give a convex lower estimate $\hat{G}_k$. Losses cannot make a design cheaper, so every leaf $(k, v, n)$ costs at least

$$B(k, v, n) = \min_{P_{min} \le P \le \min(\bar{P}, R_v n)} \hat{G}_k(P) + A_v(P, n), \tag{3.38}$$

   where $A_v(P, n)$ is the annual cost of $n$ circuits of option $v$ with capacity $P$ (investment, O&M, fee, platform and no-load losses) and $P_{min}$ the capacity every feasible design needs: the contracted grid energy (unless a pipe is built and onshore fuel cells are allowed, so that piped hydrogen can supply it), plus the electricity for the hydrogen made onshore when there is no pipe, divided by 8760 hours. When $P_{min} = 0$ a leaf without any link is added. Both terms are piecewise linear in $P$, so the minimum lies at an end or a kink.

3. **Leaves.** In order of $B$, solve the leaf LP with $k$, $v$ and $n$ fixed and its losses included, and keep the cheapest. Stop when the next bound is not below the best cost found. Pipes are visited in order of $Z_k$ and skipped when $Z_k$ cannot beat the best cost. A larger pipe can only lower the rest of the cost, so the smallest leaf bound of the largest pipe, corrected by the difference in annual pipe cost, bounds every other pipe; a pipe whose corrected bound cannot beat the best cost is skipped without new LPs.

The result is the cheapest leaf, the optimum of the mixed-integer problem up to the LP tolerance. Every LP re-solves the same model with a few parameters or bounds changed, warm-started by the dual simplex method of HiGHS [30]; the first is solved by interior point with crossover, with simplex as a fallback. With GLPK the same steps run without warm starts. A test of the code compares the result with a complete enumeration of all leaves on a short reduced year, for a design without a pipe, with a required pipe and with an optional pipe, and finds the same optimum while solving under a quarter of the leaves. On the example of this handbook the search solves 13 LPs, 5 of them leaves out of 387, in about 100 seconds on 1,460 blocks.

**Connection options.** When the electricity or hydrogen connection is set to "let the model choose", the derived parameters of Section 3.7 are computed for every combination of (direct, hub) and (direct, tie-in) (the hub only if the export cable is allowed, the tie-in only if the pipeline is allowed), the sizing problem is solved for each, and the option with the lowest $Z$ is kept. All options are solved on the same reduced year, so their costs are directly comparable.

## 3.9 Levelized costs

The island makes two products from shared assets, so its annual cost has to be divided between them before unit costs can be stated. The planner uses an energy allocation (`levelized.py`). Let $C_c$ be the annual cost of component $c$ (the summand of Equation (3.1) for $c$, with the grid fee and variable O&M as two further components), $E^{grid} = \sum_t \Delta_t p_t^{grid}$ the electricity delivered, $H = \sum_t \Delta_t h_t^{sell}$ the hydrogen delivered, and $E^{ez} = \sum_t \Delta_t (p_t^{ez,\mathrm{off}} + p_t^{ez,\mathrm{on}})$ the electricity fed to electrolysers, all on the full year. The hydrogen share of shared costs is

$$s^h = \frac{E^{ez}}{E^{ez} + E^{grid}} \qquad (s^h = 0 \text{ if } H = 0,\ s^h = 1 \text{ if } E^{grid} = 0). \tag{3.39}$$

Components used only for hydrogen (electrolysers, water treatment, compressors, pipeline, tanks) go to hydrogen when hydrogen is delivered, fuel cells and the TSO fee go to electricity when electricity is delivered, and every other component (wind, PV, cable, platform, batteries, hub access fee, variable O&M) is split by $s^h$. If no hydrogen is delivered (a power-led island that uses hydrogen only as storage), the hydrogen equipment is part of the cost of electricity; if no electricity is delivered, the fuel cells are part of the cost of hydrogen. Then

$$\mathrm{LCOE} = \frac{C^e}{E^{grid}} \ \text{[EUR/MWh]}, \qquad \mathrm{LCOH} = \frac{C^h}{H} \ \text{[EUR/kg]}, \tag{3.40}$$

and in energy terms $\mathrm{LCOH}_{\mathrm{MWh}} = 1000\,\mathrm{LCOH}/\mathrm{LHV}$ [EUR/MWh], where $C^e$ and $C^h$ are the costs allocated to each product, $C^e + C^h = \sum_c C_c$. Revenues and penalties are not part of the levelized costs. The app also reports the cost of generation, $(C_{wind} + C_{PV})/\sum_t \Delta_t g_t$, and, in the results workbook, the net unit costs of the earlier project model: $(\sum_c C_c - R^h)/E^{grid}$ and $(\sum_c C_c - R^e)/H$.

# 4

# Cost model

This chapter documents every cost in the planner: the formula, the default value, where the value comes from, and how it compares with the unit investment cost indicators that the European Union Agency for the Cooperation of Energy Regulators published on 24 April 2026 [22]. Chapter 3 states how these costs enter the optimisation; here the question is what each number is and how far it can be trusted.

## 4.1 Three kinds of source

Each default value comes from one of three places. The app shows the source of every input under its ? icon, the inputs workbook has a column `source of the default`, and Appendix A lists it.

**ACER 2026.** The median of ACER's unit investment cost indicators. ACER collects the costs of commissioned European projects from their promoters, removes inflation country by country and reports, for each asset class, the number of assets, the mean, the quartiles and the median. The planner uses the median because the samples are small and skewed. ACER covers transmission assets: cables, converter stations, substations, transformers, pipelines and compression stations. It publishes no indicator for generation, electrolysers or storage; its report says that promoters "did not submit any electrolyser".

**Engineering data and cost functions.** Where ACER gives one number for a class that the model needs to split (one cable cost for all voltages, one pipeline cost for a range of diameters), or no number at all, the value comes from manufacturer data, a published cost function or physics: cable capacitances and current ratings from the ABB XLPE guide [1], cable supply prices from NREL ORBIT [38], the HVDC cost function fitted to realised projects by Vrana and Härtel [48], a transformer cost function [33], and pipeline capacities from gas flow [10, 36, 43]. ACER's medians are then used to check the level.

**EVOLVE v9 model.** Every other value is the one used in the project's v9 model (`OffShore-Onshore_Setting.py`), unchanged. These values have no ACER counterpart. The cost values among them are marked as placeholders (• in Appendix A) until Step 3 of EVOLVE replaces them with referenced values.

ACER values are in euros adjusted for inflation up to 2025. The v9 values carry no stated price year. Mixing them is acceptable for comparing designs, but a project estimate should bring all values to one price base.

## 4.2 From investment to annual cost

Every component $c$ costs, each year,

$$C_c = \left(\mathrm{CRF}(r, n_c) + \phi_c\right) I_c, \tag{4.1}$$

with the investment $I_c$ [EUR], the lifetime $n_c$ [yr], the fixed O&M share $\phi_c$ [1/yr] and the capital recovery factor of Equation (3.2). At the default discount rate of 7 % the factor is 0.109 8 for 15 years, 0.094 4 for 20, 0.085 8 for 25, 0.080 6 for 30 and 0.075 0 for 40 years. A cable that costs 86 million EUR, lasts 30 years and has 0.5 % fixed O&M therefore costs $86 \times (0.0806 + 0.005) = 7.36$ million EUR a year. Grid fees and variable O&M are added as annual amounts. No component is replaced within the horizon: the model sizes one representative year, and the annuity spreads each investment over its own lifetime.

## 4.3 What each component includes

Table 4.1 answers a frequent question: where are the inverters, transformers, substations and water treatment? Power electronics are part of the component they serve, so they are inside its unit cost.

**Table 4.1.** What the cost of each component covers.

| Component | Included in its unit cost |
|---|---|
| Offshore wind | turbine with its converter, tower, foundation (or floater and moorings), array cables, installation (v9 value per MW) |
| Floating PV | modules, floats and anchoring, inverters, installation (v9 value per kW) |
| Electrolyser | stack and balance of plant, including rectifiers and transformers (“CAPEX, stack + balance of plant”) |
| Water treatment | desalination and purification feeding the electrolysers, per MW of electrolyser (v9) |
| Compressor | compression station per MW of electric drive (ACER) |
| Fuel cell | stack, inverter and balance of plant (v9) |
| Battery | energy part per MWh (cells, racks) and power part per MW (inverters, transformers) (v9) |
| Hydrogen tank | pressure vessels per kg stored (v9) |
| Export cable | AC: the cable circuits, supplied and installed, transformers at both ends, shunt reactors and, if chosen, a mid-route reactor platform; HVDC: the cable, the two converter stations and the offshore converter platform (Section 4.4) |
| Island platform | the structure (topside steel and jacket, or hull and moorings) that carries the island’s electrical and process equipment, installed, with engineering and contingency: a part per MW of AC export capacity and a part per MW of island equipment (Section 4.7); the equipment itself is priced in its own component. In concept C: the integration of an electrolyser in each turbine and the in-field hydrogen lines |
| Hydrogen pipeline | the pipeline, supplied and laid (ACER); a tie-in adds its own cost for the hot tap and metering |

Not modelled: onshore grid reinforcement, development and financing costs beyond the discount rate, decommissioning, and replacement of electrolyser stacks within the lifetime. Reactive compensation of AC cables is modelled (Section 4.4); reactive power demanded by the grid beyond the power factor of 0.95 is not.

## 4.4 Export link

### 4.4.1 The catalogue

The export link is chosen from a catalogue of complete link types. Each AC option is one voltage (66, 132, 220 or 275 kV), one conductor size (500, 630, 800 or 1 000 mm$^2$, three-core XLPE with copper conductors) and one compensation layout: shunt reactors at both ends, or at both ends and halfway along the route on a small platform. That gives 32 AC options. The HVDC option is a voltage-source-converter (VSC) link whose cable and converters are sized to the capacity built. For a route of length $L$ every option is evaluated before the solve: its usable rating per circuit, its costs and its losses (`evolve_planner/cables.py`). The optimiser then chooses the option, its whole number of circuits (AC) or converter units (DC) and the capacity, exactly (Section 3.8). No distance rule decides between AC and DC: the technology that wins at a given distance and size is a result.

**Table 4.2.** AC cable data used for every voltage: three-core XLPE, copper. Capacitance $C'$ in µF/km.

| | 500 mm$^2$ | 630 mm$^2$ | 800 mm$^2$ | 1 000 mm$^2$ |
|---|---|---|---|---|
| Current rating $I_r$ [A] | 655 | 715 | 775 | 825 |
| AC resistance at 90 °C, $R'$ [Ω/km] | 0.047 | 0.038 | 0.032 | 0.027 |
| $C'$ at 66 kV | 0.29 | 0.32 | 0.35 | 0.38 |
| $C'$ at 132 kV | 0.20 | 0.21 | 0.23 | 0.25 |
| $C'$ at 220 kV | 0.14 | 0.16 | 0.17 | 0.19 |
| $C'$ at 275 kV | 0.14 | 0.16 | 0.17 | 0.18 |

Current ratings and capacitances: ABB XLPE submarine cable systems guide (seabed 20 °C, burial 1 m, 1.0 K m/W, conductor 90 °C) [1]. Resistance: IEC 60287 method for round compacted copper at 90 °C.

### 4.4.2 AC physics: usable rating, compensation and losses

A cable is a long capacitor. Its charging current per phase and km is

$$i_c = \omega C' U/\sqrt{3}, \qquad \omega = 2\pi \cdot 50\,\text{Hz}, \tag{4.2}$$

7.6 A/km for a 220 kV, 1 000 mm$^2$ cable. With shunt reactors that absorb it at $m$ evenly spaced points ($m = 2$ for both ends, $m = 4$ with a mid-route reactor as well), the largest charging current in the conductor is $i_c L/m$, and it shares the thermal rating with the active current [8]. The usable active power of one circuit is

$$R_u(L) = \sqrt{3}\,U \cos\varphi \sqrt{I_r^2 - \left(i_c L/m\right)^2}, \tag{4.3}$$

with the power factor $\cos\varphi = 0.95$ that the link must provide at the grid connection. At $L = 0$ this is 89.6, 179.2, 298.6 and 373.3 MW for the four voltages at 1 000 mm$^2$. An option whose usable rating falls below 5 % of its rating at zero length is dropped: the charging current has used up the conductor. Figure 4.1 shows the derating. With reactors at both ends a 220 kV circuit keeps 97 % of its rating at 55 km and 89 % at 100 km; a mid-route reactor extends the reach to well beyond 200 km.

The charging power of one circuit, $Q = \omega C' U^2 L$ (2.9 MVAr/km at 220 kV, 1 000 mm$^2$), is fully compensated by shunt reactors, split equally between the two ends, or a quarter at each end and half mid-route.

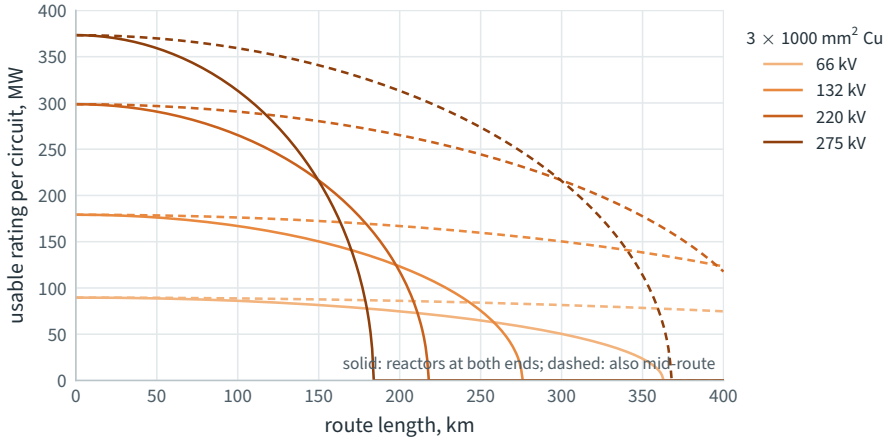



**Figure 4.1.** Usable rating of one AC circuit (1 000 mm$^2$) against route length, Equation (4.3).

Losses have a part that grows with the load and a part that does not. With the current $I = P/(\sqrt{3}U\cos\varphi)$, the load losses of one circuit are $3I^2R'k_{loss}L$, where $k_{loss} = 1.5$ adds the screen and armour losses [40]. As a share of the power at full use of the rating,

$$\rho_v = \frac{R'k_{loss}\,L\,R_v(L)}{U^2\cos^2\varphi} \qquad (U \text{ in kV},\ R_v \text{ in MW}). \tag{4.4}$$

In the model the loss of $n$ circuits carrying $f$ is $\rho_v f^2/(nR_v)$, a quadratic. It enters the energy balance as a piecewise-linear function over three equal segments of the installed rating, with the chord slopes $\rho_v/3$, $\rho_v$ and $5\rho_v/3$: exact at the segment ends and slightly above in between. Power sent from shore to the island, which is small, is charged the full-load share $\rho_v$. The no-load losses run all year:

$$P_0 = R'k_{loss}\,\frac{i_c^2L^3}{m^2} \;+\; \omega C'U^2L\tan\delta, \tag{4.5}$$

the conductor losses of the charging current and the dielectric losses ($\tan\delta = 0.001$ for XLPE). They are valued at 50 EUR/MWh, the order used in the AC/DC comparisons of Van Eeckhout et al. [47], Elliott et al. [16] and Xiang et al. [50].

### 4.4.3 AC costs

One AC circuit costs

$$I_v^{circ} = \left(c_v^{sup} + c^{inst}\right)L \;+\; (1+\mu)\,C^{tr}(S_r) \;+\; \sum_s \left(\mu_s c^Q Q_s + C_0^Q\right), \tag{4.6}$$

plus, once per link, the mid-route platform $C^{mid}$ if that layout is chosen. The sum runs over the reactor sites $s$ of one circuit, with $Q_s$ half of $Q$ at each end, or a quarter at each end and half mid-route; $\mu_s$ is the offshore premium $\mu$ at the island and 1 elsewhere.

**Cable supply $c_v^{sup}$.** The NREL ORBIT cable library, 2024 USD per km of one cable [38], converted at 0.924 EUR/USD (ECB 2024 average). ORBIT lists 66 kV 630 mm$^2$, 132 kV 500 mm$^2$, 220 kV 500 to 1000 mm$^2$ and 275 kV 1200 mm$^2$ and above; other sizes take the ECN cost function's size ratio ($0.2625S + 4.985V - 28.7$ kEUR/km, 2, 41) to the nearest ORBIT entry of the same voltage.

**Installation $c^{inst}$.** 0.91 million EUR per circuit-km, the ENTSO-E installation entry reviewed by Härtel et al. [25], at the top of the 0.23–0.98 million EUR/km range of ENTSO-E Regional Group North Sea [18]. With ORBIT's 220 kV, 1 000 mm$^2$ supply price this reproduces ACER's median for AC submarine cables, 2.30 million EUR/km (11 assets), to within 0.2 %.

**Transformers $C^{tr}$.** One at each end, sized to the circuit's apparent power $S_r = \sqrt{3}UI_r$, with the function of Lakshmanan et al. [33]: $1.15(105S^{0.447} - 122.4)$ kEUR (1.41 million EUR at 300 MVA; ACER's median for a 220/50 kV transformer is 1.68). The island one carries the offshore premium $\mu$.

**Shunt reactors.** 20 kEUR/MVAr plus 1 million EUR per site and circuit. Published values span about 10–30 kEUR/MVAr: ORBIT uses 10–14 kUSD/MVAr [38], Dicorato-type cost functions 10.5–15.8 kEUR/MVAr plus 0.8–1.2 million EUR, and ENTSO-E Regional Group North Sea [18] 2.8–3.0 million EUR for a 100 MVAr, 275 kV reactor. The per-MVAr part of the island reactor carries the offshore premium.

**Mid-route platform $C^{mid}$.** 50 million EUR, a placeholder: no public cost of a reactive compensation platform exists (Hornsea One's is inside its total transfer value). The cost of a small 300 MW AC platform in ENTSO-E Regional Group North Sea [18], about 32–48 million EUR in 2011 prices, escalated, is the basis; test 12–80 million EUR.

The structure that carries the island AC substation is part of the island platform (Section 4.7). Fixed O&M is 0.5 % a year and the lifetime 30 years.

**Dynamic section on a floating platform.** In concept B each export circuit leaves a floating hull, so its first `dyn_cable_km` (0.5 km) is a dynamic cable that hangs in the water column and follows the motion of the platform. Dynamic cables cost 30–50 % more than static cables up to 33 kV and 60–90 % more up to 66 kV (Collin et al., cited by 31); BVG's dynamic array cable is about 50 % dearer than the static one [5]. The planner adds

$$I_v^{dyn} = p^{dyn}\left(c_v^{sup} + c^{inst}\right)\min(\ell^{dyn}, L) \tag{4.7}$$

to each AC circuit, with $p^{dyn}$ = `dyn_cable_premium` (0.75, the middle of the 66 kV range), and scales the per-km cable terms of an HVDC link by $1 + p^{dyn}\min(\ell^{dyn}, L)/L$. Dynamic export cables are qualified only up to 66 kV [5, 11]: the BVG floating guide therefore uses a bottom-fixed substation in its floating reference. When the optimal design of a floating central island exports above `dyn_cable_max_kv` (66 kV) the app shows a warning; with **Floating: allow only qualified dynamic voltages** on, AC options above that voltage are not offered.

### 4.4.4 HVDC cost and losses

The HVDC link follows the cost function fitted by Vrana and Härtel [48] to 21 realised VSC projects (their QQEF set), which uses the structure of Härtel et al. [25]. For a link of capacity $p$ [GW] over $L$ km, with $u = \lceil p/2\,\text{GW} \rceil$ units,

$$I^{dc} = e\Big[\underbrace{B_{lp}Lp + u(B_l L + B_0)}_{\text{cable}} + \underbrace{2\,(N_p p + uN_0)}_{\text{two converter stations}} + \underbrace{\beta^{dc}(S_p p + uS_0)}_{\text{offshore station}}\Big], \tag{4.8}$$

with $B_{lp} = 0.98$ MEUR/(GW km), $B_l = 0.27$ MEUR/km, $B_0 = 3.63$ MEUR, $N_p = 113$ MEUR/GW, $N_0 = 23.5$ MEUR, $S_p = 723$ MEUR/GW and $S_0 = 57.3$ MEUR. The values are total project costs:

contract values plus a 25 % markup for owner's costs, at the price level of 2007–2016 contracts. $\beta^{dc}$ is the floating offshore-station premium (1.5 for concept B, a v9 value; a placeholder; the dynamic section of Equation (4.7) is added to the cable terms) and $e$ the HVDC cost escalation factor (`hvdc_cost_factor`, 1.0), which scales the whole link. Because the cable cost grows with $p$, the cable is sized to the power; because of the fixed terms, small links are dear per MW.

Table 4.3 tests the function against projects. It is within about ±25 % of each project, and its converter and platform part is within 6 % of TenneT's 2 GW programme (about 2.1 billion EUR per system for the converter stations and platforms, contracts of 2023). ACER's 2026 median of 0.21 MEUR/MW per converter station would put the converter stations of the same 2 GW link at well under half of what the fitted function and TenneT's contracts give, which is why the planner uses the fitted function and keeps an escalation factor (`hvdc_cost_factor`, 1.0) for sensitivity.

**Table 4.3.** HVDC cost function against reference costs (bottom-fixed, no floating premium).

| Project | Power [MW] | Length [km] | Function [M EUR] | Reference [M EUR] | Deviation |
|---|---|---|---|---|---|
| BorWin1 | 400 | 219 | 633 | 529 | +20 % |
| HelWin2 | 690 | 141 | 897 | 1057 | −15 % |
| DolWin1 | 800 | 188 | 1066 | 853 | +25 % |
| BorWin3 | 900 | 168 | 1156 | 1563 | −26 % |
| TenneT 2 GW system, stations only | 2000 | – | 2003 | ≈2140 | −6 % |

Reference costs: Vrana and Härtel [48], Table 1 (contracted cost ×1.25; length is the cost-equivalent length, sea cable plus 1.25 times land cable); TenneT: €30 billion for 14 systems, cables excluded (press release, 20 April 2023).

HVDC losses are 1 % per converter station (PyPSA technology data; v9) and, in the cable, 0.45 % per 100 km at full load, the middle of the values computed from ORBIT's 320 kV, 2 000 mm$^2$ (0.61 %) and 525 kV, 2 500 mm$^2$ (0.30 %) cables at 70 °C. Both are applied as fixed shares. The link's voltage is reported by its size, following practice: up to 600 MW the ±150–200 kV class, up to 1 250 MW ±320 kV, above that ±525 kV. Fixed O&M is 1 % a year.

### 4.4.5 Fees

The TSO grid fee of 42 000 EUR/MW/yr (v9, a placeholder) is charged per MW of capacity injected into the existing grid, for AC and DC alike. Power that crosses the cable only to feed onshore electrolysers is behind the meter and pays no fee. With a hub connection the hub access fee is charged per MW of the cable to the hub instead.

### 4.4.6 The options on the example route

Table 4.4 shows a selection of options on the 55 km Belle-Île route. Figure 4.2 compares the cheapest option of each family against the capacity, with losses left out. At 55 km the AC families are far cheaper than HVDC at every size; at 150 km HVDC is close to the best AC option from about 600 MW, and the losses, which favour HVDC at that length, decide.

### 4.4.7 Which technology wins: a result, not a rule

Table 4.5 gives the link the optimiser chooses for the Belle-Île conditions at route lengths from 30 to 250 km and three export sizes (sizing on 365 TDA blocks; the two larger cases are power-only

**Table 4.4.** Selected AC options on the 55 km route with reactors at both ends, and the HVDC option (concept B: floating converter-station premium and 0.5 km dynamic section included; the annual cost includes the grid fee on the full rating and the electrical platform part, $f_E = 1.518$).

| Option | Rating at 55 km [MW] | Cable [M EUR/km] | One circuit [M EUR] | Load loss at full use | No-load loss [kW] | Annual cost at full use [EUR/MW] |
|---|---|---|---|---|---|---|
| 66 kV, 500 mm$^2$ | 70.4 | 1.47 | 85.1 | 6.9 % | 57 | 168 200 |
| 66 kV, 630 mm$^2$ | 76.8 | 1.51 | 87.7 | 6.1 % | 59 | 162 362 |
| 132 kV, 500 mm$^2$ | 139.4 | 1.40 | 82.8 | 3.4 % | 127 | 115 562 |
| 132 kV, 1000 mm$^2$ | 175.6 | 1.48 | 88.0 | 2.5 % | 136 | 107 565 |
| 220 kV, 800 mm$^2$ | 272.3 | 1.77 | 106.0 | 1.6 % | 234 | 98 028 |
| 220 kV, 1000 mm$^2$ | 289.0 | 2.30 | 135.5 | 1.5 % | 256 | 104 857 |
| 275 kV, 1000 mm$^2$ | 356.2 | 2.14 | 128.8 | 1.2 % | 371 | 95 721 |
| HVDC (per 2 GW unit) | 2000 | – | 151.6 + 1.37/MW | 2.25 % | 0 | 172 549 |

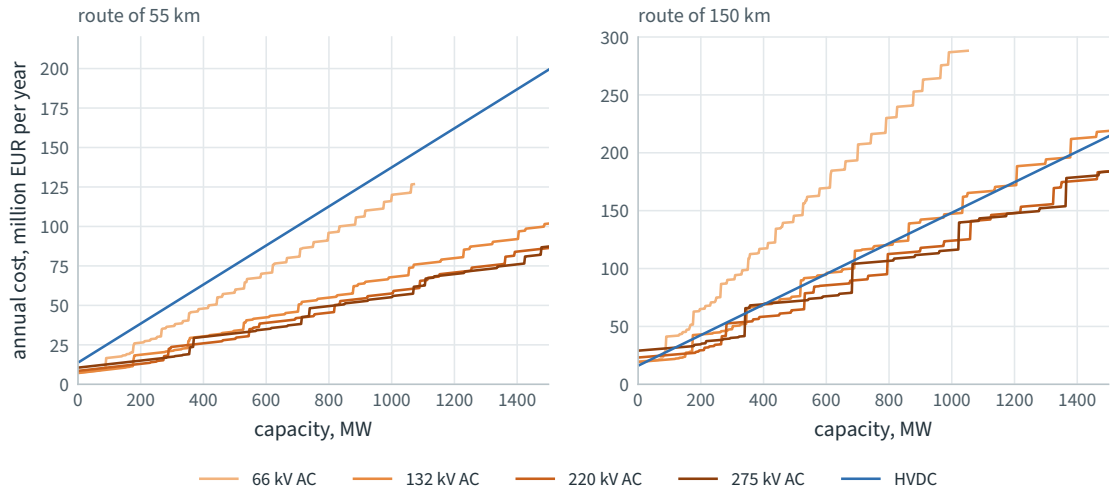



**Figure 4.2.** Annual cost (annuity, fixed O&M, grid fee, island platform share, no-load losses) of the cheapest option of each family against the capacity, on routes of 55 and 150 km. Load losses are not included; the optimiser adds them.

islands with a larger sea area and grid limit). AC wins up to 165 km for the small island and up to 180 km for the two larger ones; HVDC wins from 180 km and 190 km respectively. The link capacity is itself chosen by the optimiser, so it differs between route lengths (the column headers give the range). Below the break-even the optimiser moves to larger conductors, higher voltages and mid-route reactors as the route gets longer; HVDC wins when the AC load and no-load losses and the reactor costs outweigh the converter stations. The break-even is less sensitive to size here than in the literature because the Vrana–Härtel function charges HVDC fixed costs per 2 GW unit and has a cable cost that grows with the power, while the AC options grow by whole circuits. The published comparisons give a range rather than a number: Elliott et al. [16] find AC with compensation cheapest up to about 200 km at 250 MW and up to 120–160 km at 1 GW; Lauria et al. [35] put the break-even of a 1 GW link at 180–400 km with mid-route compensation; Xiang et al. [50], without mid-route compensation, at 87 km for 0.6 GW and 73 km for 1.4 GW; Van Eeckhout et al. [47] at 35 km for a single owner and 80 km with separate owners. None of them supports a fixed threshold; the v9 rule of 60 km lies within their range, near its low end. In the planner the answer depends on the route, the size, the cost data and

the value of losses, so the break-even should be read as a result of the inputs, and tested with the HVDC cost factor, the mid-route platform cost and the loss value.

**Table 4.5.** Export link chosen by the optimiser against route length and export size (Belle-Île weather and prices; 365 TDA blocks). AC entries are voltage [kV] / conductor [mm$^2$], '+ mid' for a mid-route reactor, and the number of circuits; the efficiency is energy delivered over energy sent.

| Route | Small island (about 60 MW) | | About 340–420 MW | | About 720–830 MW | |
|---|---|---|---|---|---|---|
| [km] | Link | Eff. | Link | Eff. | Link | Eff. |
| 30 | AC 132/500 ×1 | 99.2 % | AC 220/630 ×2 | 99.3 % | AC 220/800 ×3 | 99.3 % |
| 60 | AC 132/500 ×1 | 98.5 % | AC 275/1000 ×1 | 99.0 % | AC 220/800 ×3 | 98.6 % |
| 100 | AC 132/500 ×1 | 97.5 % | AC 275/1000 + mid ×1 | 98.3 % | AC 275/1000 + mid ×2 | 98.3 % |
| 150 | AC 132/500 ×1 | 96.3 % | AC 275/1000 + mid ×1 | 97.5 % | AC 220/800 + mid ×3 | 96.6 % |
| 165 | AC 132/500 ×1 | 96.0 % | AC 275/1000 + mid ×1 | 97.3 % | AC 220/800 + mid ×3 | 96.3 % |
| 180 | HVDC ±150–200 kV | 97.2 % | AC 275/1000 + mid ×1 | 97.0 % | AC 220/800 + mid ×3 | 96.0 % |
| 190 | HVDC ±150–200 kV | 97.1 % | HVDC ±150–200 kV | 97.1 % | HVDC ±320 kV | 97.1 % |
| 200 | HVDC ±150–200 kV | 97.1 % | HVDC ±150–200 kV | 97.1 % | HVDC ±320 kV | 97.1 % |
| 250 | HVDC ±150–200 kV | 96.9 % | HVDC ±150–200 kV | 96.9 % | HVDC ±320 kV | 96.9 % |

### 4.4.8 The v9 linear model

**Cable cost model** = **v9 linear** restores the cable cost of the v9 model, for comparison with earlier results: HVAC up to 60 km at $\left(1\,360+206\,130/\max(\bar{P}^{grid},1)\right)L$ EUR/MW with a loss of $6.7\times10^{-5}$ per km; HVDC beyond, at a converter pair of 300 000 EUR/MW (times the floating premium) plus 1 000 EUR/MW/km, with 2 % converter loss and $3.5\times10^{-5}$ per km. On the example route with a 100 MW grid limit v9 gives 188 kEUR/MW, about 12.5 million EUR for 66 MW; the catalogue's best option costs about 82 million EUR. For HVDC, v9 underestimates the reference projects of Table 4.3 by a factor of three to four.

## 4.5 Hydrogen pipeline

### 4.5.1 Classes

The pipeline is built in one of eight new sizes, 4, 6, 8, 12, 16, 20, 24 and 36 inch nominal diameter, or as a repurposed 36-inch gas pipeline when **Existing gas pipeline on the route can be repurposed** is on. The inner diameters (`pipe_id_*in`, from 0.102 to 0.876 m) are ASME B36.10 outside diameters minus a typical wall. A tie-in spur uses the same classes.

### 4.5.2 Cost per km

ACER reports two indicators for pipelines of its gas, biomethane and hydrogen blending category (GBH), natural gas assets that "can be utilised for hydrogen and low-carbon gases": a median of 2.08 million EUR/km for 12–16 inch (4 assets; quartiles 1.77 and 2.55) and 3.14 for 36–48 inch

(8 assets; quartiles 2.50 and 5.18). The planner draws a straight line through the two medians, placed at 14 and 42 inch:

$$c^{pipe}(d) = \left(c^0 + c^1 d\right) \beta^{pipe}, \qquad c^0 = 1.55 \text{ million EUR/km}, \quad c^1 = 37\,857 \text{ EUR/km per inch}, \tag{4.9}$$

with $d$ the nominal diameter in inch and a subsea factor $\beta^{pipe}$ (`pipe_offshore_factor`, 1.7). ACER's indicators do not separate subsea from onshore pipelines and are mostly onshore; the European Hydrogen Backbone puts offshore pipelines at 1.7 times the onshore cost of the same diameter [21], which the planner applies to the island-to-shore pipeline and the tie-in alike. Below 12 inch the line is an extrapolation without data; the fixed part, which stands for laying, right of way and project costs, then dominates, which is plausible but unverified. The repurposed class costs 1.18 million EUR/km times the same factor, ACER's median for hydrogen pipelines, all three of which are repurposed lines (`pipe_repurposed_capex_eur_km`). ACER warns that the sample is small. Fixed O&M is 0.9 % a year, the lifetime 40 years and the delivery efficiency 0.99 (v9).

Table 4.6 compares the line with other sources. The v9 values (1.0 and 3.7 million EUR/km) are close to the North Sea Energy cost function used by Singlitico et al. [43], 1.75 [$0.314 + 0.574 \times 10^{-3} D + 1.7 \times 10^{-6} D^2$] million EUR/km with $D$ in mm. Per-inch rules from offshore studies, 45 kEUR per inch and km [42], are lower. ACER's slope, 37.9 kEUR per inch and km, is close to those rules; the difference lies in the fixed part.

**Table 4.6.** New pipeline cost per km in million EUR.

| Source | 12 inch | 36 inch | Note |
|---|---|---|---|
| This model, Equation (4.9), subsea | 3.41 | 4.95 | ACER line ×1.7 |
| This model without the subsea factor | 2.00 | 2.91 | through ACER 2026 medians |
| European Hydrogen Backbone 2022, new | – | 2.2 / 3.7 | onshore / offshore, medium [21] |
| ACER 2026, GBH pipelines (median) | 2.08 | 3.14 | 12–16 and 36–48 inch classes |
| EVOLVE v9 | 1.00 | 3.70 | two classes only |
| North Sea Energy function [43] | 1.13 | 3.71 | with the inner diameters of this model |
| 45 kEUR per inch and km [42] | 0.54 | 1.62 | range 40–57 in the literature it cites |

### 4.5.3 Capacity from gas flow

The capacity of each class is the smaller of two limits. The *velocity limit* keeps the gas speed at the outlet, where the pressure is lowest and the gas fastest, at the design velocity $u_{max}$ and below a share $f$ of the erosional velocity of API RP 14E [36, 43]:

$$u = \min\left(u_{max},\ f\,\frac{C}{\sqrt{\rho_{out}}}\right), \qquad \rho_{out} = \frac{p_{out} M_{H_2}}{ZRT}, \qquad \dot{m}_{vel} = u\,\rho_{out}\,\frac{\pi D^2}{4}, \tag{4.10}$$

with $u_{max} = 20$ m/s (`pipe_max_velocity_ms`), the velocity at which the Danish Energy Agency's 2025 catalogue sizes hydrogen pipelines [10], $f = 0.5$, $C = 122$ (SI units; 100 in the US units of the standard), $Z = 1.05$ and $T = 288$ K. No code sets a velocity limit for hydrogen; the capacities that the European Hydrogen Backbone gives for 20 to 48-inch pipes correspond to about 20 to 30 m/s at the outlet, so 20 m/s is the cautious end. At 30 bar the erosional term is 39 m/s, so the design velocity binds. The *pressure-drop limit* is the flow that isothermal flow can push from the inlet pressure to the outlet pressure over the route,

$$p_{in}^2 - p_{out}^2 = \Lambda\,\frac{L}{D}\,\frac{ZRT}{M_{H_2}}\left(\frac{\dot{m}}{A}\right)^2, \qquad \frac{1}{\sqrt{\Lambda}} = -2\log_{10}\frac{\varepsilon}{3.7D}, \tag{4.11}$$

with the friction factor $\Lambda$ of fully rough flow and the wall roughness $\varepsilon = 0.05$ mm [43]. The inlet pressure is the compressor outlet pressure (70 bar), the outlet pressure the delivery pressure (30 bar). For a tie-in, the inlet is the host pipeline's pressure plus 5 bar and the outlet the host pressure.

Table 4.7 and Figure 4.3 show the result on the 42 km pipeline route of the example. Every class is limited by the design velocity; the pressure-drop limit is 1.4 to 5 times higher on this route and binds only on much longer routes or with a smaller pressure difference.

**Table 4.7.** Pipeline classes on a 42 km subsea route, 70 to 30 bar.

| Class | Inner diameter [m] | Cost [M EUR/km] | Route cost [M EUR] | Capacity [kg/h] | Pressure-drop limit [kg/h] |
|---|---|---|---|---|---|
| new 4 inch | 0.102 | 2.89 | 121.9 | 1 424 | 2 025 |
| new 6 inch | 0.154 | 3.02 | 127.3 | 3 230 | 5 900 |
| new 8 inch | 0.203 | 3.15 | 132.8 | 5 589 | 12 051 |
| new 12 inch | 0.305 | 3.41 | 143.6 | 12 646 | 34 860 |
| new 16 inch | 0.384 | 3.66 | 154.5 | 20 080 | 63 566 |
| new 20 inch | 0.483 | 3.92 | 165.3 | 31 682 | 114 909 |
| new 24 inch | 0.581 | 4.18 | 176.2 | 45 982 | 186 303 |
| new 36 inch | 0.876 | 4.95 | 208.7 | 104 340 | 538 777 |
| repurposed 36 inch | 0.876 | 2.01 | 84.6 | 104 340 | 538 777 |

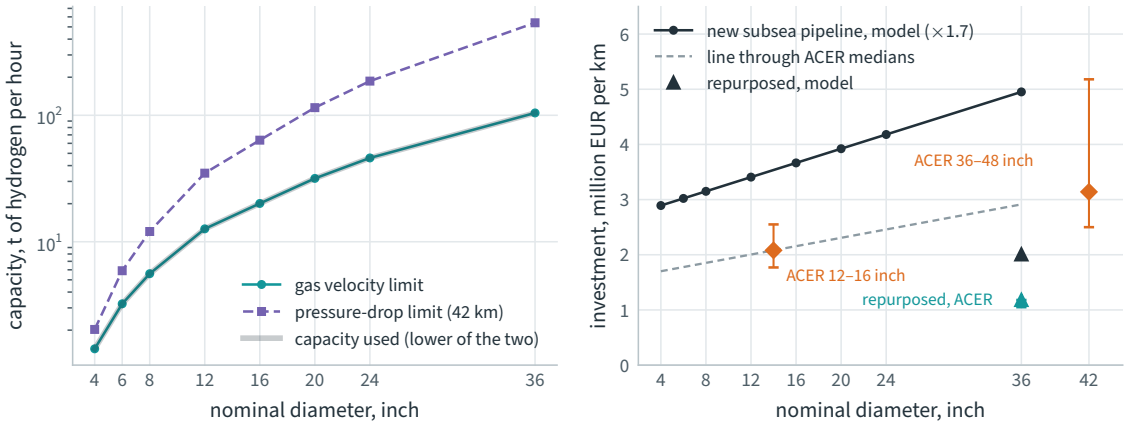



**Figure 4.3.** Left: capacity of each pipe size on the 42 km example route, as the lower of the velocity and pressure-drop limits. Right: cost per km, with the subsea factor and without it, and ACER's medians and quartile ranges.

The smallest pipe costs 122 million EUR on this route and carries 1.4 t/h, the hydrogen of about 63 MW of electrolysis. A pipeline is therefore worth building only for large hydrogen flows or short routes; below that, electrolysis onshore with the electricity sent through the cable is usually cheaper.

## 4.6 Compressors

Compressors cost 5.08 million EUR/MW of electric drive (`comp_capex_eur_mw`): ACER's median for GBH compression stations of 10–50 MW (3 assets). ACER's median for stations below 1 MW is 17.13 million EUR/MW (6 assets). An island compressor is often below 1 MW, so the default may be low for small projects; it is 1.7 times the 3 000 EUR/kW of Singlitico et al. [43] and 2.5 times the v9 value of 2 million EUR/MW. Fixed O&M is 3 % and the lifetime 20 years (v9). The work per kg follows Equation (3.35).

## 4.7 Island platform

The island platform is the structure that carries the island's equipment: topside steel and a jacket in concept A, topside steel and a floating hull with moorings in concept B. The equipment itself (transformers and reactors, electrolysers, compressors, fuel cells, batteries) is priced in its own component. Version 0.10 replaces the v9 rule, a single factor of 2 for any floating platform, with a model in which every term has a stated basis. In concept C there is no central platform; Section 4.7.3 gives the costs that take its place.

### 4.7.1 Cost of the structure

The investment is

$$I^{plat} = 1000\, k^{esc} k^{inst} \left[ f_E\, c_E\, \mathbb{1}_v^{AC} P^{cab} + f_H\, c_H\, S^H \right] + C_0, \qquad S^H = P^{ez,\mathrm{off}} + w_c P^{cmp,\mathrm{off}} + w_f P^{fc,\mathrm{off}} + w_b P^{b,\mathrm{off}}, \tag{4.12}$$

in EUR, with the capacities in MW. Table 4.9 lists the inputs.

**Coefficients $c_E$ and $c_H$.** Both come from the platform cost model of the DNV GL study for TenneT and Gasunie on power-to-hydrogen at IJmuiden Ver [12], in the linear form used by Bouma [3, Eq. 3.3]. DNV GL tabulates topside and jacket costs for platforms of 100 to 2 000 MW in 30 m of water. Per MW they are 132.6, 138.2, 138.9 and 139.4 kEUR/MW for the electrolysis platform at 100, 500, 1000 and 2 000 MW, which reproduces $c_H$ = 139.281 kEUR/MW, and 76.7, 83.5, 85.6 and 88.8 kEUR/MW for the HVDC converter platform, which brackets $c_E$ = 84.651 kEUR/MW. The coefficients therefore price *structure*: $c_E$ the structure needed to carry the export electrical equipment, per MW of export capacity, and $c_H$ the structure for electrolysis and other process equipment, per MW of that equipment. They are not the cost of the cable or of the electrolyser (the electrolysis topside of about 100 kEUR/MW at 1 GW is far below any electrolyser price, so the stacks are excluded and pricing the electrolysers separately is not double counting). The cable part applies to AC links only, because an HVDC link carries its own converter platform in Equation (4.8).

Three limits apply. First, the price year is not stated in either source; the planner assumes 2018 EUR, the date of the DNV GL report, and keeps the values with $k^{esc} = 1$. Offshore costs have risen since: the Danish Energy Agency reports about 30 % real growth of offshore wind CAPEX between its 2021/22 data and investment decisions in 2025 [9], so $k^{esc}$ between 1.2 and 1.4 is a useful sensitivity. Second, DNV GL prices steel at 2000 EUR/t throughout and quotes a mass uncertainty of +25/−30 %, so the five significant figures overstate the precision. Third, Bouma's capacity is the electrolyser's only; the planner also counts island compressors, fuel cells and

battery power with weights $w_c$, $w_f$, $w_b$ (default 1, the behaviour of v9). That extension has no source; a weight of 0 is the natural test.

**Installation, engineering and contingency $k^{inst}$.** The value 1.5 is kept from v9 and Bouma, but neither source defines it, and DNV GL puts installation at "circa 15 % of the fabrication and supply". The planner documents it as

$$k^{inst} = \underbrace{1.15}_{\text{installation (DNV GL)}} \times \underbrace{1.30}_{\text{engineering, owner's cost, contingency}} \approx 1.5, \tag{4.13}$$

where the second factor is a stated assumption after the owner's cost, design and contingency factors used by Ekhar [15]. A range of 1.15 to 1.6 should be tested.

**Fixed term $C_0$.** The linear model runs through the origin, but a real platform has items that do not scale with capacity: cranes, accommodation, standby generation, a helideck. NREL's ORBIT model carries a fixed topside mass of 285 t and ancillary items worth together about 12 million USD [38]. For a platform of a gigawatt this is small; for an island of tens of MW it can match the variable term. The default is 0; `platform_fixed_eur` adds it.

### 4.7.2 Configuration factors

The coefficients describe a jacket in 30 m. The DNV GL tables put the jacket at a share $s_H = 0.28$ of the electrolysis platform cost (0.27–0.28 above 500 MW) and $s_E = 0.22$ of the HVDC platform cost. Depth and floating change that share only; the topside stays.

**Concept A, bottom-fixed.** The jacket cost grows with depth $d$:

$$f = (1 - s) + s\left(d/d_{ref}\right)^{\alpha}, \qquad d_{ref} = 30\,\text{m}. \tag{4.14}$$

Ekhar [15] gives a jacket mass per metre of depth that is linear in the topside mass, so $\alpha \approx 1$ (default). In the BVG guides the substation foundation grows from 22 to 65 kGBP/MW between 50 and 100 m [4, 5], which implies $\alpha \approx 1.56$ (high case). Platform O&M is 2 % a year [3].

**Concept B, floating central platform.** The jacket is replaced by a hull with moorings that costs $r$ times as much, and the topside needs a motion-tolerant design that costs $u$ times as much:

$$f = (1 - s)\,u + s\,r. \tag{4.15}$$

No floating substation or floating hydrogen platform has been priced in public. The only floating substation built, the 25 MVA, 66 kV unit of the Fukushima demonstration (2013), has no published cost; DNV expects bottom-fixed substations to stay competitive to about 100 m [11] and BVG expects floating substations in the early to mid-2030s. The ratio $r$ is therefore taken from turbine substructures: a semi-submersible with six-point mooring costs about 3.9 times a monopile in the BVG guides (100 m against 50 m), and Myhr et al. [37] give a manufacturing factor of 2.0 for a semi-submersible against a monopile. The default $r = 3$ lies between. Dolphyn's barge trial found electrolyser output varying with pitch and roll and sloshing in the separator [20], which is the reason for $u = 1.1$ (stated, 1.0–1.2). The defaults give $f_H = 1.632$ and $f_E = 1.518$; the range is 1.28 ($u = 1$, $r = 2$) to 1.96 ($u = 1.2$, $r = 3.9$).

The v9 factor of 2 lies at the upper edge of that range and has no source. It was attributed to the BVG floating guide's substation of 282 kGBP/MW, but that guide uses a *bottom-fixed* jacket substation in 100 m of water, because dynamic export cables above 66 kV are not yet qualified [5]; its ratio to the substation of the bottom-fixed guide (240 kGBP/MW in 50 m) is 1.18, a depth effect on a jacket. Other models apply no floating premium at all: NREL's floating reference uses the same substation cost as its fixed one [44], ORBIT's floating substation drops the piles and adds no moorings [38], and Lanni et al. [34] use the same platform for floating and fixed cases.

Platform O&M is 3 % a year for a floating central platform, a stated value within the published 1 % [46], 2 % [3] and 7 % [34]: central floating vessels need a permanent crew [31], and the Dolphyn concept study assumed two to three manned vessels for a 4 GW central option [19]. The dynamic section of the export cable is part of the export link (Equation (4.7)).

**Where the concepts cross.** Table 4.8 gives the factors and the resulting cost per MW. A bottom-fixed process platform reaches the floating factor at about 98 m with $\alpha = 1$ and 64 m with $\alpha = 1.56$; Ekhar [15] finds the crossover between a jacket and a semi-submersible for a 1 GW hydrogen platform near 50 m, and DNV near 100 m. In the planner the concept follows the **Fixed/floating threshold depth** (60 m); in the band from 50 to 100 m it is worth running both concepts by moving that threshold.

**Table 4.8.** Platform factors and structure cost per MW with the defaults ($k^{inst} = 1.5$, $k^{esc} = 1$, $C_0 = 0$).

| Concept and depth | $f_H$ | $f_E$ | Process part [kEUR/MW] | Electrical part [kEUR/MW] |
|---|---|---|---|---|
| A, 30 m | 1.000 | 1.000 | 208.9 | 127.0 |
| A, 55 m | 1.233 | 1.183 | 257.7 | 150.3 |
| A, 60 m | 1.280 | 1.220 | 267.4 | 154.9 |
| A, 60 m, $\alpha = 1.56$ | 1.546 | 1.429 | 322.9 | 181.4 |
| A, 100 m | 1.653 | 1.513 | 345.4 | 192.2 |
| B, any depth | 1.632 | 1.518 | 341.0 | 192.8 |
| v9, floating (all designs) | 2 | 2 | 417.8 | 254.0 |

### 4.7.3 Turbine-level electrolysis (concept C)

In concept C each turbine carries its own electrolyser, as in the Dolphyn design: PEM electrolysis and desalination on each semi-submersible turbine, hydrogen at 30–50 bar through a flexible riser into in-field lines, no power export cable, turbines about 1 km apart [19, 20]. The only hardware built is a barge-mounted electrolyser of about 180 kW tested in 2024 [20], and the company publishes no costs [13]. Version 0.9 priced concept C like a central jacket platform. That cannot show the two effects that decide the comparison: the central platform and array cables are avoided, but small electrolysers lose economies of scale. Version 0.10 sets $f_E = f_H = 0$ and adds four terms:

$$\begin{aligned} I^C = {} & 1000\, k^{esc} k^{inst} f_C\, c_H\, S^H && \text{integration in each turbine} \\ & + 1000\,(k^{scale} - 1)\, c^{ez} \mu\, P^{ez,\mathrm{off}} && \text{scale penalty of small electrolysers} \\ & + c^{inf} \ell^{inf}\, P^{wt} / P^{wt}_{unit} + C^{man} && \text{in-field hydrogen lines and manifold} \\ & - c^{arr} P^{wt} && \text{array cables not needed} \end{aligned} \tag{4.16}$$

and island electrolysers make $(1 + g)$ times as much hydrogen per MWh (Equation (3.13)).

**Integration** $f_C = 0.5$. Floater and topside reinforcement, desalination, a buffer tank and standby power in each turbine, as a share of the central process-platform cost. No public source prices the extra buoyancy and steel; Lanni et al. and Travaglini et al. use 0 and version 0.9 used 1, so 0.5 is a stated middle value.

**Scale penalty** $k^{scale} = 1.3$. Singlitico et al.'s PEM cost curve makes a 15 MW unit 1.30 times the specific cost of a plant of 100 MW or more [43]; DNV GL's rule that stacks scale linearly and gas handling with an exponent of 0.7, applied to Ekhar's cost split, gives 1.24. Travaglini et al. scale the whole plant with an exponent of 0.69 and find turbine-level electrolysers three to four times dearer [46]; that is the high case.

**In-field lines.** 0.55 million EUR/km (0.2–0.9), O&M 0.5 % a year [46], over 1 km per turbine; a per-GW rate such as Lanni et al.'s 0.96 MEUR/(GW km) would price the line from one turbine at only 14 kEUR/km. The collection manifold, 50 kEUR [34], is added to the fixed cost of the export pipeline and is probably low. Flexible risers have no published cost.

**Array-cable credit** $c^{arr}$. 145 kEUR/MW, the floating intra-array connection of Lanni et al. [34, Table 3], subtracted from the wind CAPEX because that CAPEX includes array cables (Table 4.1).

**Yield gain** $g = 0.5\,\%$. The intra-array loss that Lanni et al. charge to central designs and that a turbine-level electrolyser avoids.

The integration term has a life of `platform_life` and 2 % O&M, the in-field lines the pipeline life. Both are hydrogen-only components in the levelized costs (Section 3.9).

**What decides the ranking.** Per MW of electrolyser, with as many MW of electrolyser as of wind, the defaults give for concept C an integration cost of 104.5 kEUR/MW, in-field lines of 36.7 kEUR/MW, an array credit of −145 kEUR/MW and a scale penalty of $0.3 \times 2\,000\,\text{EUR/kW} \times 1.1 = 660\,\text{kEUR/MW}$: about 656 kEUR/MW against 341 kEUR/MW of structure for a floating central platform. With the planner's electrolyser price, the scale penalty decides the comparison; with $k^{scale} = 1$ concept C would be cheaper than B, with Travaglini et al.'s whole-plant scaling it would be far dearer. The published comparisons disagree for this reason: those that find turbine-level electrolysis cheapest remove the central platform and array cables without adding a scale penalty [19, 34], and those that find it dearest add the penalty [43, 46]. A result for concept C should therefore be reported with its sensitivity to $k^{scale}$ and $f_C$.

**Table 4.9.** Inputs of the island platform (group **Island platform**).

| Input | Default | Range to test | Basis |
|---|---|---|---|
| $c_H$, process structure | 139.281 kEUR/MW | ±30 % | DNV GL 2018 via Bouma 2023; origin confirmed |
| $c_E$, electrical structure | 84.651 kEUR/MW | ±30 % | DNV GL 2018 HVDC platform via Bouma 2023; origin probable |
| $k^{inst}$ | 1.5 | 1.15–1.6 | 1.15 from DNV GL, 1.30 stated (Equation (4.13)) |
| $k^{esc}$ | 1.0 | 1.2–1.4 | 2018 EUR assumed; DEA: about +30 % real since 2021/22 |
| $C_0$ | 0 | 0–12 M | ORBIT fixed topside and ancillaries |
| $w_c, w_f, w_b$ | 1 | 0–1 | none (Bouma counts electrolysers only) |
| $s_H, s_E$ | 0.28, 0.22 | 0.24–0.28, 0.17–0.22 | DNV GL 2018 tables |

| Input | Default | Range to test | Basis |
|---|---|---|---|
| $\alpha$ | 1.0 | 1.0–1.56 | Ekhar 2021; BVG guides |
| $u$ | 1.1 | 1.0–1.2 | stated (motion effects in the Dolphyn trial) |
| $r$ | 3 | 2.0–3.9 | Myhr et al. 2014; BVG guides (turbine substructures) |
| O&M, A and C / B | 2 % / 3 % | 1–7 % | Bouma 2023 / stated within Travaglini 2025 and Lanni 2025 |
| Dynamic section | 0.5 km, +75 % | +30–90 % | Collin et al. via Ibrahim et al. 2022 |
| Qualified dynamic voltage | 66 kV | – | DNV 2021; BVG 2024 |
| $f_C$ | 0.5 | 0–1 | stated; no source exists |
| $k^{scale}$ | 1.3 | 1.0–3.7 | Singlitico 2021; DNV GL with Ekhar 2021; Travaglini 2025 |
| In-field lines | 0.55 M EUR/km, 1 km per turbine | 0.2–0.9 M EUR/km | Travaglini 2025; ERM 2019 |
| Manifold | 50 kEUR | – | Lanni 2025 (probably low) |
| Array credit | 145 kEUR/MW | ±50 % | Lanni 2025, Table 3 |
| Yield gain $g$ | 0.5 % | 0–1 % | Lanni 2025 |

## 4.8 Components without an ACER indicator

Table 4.10 lists the cost values that keep the v9 value (or, for floating wind, the battery power and the platform, replace it), with a comment where the value looks doubtful. They are the placeholders of Step 3. Public catalogues that can replace them include the Danish Energy Agency technology data [9] and the NLR Annual Technology Baseline [39].

**Table 4.10.** Cost values kept from the v9 model.

| Component | Value | Comment |
|---|---|---|
| Offshore wind, bottom-fixed | 2.54 million EUR/MW, 2.5 %/yr, 25 yr | Plausible for bottom-fixed wind. · |
| Offshore wind, floating | 3.429 million EUR/MW, 3.0 %/yr | Not a v9 value since version 0.10: 1.35× the bottom-fixed value. Floating wind costs 1.33× bottom-fixed in Lanni et al. [34, Table 3], 1.36× in NREL's 2024 cost review [44] and 1.56–1.70× in the BVG guides [4, 5]; test 3.30–4.32 million EUR/MW. The Lanni build-up (turbine and floater 1.73, moorings 0.12, array 0.145 million EUR/MW, installation 8.4 %, development 3.8 %) sums to 2.24–2.27 million EUR/MW, so it does not support the v9 value of 2.54 for floating. · |
| Floating PV | 1 500 EUR/kW, 2 %/yr, 25 yr | Offshore floating PV is immature; wide uncertainty. · |
| Electrolyser | 2 000 EUR/kW, 1.5 %/yr, 15 yr | Includes balance of plant. ACER's "processing plant" (4.99 million EUR/MW) is not an electrolyser. · |
| Water treatment | 20 750 EUR/MW of electrolyser, 10 %/yr, 20 yr | Small next to the electrolyser. |

| Component | Value | Comment |
|---|---|---|
| Fuel cell | 1 170 EUR/kW, 3 %/yr, 15 yr | |
| Battery, energy | 160 000 EUR/MWh, 2 %/yr, 15 yr | |
| Battery, power | 100 000 EUR/MW | The v9 value, 5 343 913 EUR/MW, made a 4-hour battery cost about 1 500 EUR/kWh. Version 0.9.1 uses 100 kEUR/MW for power conversion and grid interface: a 4-hour battery then costs 185 EUR/kWh, close to BNEF's 2025 European turnkey average of 177 USD/kWh [17]. |
| Hydrogen tank | 650 EUR/kg, 2 %/yr, 20 yr | |
| Island platform | 84.651 and 139.281 kEUR/MW, $k^{inst} = 1.5$, factors $f_E$, $f_H$ by concept and depth | Revised in version 0.10 (Section 4.7); v9 used $\times 2$ for any floating platform, a factor with no source. Without a fixed term $C_0$ the cost of a small island is lower than any real platform. • |
| Offshore premium | $\times 1.10$ on island equipment | • |
| Floating HVDC station | $\times 1.5$ on the offshore converter platform | The v9 premium for a floating converter station, applied to the Vrana–Härtel platform terms. • |
| Mid-route reactor platform | 50 million EUR | Not a v9 value: no public cost exists (Section 4.4); test 12–80 million EUR. • |
| Island batteries and tanks | +0.5 % fixed O&M | 2.5 % instead of 2.0 %, as in v9. |
| TSO grid fee, hub access fee | 42 000 EUR/MW/yr | Depends on the national tariff and the hub owner. • |
| Tie-in cost | 5 million EUR | Assumption for the hot tap and metering; ACER's metering station median is 0.81 million EUR. • |

## 4.9 Comparison with ACER 2026

Table 4.11 puts the model's defaults next to the ACER medians, with the v9 values that the planner used before version 0.8.0.

**Table 4.11.** Model defaults, v9 values and ACER 2026 medians.

| Asset | Model (0.9.0) | v9 | ACER 2026 median (P25–P75; assets) | Use |
|---|---|---|---|---|
| Offshore RES export cable | AC 1.40–2.30 M EUR/km per circuit, supplied and installed | 0.21 M EUR/km + 1.36 kEUR/MW/km | 3.67 M EUR/km (2.98–4.61; 18) | check only |
| AC submarine cable | 2.30 M EUR/km at 220 kV, 1000 mm$^2$ | – | 2.30 M EUR/km (1.49–2.67; 11) | sets the installation cost |
| DC submarine cable | $0.98p + 0.27$ M EUR/km per 2 GW unit ($p$ in GW) | 1 kEUR/MW/km | 1.38 M EUR/km (1.01–1.56; 8) | check only |

| Asset | Model (0.9.0) | v9 | ACER 2026 median (P25–P75; assets) | Use |
|---|---|---|---|---|
| HVDC converter station | 113 M EUR/GW + 23.5 M EUR per station; offshore platform 723 M EUR/GW + 57.3 M EUR | 0.30 M EUR/MW per pair | 0.21 M EUR/MW (0.16–0.24; 10) | check only (ACER is low against recent contracts) |
| Shunt reactor | 20 kEUR/MVAr + 1 M EUR per site | – | none | – |
| Transformer | 1.41 M EUR at 300 MVA (function) | – | 1.68 M EUR, 220/50 kV (1.55–2.45; 7) | check only |
| Offshore AC substation | in the platform | in the platform | 168 M EUR/asset (129–178; 11) | check only |
| Hydrogen pipeline, new | $(1.55 + 0.0379d) \times 1.7$ M EUR/km | 1.0 (12 in), 3.7 (36 in) | GBH: 2.08 (12–16 in), 3.14 (36–48 in) | line through medians, times the subsea factor |
| Hydrogen pipeline, repurposed | $1.18 \times 1.7$ M EUR/km | 0.40 | 1.18 M EUR/km (1.05–1.18; 3) | used, times the subsea factor |
| Compression station | 5.08 M EUR/MW | 2.0 | 5.08 (10–50 MW; 3); 17.13 (<1 MW; 6) | used (10–50 MW) |
| Metering station | – | – | 0.81 M EUR/asset | check of the tie-in cost |
| Electrolyser, wind, PV, batteries, fuel cells, tanks | v9 | v9 | none | placeholders |

## 4.10 Effect on the example

The costs of this chapter raise the cost of the Belle-Île example (Chapter 8). With the v9 cable and compressor costs (**Cable cost model = v9 linear**) and the other defaults of version 0.10, the island runs at a net cost of 7.21 million EUR a year, with an LCOE of 128.8 EUR/MWh and an LCOH of 6.73 EUR/kg. With the costs of this chapter it runs at a net cost of 13.72 million EUR a year, with an LCOE of 151.0 EUR/MWh and an LCOH of 7.75 EUR/kg. Almost all of the difference is the export link: the optimiser chooses one 132 kV, 500 $mm^2$ circuit with reactors at both ends (139 MW usable, 82.8 million EUR with the dynamic section), which costs 7.14 million EUR a year against 1.03 million EUR with the v9 cost. It carries at most 65.3 MW, less than half its rating, yet it is the cheapest choice: the 66 kV circuits that would just fit cost more, because the cost per km hardly falls with the voltage, and lose twice as much at full load. Its losses, 5.4 GWh a year (load losses of 1.5 % of the energy sent, plus 1.1 GWh of no-load losses), raise the wind capacity from 76.6 to 77.9 MW. The hub option, with a 9 km cable, costs 8.49 million EUR a year, 5.2 million EUR less than the island's own link.

Between versions 0.9 and 0.10 the net cost of the example rose from 6.35 to 13.72 million EUR a year. Most of the rise is the floating wind CAPEX (3.43 instead of 2.54 million EUR/MW, about 8.0 million EUR a year on 78 MW); the platform is cheaper than before (1.46 instead of 1.78 million EUR a year, factor 1.52 instead of 2 on the electrical part), and the dynamic section adds 0.5 million EUR of investment.

# 5

# Time-series reduction with topological data analysis

Solving the model of Chapter 3 on all 8 760 hours is possible but slow, and the planner may have to solve it several times in one run (once per connection option, and for many link options and pipeline classes). The planner therefore sizes the system on a reduced year and checks the result on the full year. This chapter explains how the year is reduced, what the reduction guarantees, and how the full-year check repairs the design.

## 5.1 Why keep the chronology

The common way to shorten a year is to cluster days or weeks into a few representative periods [28]. Storage then needs extra modelling to carry energy from one period to the next [32], and the calendar order of weather is lost inside the representatives. An island that stores hydrogen through a calm month depends on exactly that order. The planner keeps the year in calendar order and only merges neighbouring hours into blocks of variable length. Where to cut the blocks is decided by the shape of the cumulative energy balance, analysed with persistent homology [14].

## 5.2 The cumulative energy balance

Let $x_a(t)$, $t = 1, \dots, 8760$, be an hourly attribute (the wind capacity factor, say) with annual mean $\bar{x}_a$. Its cumulative balance is

$$S_a(t) = \sum_{\tau=1}^{t} \bigl(x_a(\tau) - \bar{x}_a\bigr), \qquad S_a(0) = S_a(8760) = 0. \tag{5.1}$$

$S_a$ rises during hours above the mean and falls during hours below it (Figure 5.1). A deep trough is a long deficit: a lull for wind. Consider a design with $P^{wt}$ of wind that must deliver a flat power equal to its mean output. The energy its storage must shift is $P^{wt}$ times the largest drop of $S_{wt}$ from a peak to a later trough. More generally, for wind, PV and a constant demand $D$ the net balance is

$$B(t) = P^{wt} S_{wt}(t) + P^{pv} S_{pv}(t) + \bigl(P^{wt}\bar{x}_{wt} + P^{pv}\bar{x}_{pv} - D\bigr)\, t, \tag{5.2}$$

a linear combination of the cumulative balances plus a straight line. The surplus and deficit episodes that size storage are the peaks and troughs of $B$.

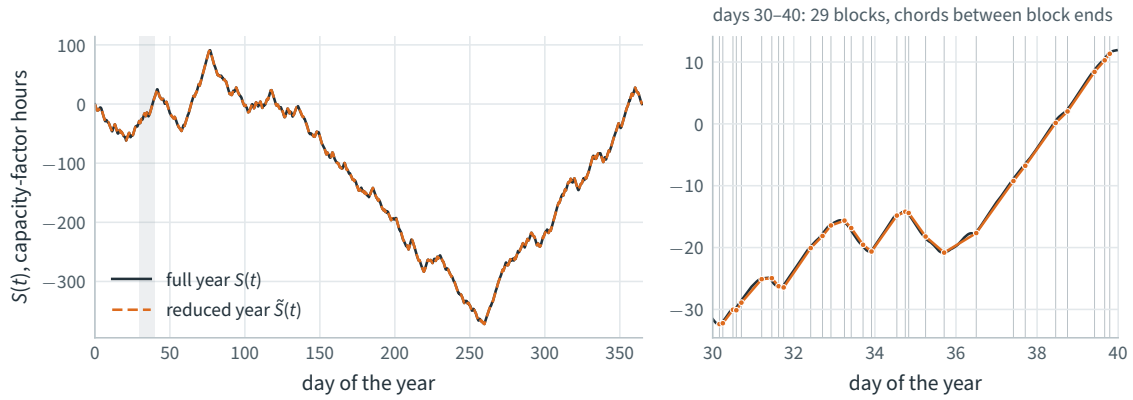


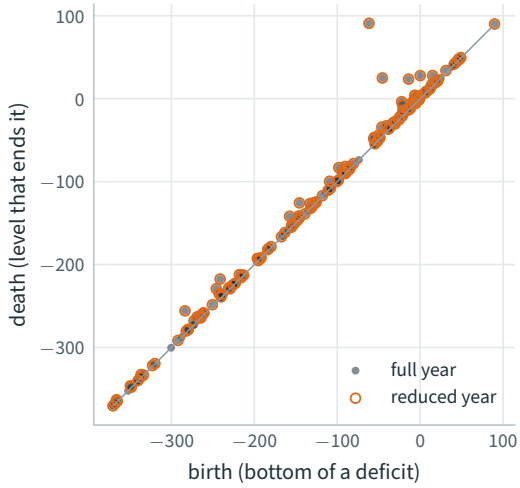

**Figure 5.1.** Cumulative balance of the wind capacity factor at Belle-Île in 2019, for the full year (dark) and the reduced year of 1 460 blocks (dashed). Right: days 30–40, where the reduced balance is the chord between block ends (dots); vertical lines are block boundaries.

## 5.3 Persistence of peaks and troughs

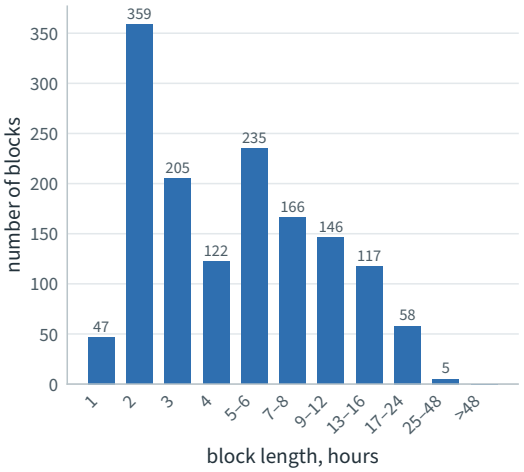
Persistent homology measures how deep each trough of a function is, relative to its surroundings, without choosing a smoothing scale.

**Definition 5.1** (Sublevel-set persistence on a line). *For a function $S$ on the hours $1, \dots, T$, sweep a level $\alpha$ upward and watch the set of hours with $S(t) \le \alpha$. A new interval of hours appears (is* born*) at each local minimum; when two intervals join at a local maximum, the one with the higher minimum* dies *(the elder rule). Each local minimum $t_b$ is paired with the level $d$ at which its interval dies, giving a point $(b, d) = (S(t_b), d)$ of the persistence diagram* $\mathrm{Dgm}(S)$. *Its* persistence $d - b$ *is the depth of the trough. The global minimum never dies and is paired with the maximum of $S$.*

For the cumulative balance, the persistence of a trough is the energy (in capacity-factor hours) that a store must provide to ride through the deficit episode before the balance recovers. Applying the same construction to $-S$ gives the peaks, the surplus episodes. The planner computes both diagrams with a union–find sweep in $O(T \log T)$ (`tda.py`, `persistence_pairs`). Figure 5.2 (left) shows the diagram of the wind balance: points far above the diagonal are the long deficits.




**(a)** Persistence diagram of the wind balance (troughs), full year and reduced year.




**(b)** Length of the 1 460 blocks. Short blocks sit where the balance turns; calm, steady spells merge into long ones.

**Figure 5.2.** What the reduction keeps, Belle-Île 2019.

## 5.4 What a block reduction preserves

A reduction cuts the year into $N$ consecutive blocks $[s_j, e_j]$ and replaces each attribute by its mean over the block. Write $\tilde{S}_a$ for the cumulative balance of the reduced series, expanded to hours.

**Proposition 5.1** (Chord property). *$\tilde{S}_a(e_j) = S_a(e_j)$ at every block end, and $\tilde{S}_a$ is linear within each block. Hence*

$$\|S_a - \tilde{S}_a\|_\infty = \max_j \max_{s_j \le t \le e_j} \left|S_a(t) - \chi_j(t)\right| =: e_a,$$

*where $\chi_j$ is the chord from $S_a(s_j - 1)$ to $S_a(e_j)$.*

*Proof.* A block mean keeps the sum of the attribute over the block, so the balance after the block is unchanged. Within the block every hour adds the same amount, so $\tilde{S}_a$ is a straight line between the two block ends. □

**Proposition 5.2** (Topology is preserved up to the chord error). *The bottleneck distance between the persistence diagrams of the full and reduced balances is at most the chord error: $d_B\big(\mathrm{Dgm}(S_a), \mathrm{Dgm}(\tilde{S}_a)\big) \le e_a$. For the net balance Equation (5.2) of any design with constant demand, $d_B\big(\mathrm{Dgm}(B), \mathrm{Dgm}(\tilde{B})\big) \le P^{wt} e_{wt} + P^{pv} e_{pv}$.*

*Proof.* The first statement is the stability theorem of persistence diagrams [7] with Proposition 5.1. For the second, $B - \tilde{B} = P^{wt}(S_{wt} - \tilde{S}_{wt}) + P^{pv}(S_{pv} - \tilde{S}_{pv})$ because the straight-line term in Equation (5.2) is reproduced exactly by chords; the triangle inequality and the stability theorem give the bound. □

In words: if every chord stays within $e_a$ of the true balance, no surplus or deficit episode of any design is misjudged by more than that amount, and no episode deeper than $2e_a$ is lost. The reduction is built to make $e_a$ small where it matters.

## 5.5 The reduction algorithm

The planner's reduction, a topology-preserving chronological aggregation, has two parts (Algorithm 1). First, both ends of every persistence pair of $S_a$ and of $-S_a$ with persistence of at least a share $\tau$ of the range become fixed boundaries: each such hour forms its own one-hour block, so the turning points of the largest episodes stay exact. Second, starting from 8 760 one-hour blocks, neighbouring blocks are merged in order of increasing Ward cost [49],

$$w(i,j) = \frac{n_i n_j}{n_i + n_j} \left\| \bar{z}_i - \bar{z}_j \right\|^2, \tag{5.3}$$

where $n_i$ is the number of hours in block $i$ and $\bar{z}_i$ the block mean of the standardised attributes. Merging the most similar neighbours first keeps power levels and duration curves. A merge is accepted only if the chord error of every attribute stays below $\varepsilon$ times its range; otherwise it is set aside. When no acceptable merge is left and there are still more than $N$ blocks, $\varepsilon$ is relaxed by a factor $\rho$ and the set-aside merges are tried again.

**Algorithm 1.** Topology-preserving chronological aggregation (`tda.py`, `tpca`).

**Input:** hourly attributes $x_a$, $a \in \mathcal{A}$; target $N$; $\tau = 0.02$; $\varepsilon_0 = 0.002$; $\rho = 1.5$
**Output:** $N$ blocks in calendar order, their lengths $\Delta_j$ and attribute means

compute $S_a$ by Equation (5.1), its range $r_a = \max S_a - \min S_a$, and standardised $z_a$;
$\mathcal{F} \leftarrow \bigcup_a \{$both ends of pairs of $S_a$ and $-S_a$ with persistence $\geq \tau r_a\}$; `// fixed cuts`
make every hour a block; forbid merges across hours in $\mathcal{F}$;
push every allowed neighbouring pair $(i, j)$ on a heap keyed by $w(i, j)$;
$\varepsilon \leftarrow \varepsilon_0$; set-aside list $\mathcal{B} \leftarrow \emptyset$;
**while** *number of blocks* $> N$ **do**
    **if** *heap empty* **then**
        **if** $\mathcal{B} = \emptyset$ **then break** `// only fixed cuts remain;`
        $\varepsilon \leftarrow \rho\,\varepsilon$; move $\mathcal{B}$ back to the heap;
    pop the pair $(i, j)$ with the smallest $w$; skip it if either block has changed;
    **if** $\max_a e_a(i \cup j)/r_a > \varepsilon$ **then**
        add $(i, j)$ to $\mathcal{B}$;
    **else**
        merge $j$ into $i$; push the new pairs with the neighbours of $i$;
report block means and lengths, the achieved $\max_a e_a/r_a$ and the bottleneck distances;

**Which attributes.** The cut attributes are the wind capacity factor, the PV capacity factor (if PV is allowed), the price (for merchant or CfD revenue) and the demand profiles (for profile delivery). All uploaded series are averaged over the same blocks.

**Settings.** The number of blocks is chosen on the **Run** step: 730 (quick), 1 460 (standard) or 2 190 (fine). The thresholds $\tau$, $\varepsilon_0$ and $\rho$ are fixed in the code. The reduction takes about one second.

**What it reports.** The **TDA check** tab of the results shows the achieved shape error $\max_a e_a/r_a$, the number of fixed cuts, the cost change in the full-year check, and the reduced and full balances with their persistence diagrams. The bottleneck distances, measured with GUDHI [45], are computed with every run and stored with the result. For Belle-Île 2019 with 1 460 blocks the shape error is 0.14 % of the range of the balance and the bottleneck distance 0.06 % for wind and 0.05 % for PV; blocks last from 1 to 41 hours, with a median of 4.

## 5.6 Full-year check and repair

A design sized on blocks has seen only block means; within a block, wind may rise above the cable rating or fall below what a firm contract needs. The planner therefore operates the design on all 8 760 hours (`workflow.py`, `verify_and_repair`). The model of Chapter 3 is built on the full year with three changes:

1. Capacities are tied to the sized design $\hat{P}_x$ through one scale factor per group,

$$P_x = \lambda_g \hat{P}_x \quad \forall x \in g, \qquad 1 \le \lambda_g \le 3, \tag{5.4}$$

   with groups generation $\{P^{wt}, P^{pv}\}$, conversion and export $\{P^{ez,\ell}, P^{cmp,\ell}, P^{fc,\ell}, P^{cab}\}$ and storage $\{E^{b,\ell}, P^{b,\ell}, M^{tk,\ell}\}$.

2. The pipeline class, the link option and its number of circuits $\hat{n}^{cab}$ are fixed to those chosen, so the capacity used $P^{cab}$ can grow up to their rating. If the rating binds and the check still wants more (a shortfall, or a conversion factor above 1), the check is run again with one more circuit (if the circuit limit allows) and the result with the lower objective, penalties included, is kept.

3. Every delivery constraint Equations (3.25) to (3.27) receives a non-negative shortfall (per hour for firm and profile contracts, per year for annual ones), penalised in the objective by

$$\Pi = \pi^{pen,e}\Big(\sum_t u_t^e + U^e\Big) + \pi^{pen,h}\Big(\sum_t u_t^h + U^h\Big). \tag{5.5}$$

The check is one linear programme (two if a circuit was tried): it chooses the three factors $\lambda_g$ and the hourly operation that minimise the net cost Equation (3.1), penalties included. It may raise several groups at once, and may raise a group beyond what delivery needs when that pays (for example under merchant revenue). With the default penalties (3 000 EUR/MWh, 100 EUR/kg) short hours are normally removed by enlarging the design; any shortfall left is shown as a warning with the undelivered energy. The final design is $\lambda_g \hat{P}_x$, and the three factors are reported as *repair factors*. All results the app shows (capacities, costs, LCOE, LCOH, hourly operation) come from this full-year solution.

**Accuracy.** In the project's validation (Step 2: Belle-Île, years 2019 and 2024, three use cases), the cost of the final design stayed within 0.3 % of the full-year optimum and every contract was delivered. In the example of Chapter 8, the conversion group was scaled by 1.022 and the total annual cost rose by 0.4 % from the reduced to the full year; the cable stayed within its one circuit.

# 6

# Using the app

This chapter walks through the app from opening it to downloading the results. Each section matches one step of the app. Chapter 7 covers the data step in more depth and Chapter 8 repeats the whole sequence on the Belle-Île example.

## 6.1 Opening the app

**Hosted version.** Partners open the link of the project's Streamlit app in any recent browser. Nothing needs to be installed, and each visitor works in a private session: inputs typed by one partner are not seen by another. The free hosting tier runs one optimisation at a time; if another partner is solving, the run waits and starts automatically.

**On your own computer.** With Python 3.11 or 3.12:

```
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
streamlit run app.py
```

The app opens at `http://localhost:8501`. HiGHS is installed with the requirements; GLPK is optional. Appendix C explains how to publish your own copy from GitHub and how to run it with Docker.

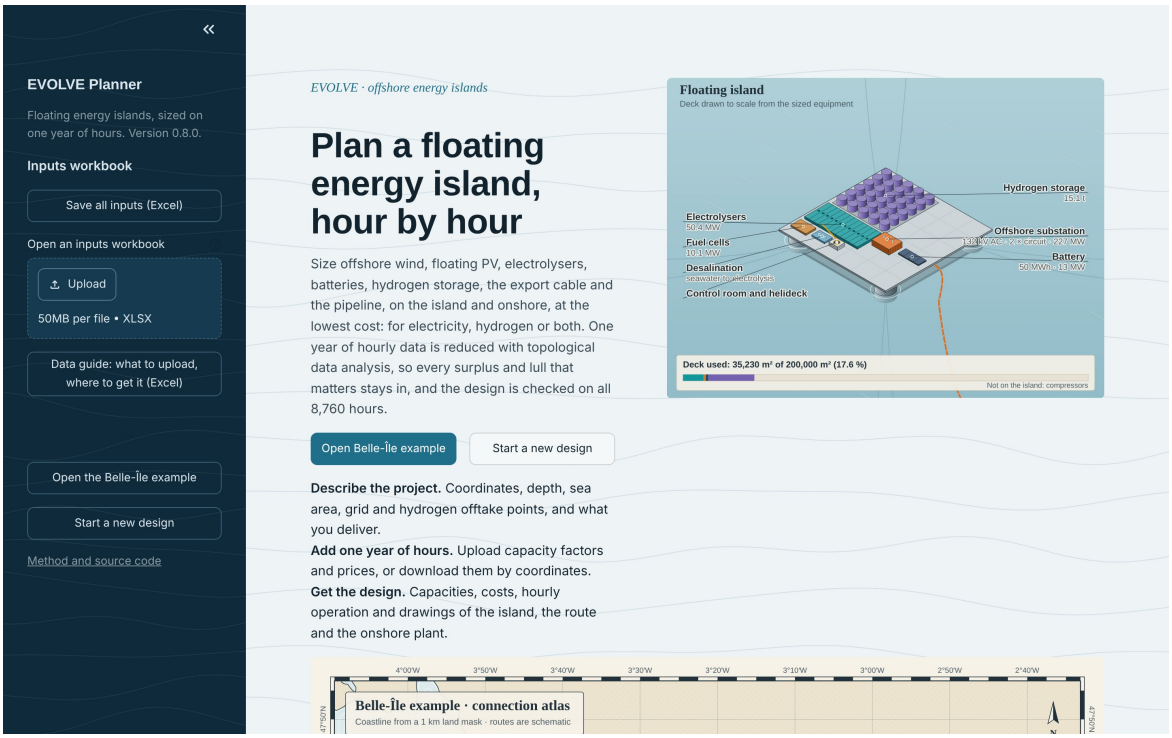



**Figure 6.1.** The start page. **Open Belle-Île example** loads the example of Chapter 8; **Start a new design** opens the first step with default values.

## 6.2 Layout and conventions

**Step selector.** Six steps at the top of the page: **Site**, **Use case**, **Data**, **Technologies**, **Run**, **Results**. They can be visited in any order; the **Next** button at the bottom of each step follows the usual order.

**Sidebar.** A summary of the project (foundation, export link, data loaded, last result), the inputs workbook (**Save all inputs** and **Open an inputs workbook**), the data guide, the example and a new design.

**Inputs.** Every number has a box with − and + buttons. Type a value, or click the buttons, which step by about a tenth of the value's order of magnitude (0.01° for coordinates, 1 m for depth, 1 000 EUR for the TSO fee). The question-mark icon explains the input. Placeholder costs are flagged in their help text.

**Drawings.** The route section, the island and the connection atlas are redrawn from the inputs on the **Site** and **Technologies** steps and from the optimised design on the **Results** step.

## 6.3 Step 1: Site

**Where things are.** Enter a name and coordinates (decimal degrees, west and south negative) for the island, the grid connection point and the hydrogen offtake point (Figure 6.2). The grid point is where the export cable meets the transmission grid, usually a coastal substation or a landfall; the offtake point is where hydrogen is delivered: a port, an industrial user or an entry point of a hydrogen network.

**Depth and areas.** **Water depth at the island** decides between bottom-fixed and floating foundations (Section 2.2). **Usable sea area** limits wind and PV; **Island deck area** and **Onshore plot area** limit the equipment at each end (Section 2.8).

**Route lengths.** Choose how lengths are found (Section 2.7). The table under the inputs shows, for the export cable and the pipeline, the length the model will use, how it was found and the straight-line distance. Enter a length from a route study in **Cable length, if known** or **Pipeline length, if known** to override the computed one. A warning appears if the island falls on land on the coastline map, or if a route is longer than 300 km, which usually means the onshore points were not moved with the island.

**Finding connection points.** On the right are links to OpenInfraMap centred on the island, to EMODnet Human Activities and to the ENTSO-E grid map. For a site with no known substation, the two buttons place the grid point or the offtake on the coast nearest the island as a first guess.

**Densities and foundations** (expander). Wind and PV power densities, the fixed/floating threshold depth, the turbine rating used in the drawings, turbine-level electrolysis (concept C) and whether an existing gas pipeline on the route can be repurposed.

**Connections to existing infrastructure.** Choose the electricity connection (own cable, a nearby hub, or let the model choose) and the hydrogen connection (own pipeline, a tie-in, or let the model choose) (Figure 6.3). For a hub, enter its coordinates, spare capacity, voltage and access fee; for a tie-in, its coordinates, spare capacity, operating pressure and the tie-in cost. For the island's own link, keep **Export link** on **auto** to let the model choose AC or HVDC, the voltage, conductor, compensation and number of circuits, or restrict it to one AC voltage or to HVDC. Comparing connection options multiplies the sizing time by the number of options. Before a run the export-link badge names the option that is cheapest at the grid injection limit, before load losses; the link actually built is decided by the run.

**Atlas and interactive map.** Below the badges, two tabs show the site. The **Connection atlas** is a chart drawn from the coastline with the wind-farm zone, the routes, waterlines at 0.5 to 4 km from shore, a graticule and a scale bar (Figure 8.3 in Chapter 8). The **Interactive map** can be zoomed and panned; it draws every candidate link with the length the model measures, and hovering any element shows its numbers. After a run the same map, under **Results**, shows what was built.

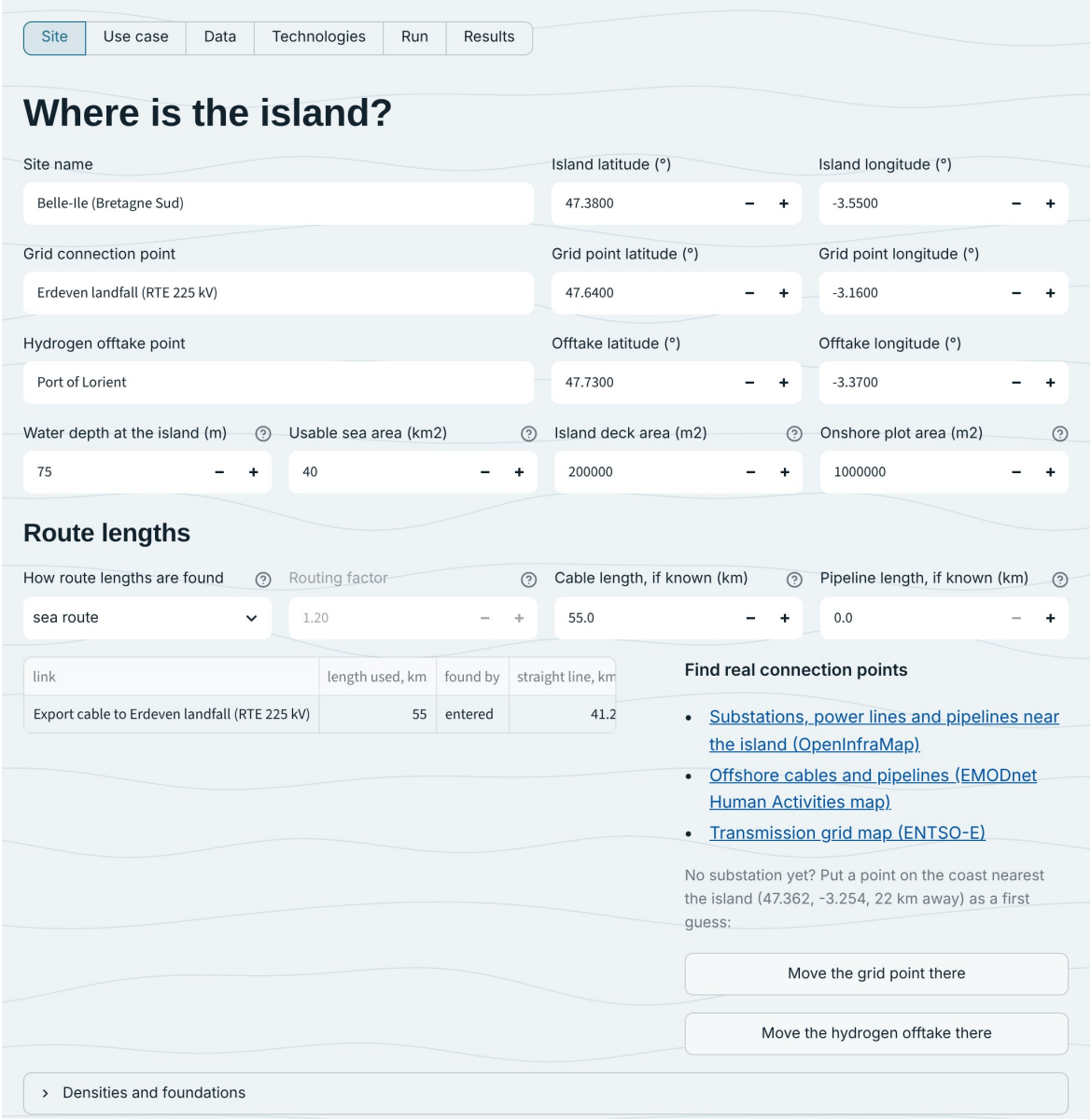



**Figure 6.2.** The **Site** step: coordinates, depth and areas, then the route lengths with the length the model will use, how it was found, and links to maps of real connection points.

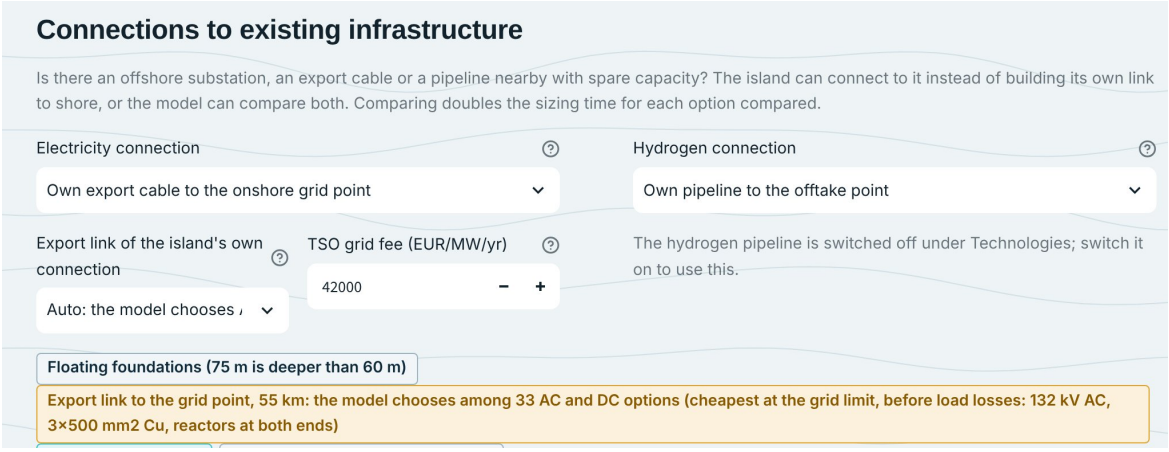



**Figure 6.3.** Connections to existing infrastructure, the TSO fee, and the summary badges: foundation, export link, pipeline route and the largest wind capacity the sea area allows.

## 6.4 Step 2: Use case

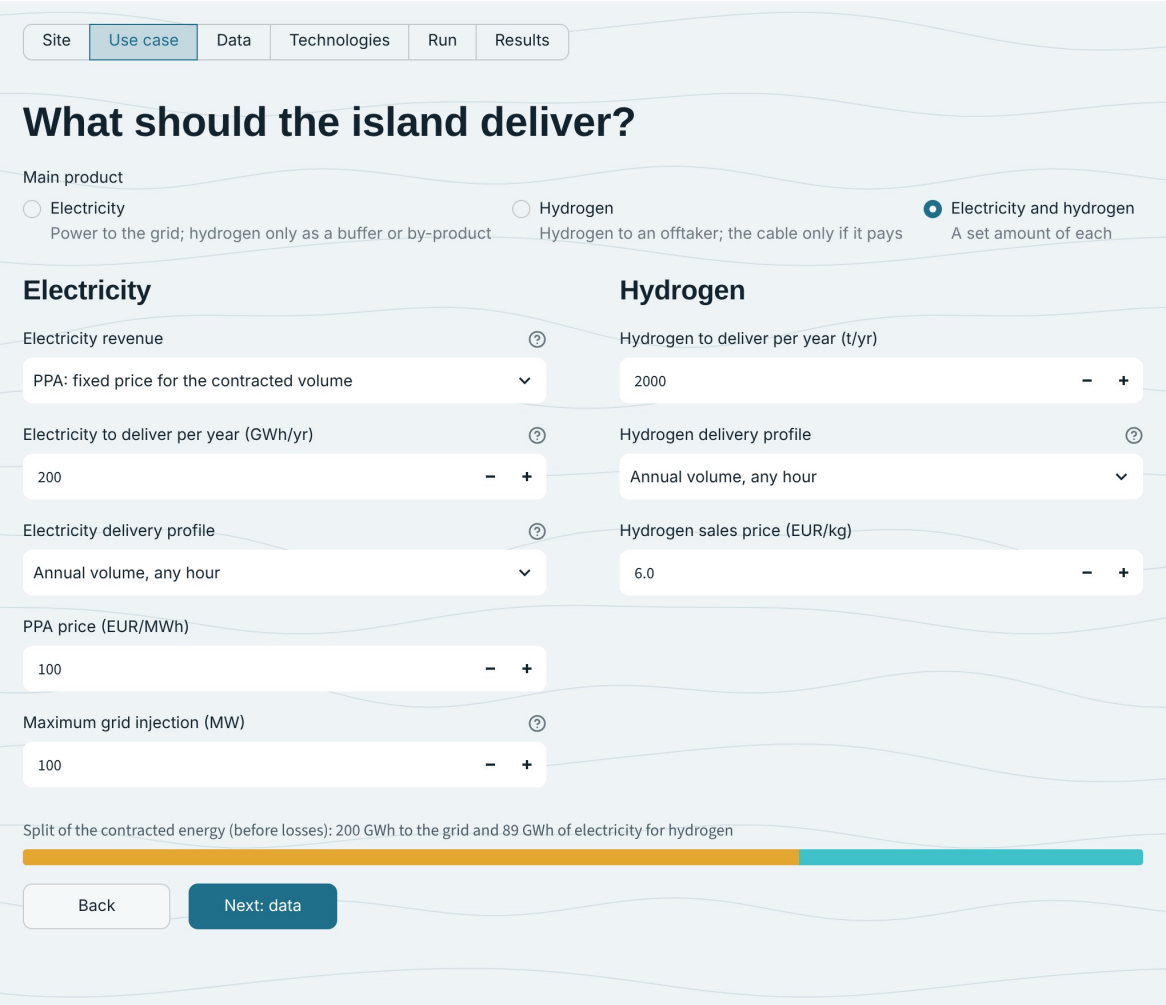



**Figure 6.4.** The **Use case** step for a hybrid island. The bar at the bottom splits the contracted energy between the grid and the electrolysers.

Choose the main product: **Electricity**, **Hydrogen** or **Electricity and hydrogen** (Figure 6.4). For electricity, choose the revenue (PPA, merchant or CfD); with a PPA, enter the annual volume (in GWh), the delivery profile and the price. Set the **Maximum grid injection**. For hydrogen, enter the annual volume (in tonnes), the delivery profile and the price. Under a single product, the other carrier can be sold as a by-product (Section 2.3).

## 6.5 Step 3: Data

Load one year of hourly series in one of three ways (Figure 6.5):

1. **Upload a file**: a CSV or Excel file with recognised column names (Section 7.2).
2. **Download by coordinates**: wind and solar capacity factors from ERA5 for the island's coordinates and any year from 2000, and day-ahead prices for a bidding zone (Section 7.4).
3. **Belle-Île example**: the 2019 or 2024 series of the example.

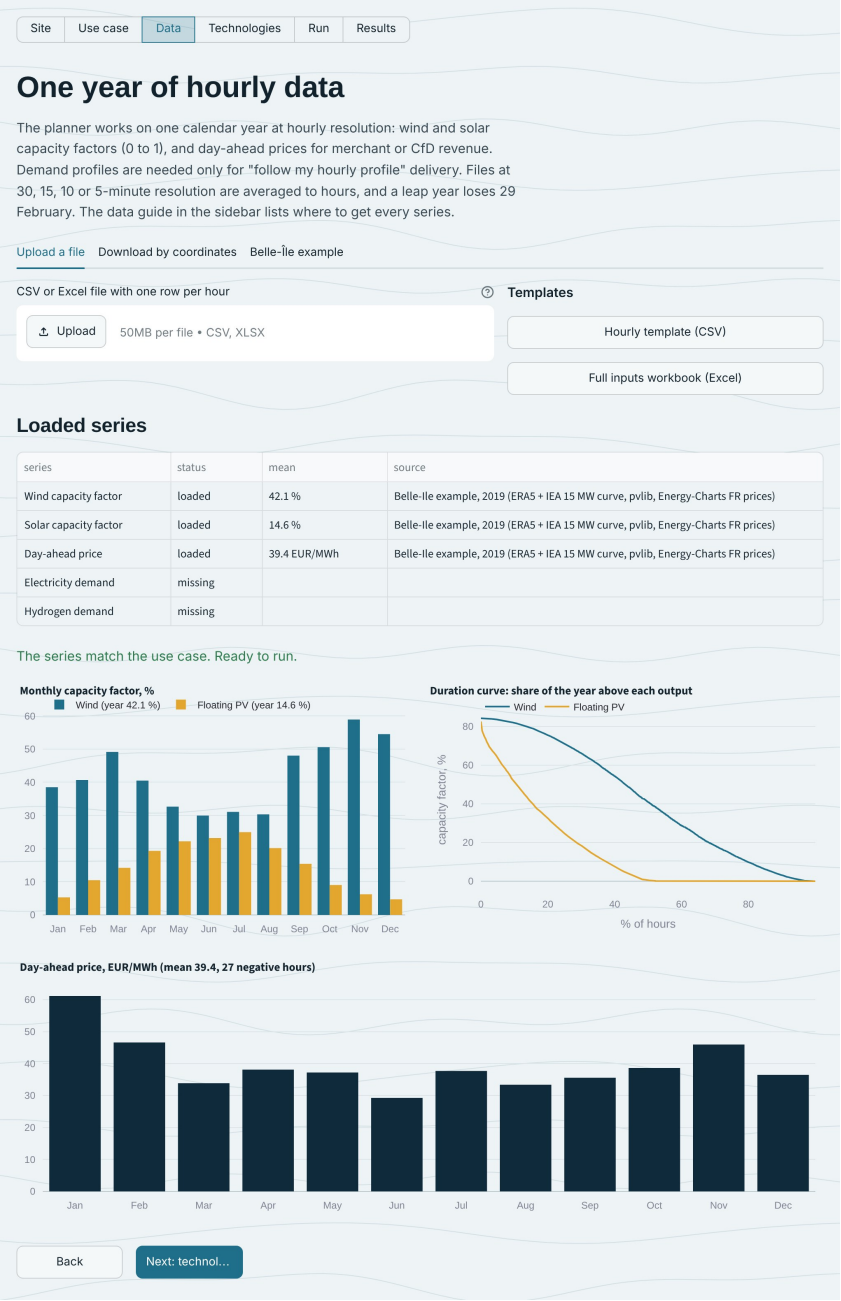



**Figure 6.5.** The **Data** step after loading the Belle-Île 2019 series: the loaded series with their source, a check against the use case, and monthly and duration views.

The table lists each series with its mean and source, and a message confirms that the series match the use case or says what is missing (for example, a price series for merchant revenue). The charts show monthly capacity factors, duration curves and monthly prices, which is a quick plausibility check.

## 6.6 Step 4: Technologies

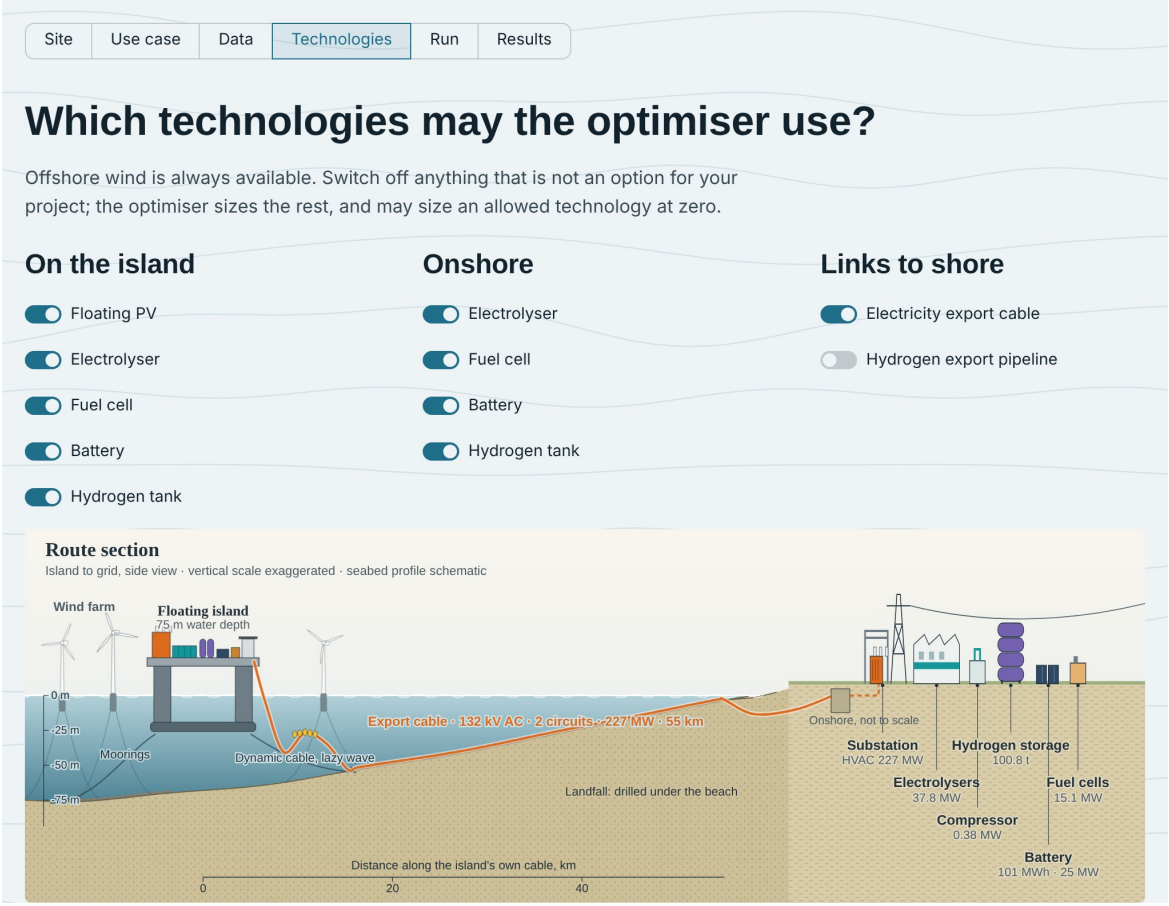



**Figure 6.6.** Technology switches and the live preview of the route section.

**Switches.** Switch off any technology that is not an option for the project (Figure 6.6). The optimiser sizes the rest and may still choose zero for an allowed technology. The previews show every allowed technology at illustrative sizes, and the deck bar on the island drawing shows how much of the deck the equipment would use (Figure 6.7).

**Costs and performance.** One tab per group (finance, offshore wind, floating PV, electrolysis, compression, fuel cell, battery, hydrogen tank, export cable, connections, hydrogen pipeline, island platform, solver) holds every number the model uses (Figure 6.8). Further parameters, such as footprints, lifetimes of water treatment, the cable loss factors or pipe diameters, are listed under each tab. The ? icon of each input gives its meaning and the source of its default (Section 4.1); values without an ACER indicator are flagged as placeholders. The **Export cable**

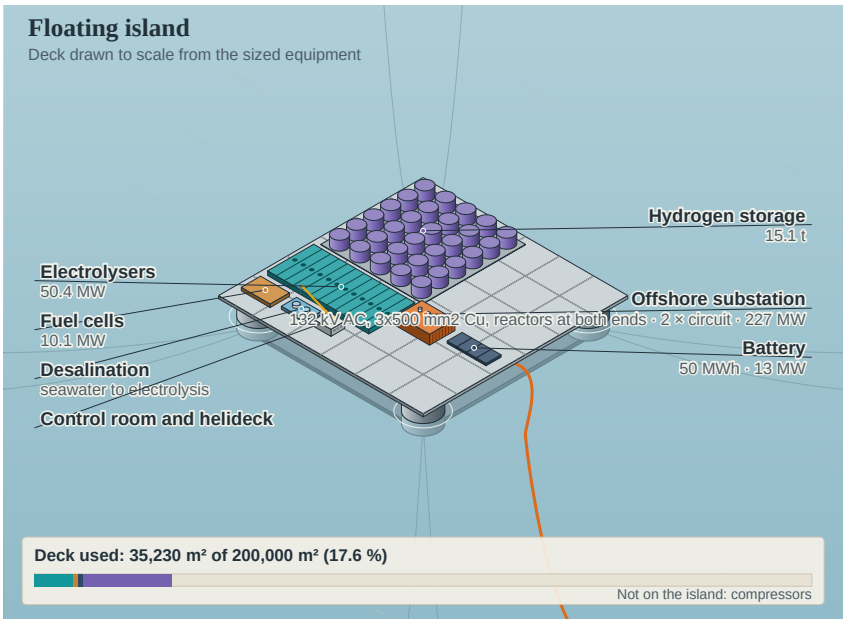



**Figure 6.7.** Island preview on the **Technologies** step with every island technology allowed, at illustrative sizes. Footprints follow the area inputs, so an undersized deck shows up before the run.

tab holds the AC installation cost, the power factor, the loss factors, the reactor and mid-route platform costs, the value of no-load losses, the seven parameters of the HVDC cost function with its losses and floating premium, the grid fee and the **Cable cost model** switch (the cable data of Table 4.2 and the ORBIT prices are fixed in `cables.py`); the **Hydrogen pipeline** tab holds the pipeline cost line, the repurposed cost and the gas-flow inputs that set each pipe's capacity. A **Reset** button appears when a group has values changed from the defaults. The screenshot shows the **Export cable** tab of version 0.9.0; Appendix A lists every input.

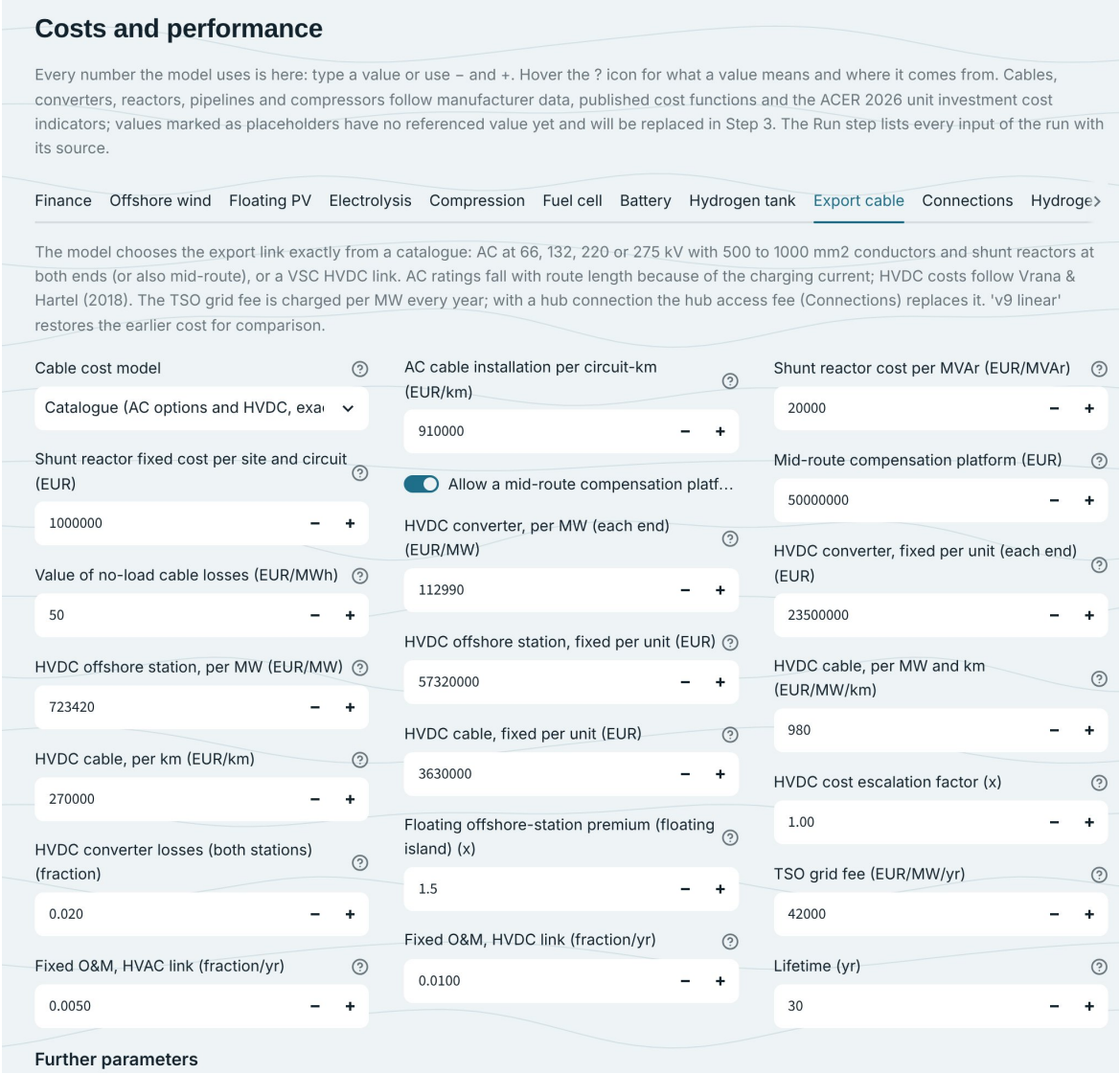



**Figure 6.8.** Cost and performance inputs, one tab per group.

## 6.7 Step 5: Run

The step summarises the project and explains the three stages of a run (Figure 6.9). Choose the **Detail of the reduced year** (730, 1 460 or 2 190 blocks) and the solver. Before running, open **Every input of this run**: it lists all inputs, can be filtered by group or to the values changed from the defaults, with the source of each default, and shows what the model derives from them: the island concept, the number of link options offered and the one cheapest at the grid limit (and, after a run, the link and circuits built), route lengths and how they were found, the link cost per MW, the fee applied, the pipeline classes allowed, the hydrogen yield and the compression work.

Press **Find the lowest-cost design**. The status box reports the reduction, the sizing (once per connection option compared) and the full-year check. With HiGHS and the standard detail a run takes two to five minutes; GLPK is much slower, because it cannot warm-start the branch and bound (Section 3.8) and is slow on the full-year check.

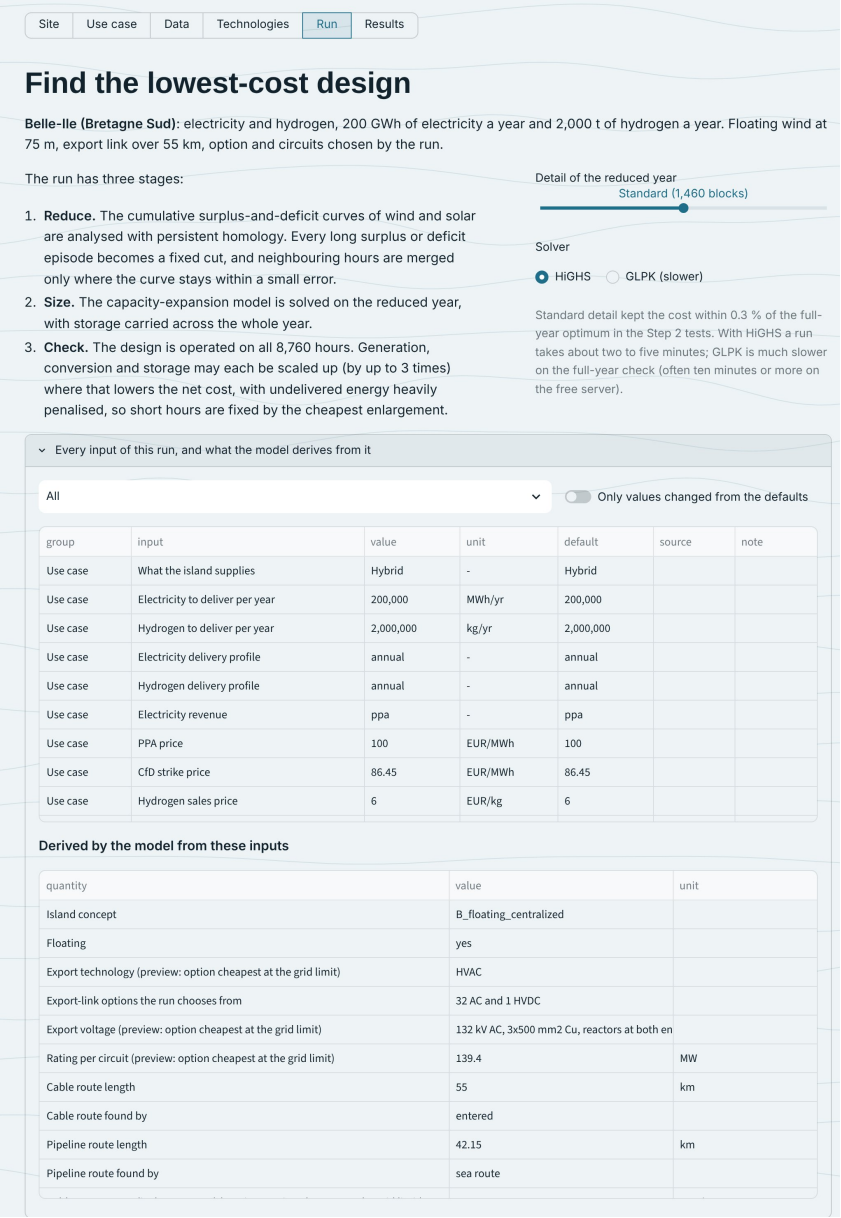



**Figure 6.9.** The **Run** step with the review of every input and of the quantities derived from them.

## 6.8 Step 6: Results

The top row gives the net annual cost or surplus, the investment, the energy delivered, LCOE and LCOH, and the share of generation curtailed (Figure 6.10). A warning appears if anything was left undelivered after the full-year check, or if a floating central island exports above the highest qualified dynamic cable voltage (Section 4.4). Below come:

- the *route section* and the *island* drawn from the sized design, with the capacities on the island and onshore, and the *connections* table (route, link built, circuits or converter units, capacity used, investment per MW, energy delivered over energy sent, losses with the no-load part, reactive compensation, length and how it was found, fee, pipeline class with its capacity, limit and cost per km, use of a hub's or pipeline's spare capacity);
- the comparison of connection options, if several were sized;

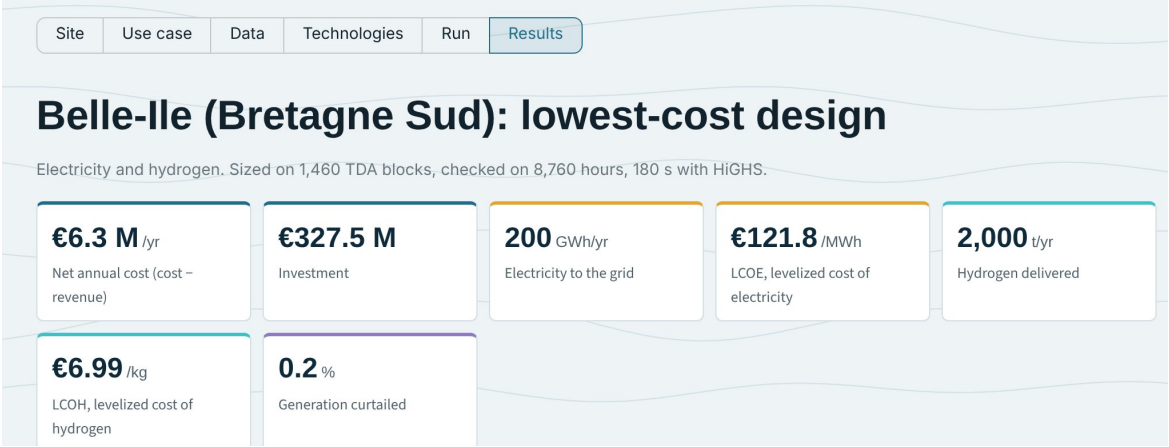



**Figure 6.10.** Key figures at the top of the **Results** step.

- the *connection atlas* of the design;
- tabs: **Levelized costs** (LCOE and LCOH with their breakdown by component), **Energy flows** (a Sankey diagram of the year), **Hour by hour** (a chosen week and a heat map of the year), **Storage** (levels over the year), **Costs and revenues**, **TDA check** (Section 5.5), **Map of the design** and **Download**.

**Downloads.** The **Download** tab gives the results workbook (Appendix B), the inputs workbook of this run, and the three drawings as HTML files that open in any browser and print to PDF.

## 6.9 Saving and resuming work

**Save all inputs (Excel)** in the sidebar writes one workbook with every input and the hourly series. Open it later with **Open an inputs workbook** to restore everything; the same workbook can be edited in Excel and shared with partners. The session itself is not stored on the server: closing the browser tab ends it.

## 6.10 Troubleshooting

**Table 6.1.** Common messages and what to do.

| Message or symptom | What to do |
|---|---|
| *wind_cf: 8,760 hourly values are needed* | Load a full year of data on the **Data** step (Chapter 7). |
| *Merchant and CfD revenue need an hourly price series* | Download prices or add a `price_eur_mwh` column. |
| *The solver stopped without an optimal solution* | The contracts cannot be met with the technologies and space allowed. Raise the sea area, deck area or grid injection limit, allow more technologies, or lower the volumes. |
| *Not delivered after the full-year check* | A limit binds only in some hours. Usually the sea area, the deck or the grid limit; relax it and run again, or use more blocks. |

| | |
|---|---|
| Nothing is built | With the costs and prices given no MWh pays for itself and no volume is contracted. Lower costs, raise prices, or contract a volume under a PPA. |
| *The floating central platform exports at … kV AC* | Dynamic export cables are qualified up to 66 kV; the cost includes the dynamic section, but its availability is not proven. Turn on **Floating: allow only qualified dynamic voltages** to exclude higher voltages, or treat the result as conditional. |
| *Input 'platform_float_factor' is no longer used* | A workbook from before version 0.10. The floating platform factor now follows from the **Island platform** inputs (Section 4.7); nothing to do. |
| Long-route warning on the **Site** step | The onshore points are far from the island: move them, or use the nearest-coast buttons. |
| The interactive map has no background | The basemap is loaded from the internet by your browser; the coastline drawn by the app still shows. |
| A drawing shows a short note instead | The drawing failed for these inputs; the results are not affected. Report the coordinates to the maintainers. |

# 7

# Preparing and importing data

The planner needs three kinds of data: one year of hourly series (weather, prices, demand), the project inputs (contracts, costs, technical parameters), and facts about the site and the infrastructure around it. This chapter explains what each one is, the format the app reads, and where to get it. The same information, with a filled example, is in the data guide workbook `examples/EVOLVE_data_guide.xlsx`, which can also be downloaded from the app's sidebar.

## 7.1 Hourly series

Table 7.1 lists the five series. Wind is always needed; the others depend on the choices made on the **Use case** and **Technologies** steps.

**Table 7.1.** Hourly series the planner reads.

| Series | Column | Unit | Needed when |
|---|---|---|---|
| Wind capacity factor | `wind_cf` | 0–1 | always |
| Floating PV capacity factor | `pv_cf` | 0–1 | floating PV allowed (otherwise set to zero) |
| Day-ahead price | `price_eur_mwh` | EUR/MWh | merchant or CfD electricity revenue |
| Electricity demand profile | `elec_demand_mw` | MW | electricity delivery "follow my hourly profile" |
| Hydrogen demand profile | `h2_demand_kgh` | kg/h | hydrogen delivery "follow my hourly profile" |

A capacity factor is the output of 1 MW of installed capacity, after all losses, divided by 1 MW: the model multiplies it by the capacity it chooses (Equation (3.5)). Prices keep their sign; negative prices matter for merchant revenue and for the CfD rule (Equation (3.4)). A demand profile is the power or flow to deliver in each hour; for a profile with a given shape and annual volume, divide each hour by the annual total of the shape and multiply by the volume.

The three rules for hourly data

1. **One calendar year.** The planner reduces that one year with TDA and checks the design on its 8 760 hours. To test other years, run them one at a time.
2. **Hourly, in UTC, from 1 January 00:00.** Files at 30, 15, 10 and 5 minutes are averaged to hours; a leap year loses 29 February.
3. **Units as in Table 7.1.** Capacity factors between 0 and 1; a column whose largest value

lies between 1 and 100 is read as percent and divided by 100.

## 7.2 File format

Upload one CSV file, or one Excel file whose sheet is named `TimeSeries` (otherwise the first sheet is read), on the **Data** step. The rules the app applies (`weather.py`) are:

**Header.** One header row. Column names are matched without regard to case or surrounding spaces; the names in Table 7.2 are recognised and any other column is ignored (an `hour` or a timestamp column can stay for your own reference). Columns can be in any order, and a column that is present but empty is ignored.

**Separator.** Comma or semicolon in CSV files, detected automatically. Use a point as the decimal separator.

**Rows.** One row per time step of one year. The resolution is recognised from the number of rows (Table 7.3).

**Gaps.** Missing values are filled by linear interpolation and the number filled is reported. Do not fill gaps with zeros.

**Range.** If the largest value of a capacity-factor column lies above 1 and at most 100, the whole column is read as percent and divided by 100; capacity factors are then limited to $[0, 1]$.

**Table 7.2.** Recognised column names.

| Series | Accepted names |
|---|---|
| Wind capacity factor | `wind_cf`, `wt_cp`, `wind`, `wind cf` |
| PV capacity factor | `pv_cf`, `pv_cp`, `pv`, `solar`, `solar_cf`, `pv cf` |
| Price | `price`, `price_eur_mwh`, `el_price`, `day_ahead`, `dayahead` |
| Electricity demand | `elec_demand`, `elec_demand_mw`, `demand_mw`, `load_mw` |
| Hydrogen demand | `h2_demand`, `h2_demand_kgh`, `h2_kgh` |

**Table 7.3.** Rows of a one-year file at each resolution. Sub-hourly values are averaged within each hour, which is right for power, capacity factors and prices.

| Time step | Steps per hour | Rows, normal year | Rows, leap year |
|---|---|---|---|
| 60 minutes | 1 | 8 760 | 8 784 |
| 30 minutes | 2 | 17 520 | 17 568 |
| 15 minutes | 4 | 35 040 | 35 136 |
| 10 minutes | 6 | 52 560 | 52 704 |
| 5 minutes | 12 | 105 120 | 105 408 |

A minimal CSV file with wind, PV and price looks like this (the Belle-Île 2019 series shipped with the app; the `hour` and `time_utc` columns are for reference):

```
hour,time_utc,wind_cf,pv_cf,price_eur_mwh
1,2019-01-01 00:00:00,0.04315,0.0,46.27
2,2019-01-01 01:00:00,0.05279,0.0,39.78
3,2019-01-01 02:00:00,0.05705,0.0,27.87
...
8760,2019-12-31 23:00:00,0.15439,0.0,41.88
```

The **Hourly template (CSV)** button on the **Data** step downloads an empty file with all five columns and the `hour` column filled.

## 7.3 Preparing the series

**Wind.** From hourly wind speed at or near hub height: bring the speed to hub height, apply the power curve of the turbine (or of the farm, smoothed for the spread of speeds across the farm), divide by the rated power, and subtract wake, electrical and availability losses. Reanalysis wind (ERA5 at about 25 km) smooths peaks; for a project, a mesoscale or measured series is better.

**Floating PV.** From irradiance: transpose global and diffuse irradiance to the module plane, model cell temperature (water cooling lowers it), apply a PV model, divide by the rated power and subtract losses. Hourly PV series can also be taken from PVGIS (2005–2023, along the coast and up to 25 km offshore) or Renewables.ninja.

**Prices.** Since 1 October 2025 the European day-ahead market clears in 15-minute steps. Upload the 15-minute file as it is: the app averages it to hours. Earlier years are hourly.

**Demand profiles.** Hourly demand comes from the offtaker. For electricity, a national load shape (for example from the ENTSO-E Transparency Platform) can be scaled to the contracted volume. No public hourly hydrogen demand exists; a flat profile is $D^h/8760$ kg/h, which is what the *firm* delivery option does without a file. `examples/belle_ile/EVOLVE_hourly_demand_profiles_2019.xlsx` holds synthetic profiles for testing: a regional electricity load of 200 GWh/yr (mean 22.8 MW, peak 38.1 MW) and a hydrogen demand of 2 000 t/yr from an industrial user and port mobility (mean 228 kg/h, peak 401 kg/h); its README sheet says how they were built. Uploaded on the **Data** step with both deliveries set to **Follow my hourly profile**, the Belle-Île case must meet every hour: the optimiser then builds about 122 MW of wind, 18 MW of PV, 39 MW of onshore electrolysis, 34 MW of onshore fuel cells and 243 t of hydrogen storage, at a net cost of about 53 million EUR a year, against 6.3 million EUR for the same volumes as annual contracts.

## 7.4 Built-in downloads

The **Download by coordinates** tab of the **Data** step fetches series without a file. The hosted app has internet access; a local copy needs it too.

**Wind and solar.** ERA5 [27] from the Open-Meteo archive API at the island's coordinates for a chosen year (2000 onwards). Wind: 100 m wind speed is extrapolated to a 150 m hub with a power law (exponent 0.11), converted with the IEA 15 MW reference turbine [24] smoothed to farm level with a Gaussian of standard deviation $0.6 + 0.2v$ m/s, and reduced by 12.5 % losses. Solar: global and diffuse irradiance are transposed with pvlib [29] to a 10° tilt facing the equator (albedo 0.06), cell temperature follows the Faiman model [23] with water cooling ($U_0 = 35$, $U_1 = 8$), output follows PVWatts with a temperature coefficient of −0.35 %/K, and 14 % losses are subtracted.

**Prices.** Day-ahead prices for a bidding zone and year from the Energy-Charts API of Fraunhofer ISE, averaged to hours, with gaps interpolated and 29 February removed. The zones offered are FR, DE-LU, NL, BE, DK1, DK2, NO2, SE4, PL, AT, CH, CZ and IT-North. The service limits how often it can be called; if it is busy the app retries for about two minutes, after which you can wait a minute and try again.

These downloads give a first estimate. For a project, replace them with the developer's own wind and PV series.

## 7.5 The inputs workbook

Every input of a design, and its hourly series, fit in one Excel workbook (`workbook.py`). Save it from the sidebar (**Save all inputs**), edit it in Excel, and open it again (**Open an inputs workbook**) to restore everything. A blank copy is `examples/EVOLVE_inputs_blank_template.xlsx`, and the Belle-Île inputs are in `examples/belle_ile/EVOLVE_inputs_BelleIle_hybrid_2019.xlsx`.

**README** explains the file.

**Inputs** has one row per input with the columns `key`, `group`, `label`, `unit`, `value`, `default`, `placeholder until Step 3`, `options`, `help` and `source of the default`. The app reads only `key` and `value`; labels can be edited freely. Numbers are plain numbers in the stated unit; switches are `TRUE` or `FALSE` (any text other than true, 1, yes or y counts as false); choices must be one of the strings in `options`, which is not checked on reading. An empty value keeps the default; a value that cannot be read as a number keeps the default and is reported. Rows can be deleted: missing inputs take their default.

**TimeSeries** has 8 760 rows with `hour`, `wind_cf`, `pv_cf`, `price_eur_mwh`, `elec_demand_mw` and `h2_demand_kgh`; unused columns stay empty. The rules of Section 7.2 apply.

> **Sharing a case**
>
> The workbook is the simplest way to hand a case to a partner: they open the app, load the workbook from the sidebar, and see exactly the inputs and series you used. The results workbook (Appendix B) also contains an `Inputs used` sheet for the record.

## 7.6 Project inputs and costs

The 180 inputs, their defaults and the source of each default are listed in Appendix A, and Chapter 4 documents every cost. Cables, converters, transformers, reactors, pipelines and

compressors already follow manufacturer data, published cost functions and ACER's indicators. Costs marked as placeholders have no ACER indicator and will be replaced in Step 3 of EVOLVE. Until then, three public sources cover most of them:

- the Danish Energy Agency technology catalogues (generation and district heating, June 2026; renewable fuels, February 2026; energy storage, April 2024) [9];
- the Annual Technology Baseline 2025 of NLR (formerly NREL), including fixed and floating offshore wind [39];
- ACER's unit investment cost indicators for electricity, hydrogen and gas infrastructure (report of 24 April 2026) [22].

Developers' and TSOs' figures replace them for a real site: the TSO's connection offer gives the grid fee and the injection limit, and a cable or pipeline route study gives the route lengths.

## 7.7 Site and infrastructure

**Table 7.4.** Where to find site and infrastructure data (links checked 8 October 2026).

| Item | Source | Link |
|---|---|---|
| Water depth | EMODnet Bathymetry DTM 2024 (about 115 m); GEBCO_2026 grid (15 arc-seconds) | https://emodnet.ec.europa.eu/geoviewer/, https://download.gebco.net |
| Usable sea area | national maritime spatial plans; EMODnet Human Activities (wind-farm polygons) | https://emodnet.ec.europa.eu/en/human-activities |
| Grid connection point | the TSO's connection studies; OpenInfraMap (OpenStreetMap); ENTSO-E grid map (schematic positions) | https://openinframap.org, https://www.entsoe.eu/data/map/ |
| Hub: substation or cable | EMODnet Human Activities; the TSO or wind-farm owner for spare MW, voltage and fee | as above |
| Hydrogen offtake or network | European Hydrogen Backbone maps; ENTSOG TYNDP 2024 | https://ehb.eu, https://www.entsog.eu/tyndp |
| Pipeline for a tie-in | ENTSOG TYNDP; EMODnet pipelines; the network operator for spare capacity and pressure | as above |
| Constraints (not modelled) | Natura 2000 viewer (EEA); EMODnet vessel density | https://natura2000.eea.europa.eu |

## 7.8 Checklist before a run

- The series cover one calendar year, start on 1 January 00:00 UTC and have the right number of rows.
- Wind and PV are capacity factors after losses, not power in MW and not wind speed.
- A price series is loaded if the electricity revenue is merchant or CfD; demand profiles are loaded if a delivery follows a profile.

- The coordinates put the island at sea and the grid and offtake points on or near the coast; the route lengths on the **Site** step look plausible.
- The usable sea area can hold enough wind for the contracts: on the **Site** step the badge “Wind up to … MW” shows the limit.
- Placeholder costs have been replaced where better values are known.

# 8

# Worked example: Belle-Île

This chapter runs the planner on the Belle-Île case of EVOLVE, the floating wind zone south of Belle-Île in southern Brittany, and explains the result. Every number below can be reproduced with the files shipped with the app. All costs are the default values of Chapter 4: ACER 2026 medians for the cable and the compressors, the island platform and floating wind of Section 4.7, v9 placeholders for the rest. The result illustrates the method and is not a cost estimate for the site.

> **Version of the results**
>
> The numbers in this chapter were computed with version 0.10.0. Compared with the edition for version 0.9, floating wind costs 3.43 instead of 2.54 million EUR/MW, the floating platform factors are 1.63 and 1.52 instead of 2, the export cable has a dynamic section and the TSO fee is charged on the capacity injected into the grid. The design is almost the same; the costs, LCOE and LCOH are higher, mainly because of the wind CAPEX.

## 8.1 The case

The island floats in 75 m of water, exports to a 225 kV landfall at Erdeven and can deliver hydrogen in the Lorient area. The project serves two contracts: 200 GWh of electricity a year under a PPA at 100 EUR/MWh and 2 000 t of hydrogen a year at 6 EUR/kg, both as annual volumes. The hydrogen pipeline is switched off in this case, so hydrogen can only be made onshore. Table 8.1 lists the inputs that differ from the defaults.

**Table 8.1.** Inputs of the example that differ from the defaults (preset `hybrid_ppa`).

| Input | Value |
|---|---|
| Island | Belle-Île (Bretagne Sud), 47.38° N, 3.55° W |
| Grid connection point | Erdeven landfall (RTE 225 kV), 47.64° N, 3.16° W |
| Hydrogen offtake point | Port of Lorient, 47.73° N, 3.37° W |
| Water depth; usable sea area | 75 m; 40 $km^2$ |
| Export cable route length (entered) | 55 km |
| Hydrogen export pipeline | off |
| Use case | hybrid: electricity and hydrogen |
| Electricity contract | 200 GWh/yr, annual volume, PPA at 100 EUR/MWh |
| Hydrogen contract | 2 000 t/yr, annual volume, 6 EUR/kg |
| Maximum grid injection | 100 MW |
| Hourly data | 2019: ERA5 wind with the IEA 15 MW curve, pvlib PV, French day-ahead prices |

## 8.2 Reproducing it

1. Open the app and press **Open Belle-Île example** (or **Open the Belle-Île example** in the sidebar). This loads the inputs of Table 8.1 and the 2019 series. Alternatively, open `examples/belle_ile/EVOLVE_inputs_BelleIle_hybrid_2019.xlsx` with **Open an inputs workbook**.
2. Check the **Site** step: the badges read *Floating foundations*, *Export link to the grid point, 55 km: the model chooses among 33 AC and DC options (cheapest at the grid limit, before load losses: 132 kV AC, 3x500 mm2 Cu, reactors at both ends)* and *Wind up to 252 MW in the sea area*.
3. On the **Data** step the table shows wind at 42.1 %, PV at 14.6 % and a mean price of 39.4 EUR/MWh, and the message "The series match the use case".
4. On the **Run** step keep **Standard (1,460 blocks)** and HiGHS, and press **Find the lowest-cost design**. The run takes about three minutes.

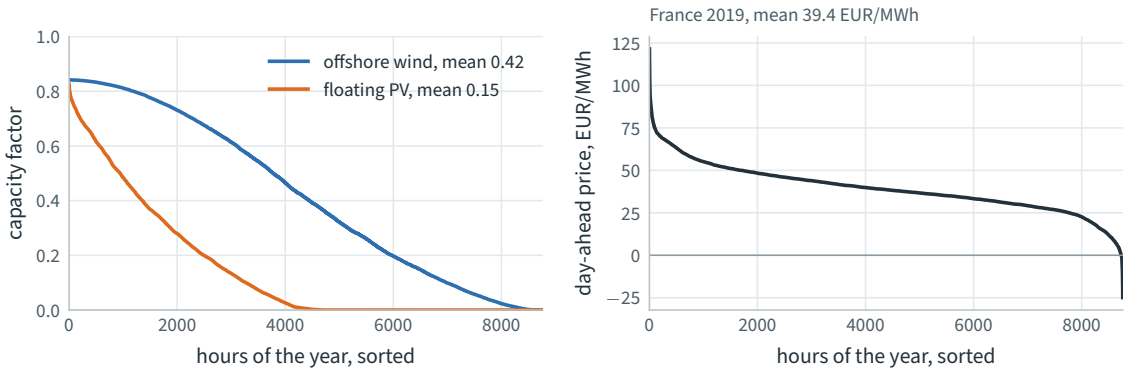



**Figure 8.1.** Hourly inputs of the example: duration curves of the wind and PV capacity factors (left) and of the day-ahead price (right), Belle-Île 2019.

## 8.3 The run

The year was reduced to 1 460 blocks in under a second, with a shape error of 0.14 % (Section 5.5); 36 extrema of the cumulative balances were kept as fixed cuts. Sizing took 114 s: the branch and bound of Section 3.8 bounded all 387 combinations of link option and circuits with 7 LPs and then solved 6 of them exactly (plus one re-solve to load the best); the first, one 132 kV circuit, was the cheapest. The full-year check took about 86 s. Both are times on a two-core cloud machine. The check scaled the conversion group (electrolyser, compressor, cable capacity used) by 1.022 and left generation and storage unchanged; the cable stayed within its one circuit, and the total annual cost rose from 45.59 to 45.72 million EUR (0.3 %).

## 8.4 The design

Table 8.2 gives the capacities. The island carries the wind farm, a small PV field and the export substation; all hydrogen is made onshore, because island equipment carries the offshore premium and platform cost, and without a pipeline island hydrogen could not reach the offtaker anyway. No battery or tank is built: with annual volumes, the contracts can be served in any hour, so storage has nothing to do. 6.0 MW of PV is built: at the floating wind price, a little PV, which produces in other hours than the wind, lets the same cable and electrolyser carry the volumes with less wind (77.9 instead of 80.0 MW; with the bottom-fixed wind price of version 0.9 no PV was built). The export link is one 132 kV AC circuit of $3 \times 500\ \text{mm}^2$ copper with shunt reactors at both ends (60 MVAr in all). Its usable rating on the 55 km route is 139 MW; it is used up to 65.3 MW. The 66 kV options, which would just fit, cost more per circuit (85.1 million EUR for $500\ \text{mm}^2$ against 82.8, both with the dynamic section) and lose twice as much at full load (6.9 % against 3.4 %, Table 4.4); the 220 kV options cost about 98 million EUR or more. Because the island floats, the app warns that a dynamic export cable at 132 kV is not yet qualified (Section 4.4); with **Floating: allow only qualified dynamic voltages** on, the model would have to use 66 kV. A larger conductor at 132 kV would lose less but costs more than the losses it saves. The cable capacity used is smaller than the wind farm (77.9 MW) and PV: clipping the rare highest hours (0.2 % of generation is curtailed) is cheaper than a larger platform and grid connection.

**Table 8.2.** Capacities of the example: as sized on the reduced year and after the full-year check.

| Component | Unit | Reduced year | Final design |
|---|---|---|---|
| Offshore wind (floating) | MW | 77.9 | 77.9 |
| Floating PV | MW | 6.0 | 6.0 |
| Export cable, 132 kV AC, 1 circuit of 139 MW: capacity used | MW | 63.9 | 65.3 |
| Electrolyser, onshore | MW | 11.5 | 11.8 |
| Compressor, onshore | MW | 0.12 | 0.12 |
| Electrolyser, fuel cell, battery and tank on the island | – | 0 | 0 |
| Battery and tank onshore | – | 0 | 0 |

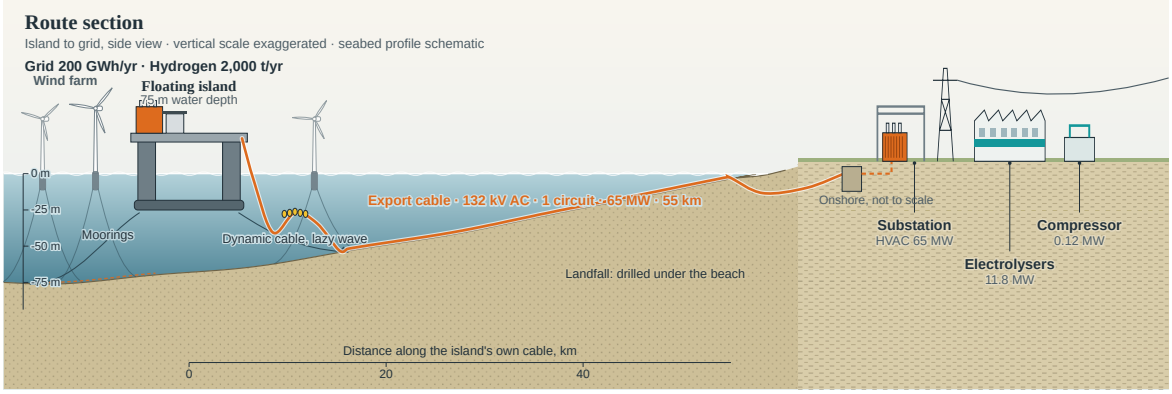



**Figure 8.2.** Route section of the design, as drawn by the app: the floating island, the dynamic export cable, the 55 km route to the landfall and the onshore substation and electrolysers with their sizes.

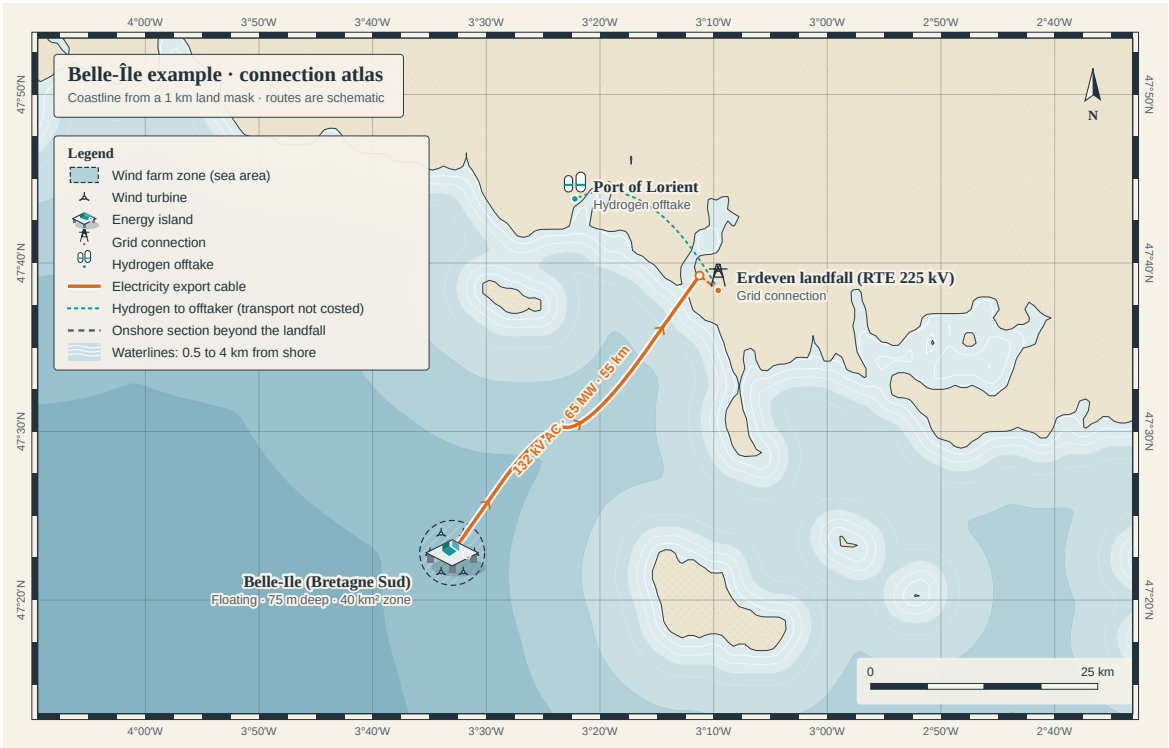



**Figure 8.3.** Connection atlas of the design: the coastline from the 1 km land mask, waterlines, the wind-farm zone with its turbines, the 132 kV export cable to Erdeven and the hydrogen offtake at Lorient.

## 8.5 Operation

The full-year operation shows how the design serves both contracts (Figure 8.4). The onshore electrolyser is sized to run most of the year, about 7 550 full-load hours: a small electrolyser that runs almost constantly is cheaper than a large one that follows the wind. It therefore takes the first 12 MW that reaches shore, and the PPA volume is delivered with the rest. In calm hours the grid delivery stops first (no power reaches the grid in 2 195 hours, which an annual contract allows); the electrolyser is idle in only 16 hours of the year. Over the year the system delivers exactly 200 GWh to the grid and 2 000 t of hydrogen, from 287 GWh of wind generation (3 684 full-load hours) and 7.6 GWh of PV; 5.4 GWh is lost in the cable, 1.1 GWh of it as no-load losses.

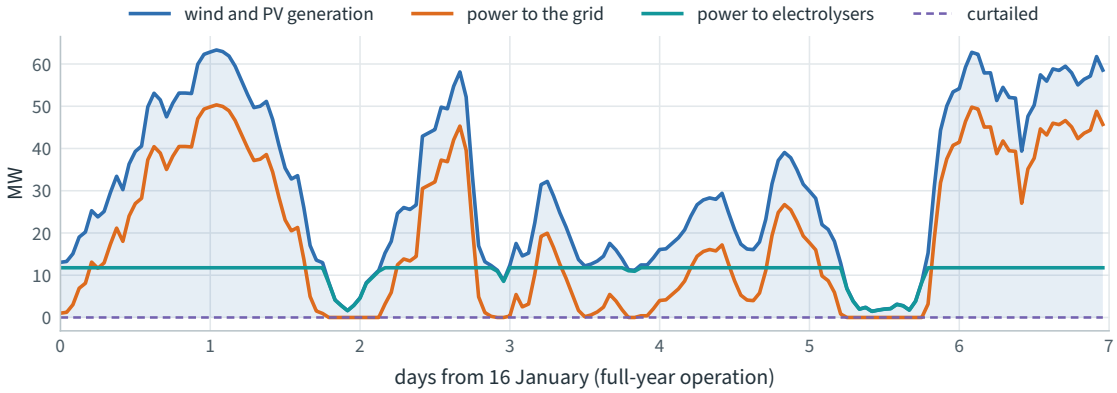



**Figure 8.4.** One week of the full-year operation, from 16 January 2019.

## 8.6 Costs, LCOE and LCOH

The investment is 395.8 million EUR, of which 82.8 million EUR is the export cable. The annual cost of 45.72 million EUR is covered by 20.0 million EUR of electricity revenue and 12.0 million EUR of hydrogen revenue, leaving a net cost of 13.72 million EUR a year. Electrolysers used 88.8 GWh, so hydrogen carries $s^h = 88.8/(88.8 + 200) = 30.7\,\%$ of the shared costs (Equation (3.39)). Table 8.3 and Figure 8.5 show the allocation.

**Table 8.3.** Annual cost by component and its allocation, million EUR per year.

| Component | Rule | Annual cost | To electricity | To hydrogen |
|---|---|---|---|---|
| Offshore wind | shared | 30.93 | 21.42 | 9.51 |
| Floating PV | shared | 0.95 | 0.66 | 0.29 |
| Export cable | shared | 7.14 | 4.95 | 2.20 |
| TSO fee | electricity | 2.17 | 2.17 | – |
| Island platform | shared | 1.46 | 1.01 | 0.45 |
| Electrolyser, onshore | hydrogen | 2.94 | – | 2.94 |
| Water treatment | hydrogen | 0.05 | – | 0.05 |
| Compressors | hydrogen | 0.08 | – | 0.08 |
| Total | | 45.72 | 30.21 | 15.51 |

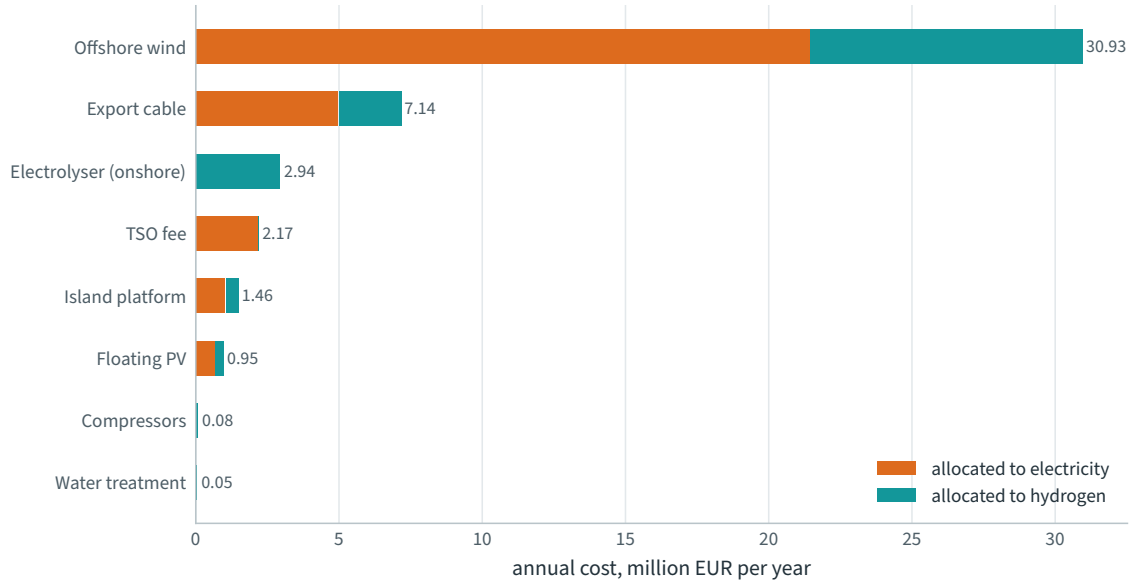



**Figure 8.5.** Annual cost by component, split between the two products.

This gives

$$\text{LCOE} = \frac{30.21 \text{ million EUR}}{200 \text{ GWh}} = 151.0 \text{ EUR/MWh},$$

$$\text{LCOH} = \frac{15.51 \text{ million EUR}}{2\,000 \text{ t}} = 7.75 \text{ EUR/kg} \quad (233 \text{ EUR/MWh on an LHV basis}),$$

both above the contract prices (100 EUR/MWh and 6 EUR/kg), which is why the island runs at a net cost. With annual volumes fixed by contract, the optimiser can only minimise the cost of meeting them. The cost of generation is 108.2 EUR/MWh; the rest of the LCOE is mostly the export cable (24.7 EUR/MWh), the grid fee (10.9 EUR/MWh) and the platform (5.0 EUR/MWh). With the v9 cable and compressor costs the same case gives an LCOE of 128.8 EUR/MWh and an LCOH of 6.73 EUR/kg (Section 4.10).

## 8.7 Variant: connecting to a nearby hub

The preset `hybrid_hub` asks the model to compare the island's own 55 km cable with a short cable to a shared offshore substation at 47.42° N, 3.45° W with 100 MW of spare capacity at 225 kV and an access fee of 42 000 EUR/MW/yr. The hub position and fee are illustrative. The sea route to the hub is 9.4 km.

The hub saves 5.25 million EUR a year on the reduced year (Table 8.4); after the full-year check the net cost is 8.49 million EUR a year. The saving is almost all in the cable: one 220 kV, 500 mm$^2$ circuit (the voltage nearest the hub's 225 kV) over 9.4 km costs 1.83 instead of 7.14 million EUR a year. The access fee, equal to the TSO fee per MW, is paid on the cable to the hub (2.70 million EUR a year) instead of on the grid injection (2.17). The LCOE and LCOH of the own-cable option in the table are those of the main example after its full-year check. The comparison is only as good as the hub inputs: a higher access fee or less spare capacity can reverse it, and the app reports both options so the difference can be discussed with the hub owner.

**Table 8.4.** Connection options compared on the reduced year, and the chosen option after the full-year check.

| Electricity connection | Net annual cost, reduced year | Cable length | LCOE | LCOH |
|---|---|---|---|---|
| | M EUR/yr | km | EUR/MWh | EUR/kg |
| Own cable to Erdeven | 13.59 | 55.0 | 151.0 | 7.75 |
| Cable to the hub (chosen) | 8.34 | 9.4 | 133.8 | 6.87 |

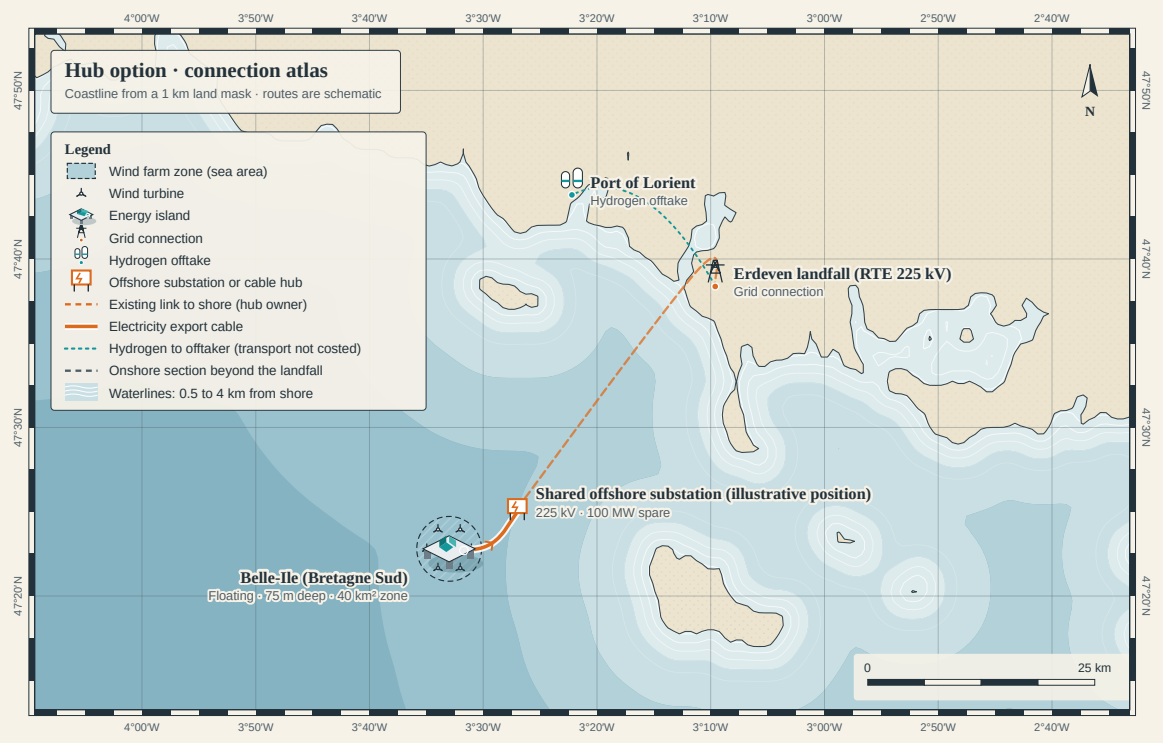



**Figure 8.6.** Connection atlas of the hub variant: the short cable to the hub, and the hub owner's existing link to shore (dashed), which is not costed.

## 8.8 What to try next

- Run 2024 instead of 2019 (**Data** step, **Belle-Île example**) to see how much the design depends on the weather year.
- Switch the hydrogen pipeline on and let the model choose between onshore and island electrolysis.
- Change the electricity delivery to *firm*: storage and fuel cells then become necessary in calm hours, and the reduced year's storage tracking matters.
- Use CfD revenue (preset `power_cfd`) or merchant revenue to let the optimiser choose the volume, and compare with the fixed-volume PPA of preset `power_ppa`.
- Set **Cable cost model** to **v9 linear** to reproduce the earlier cable cost, or set **Export link** to **HVDC** or **220 kV** to see what that link would cost here. Increase the grid injection limit, the demand and the sea area, or move the grid point further away, to see the choice move to higher voltages, mid-route compensation and HVDC (Table 4.5).

- Change the HVDC cost factor, the mid-route platform cost or the value of no-load losses, the most uncertain inputs of the link choice.
- Test the island platform: set the depth below the 60 m threshold to compare a bottom-fixed island, switch on turbine-level electrolysis (concept C) with the pipeline, and vary the hull ratio, the scale penalty and the integration factor within the ranges of Table 4.9.
- Replace the placeholder costs (Table 4.10) with referenced values for the site.

# A

# All inputs

The table lists every input of version 0.10.0 with its key (as in the `Inputs` sheet of the inputs workbook), its label in the app, its unit, its default value and the source of the default (Section 4.1): *ACER* for the ACER 2026 medians, *v9* for values kept from the project's v9 model, an equation or a reference for values computed in Chapter 4 or taken from a publication, *literature* for values chosen within a published range (Section 4.4), and no entry for project inputs that the user supplies (site, contracts, switches). A • marks cost values without a referenced source that Step 3 of EVOLVE will replace. The table is generated from `evolve_planner/params.py`, so it matches the software exactly.

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| **Use case** | | | | |
| `use_case` | What the island supplies | | `Hybrid` | |
| `annual_demand_mwh` | Electricity to deliver per year | MWh/yr | 200 000 | |
| `annual_h2_demand_kg` | Hydrogen to deliver per year | kg/yr | 2 000 000 | |
| `elec_delivery` | Electricity delivery profile | | `annual` | |
| `h2_delivery` | Hydrogen delivery profile | | `annual` | |
| `elec_revenue` | Electricity revenue | | `ppa` | |
| `ppa_price_eur_mwh` | PPA price | EUR/MWh | 100 | |
| `cfd_strike_eur_mwh` | CfD strike price | EUR/MWh | 86.45 | |
| `h2_price_eur_kg` | Hydrogen sales price | EUR/kg | 6 | |
| `sell_surplus` | Sell the other carrier as a by-product | | off | |
| `grid_injection_cap_mw` | Maximum grid injection | MW | 100 | |
| **Site** | | | | |
| `site_name` | Site name | | *My floating island* | |
| `site_lat` | Island latitude | deg | 47.38 | |
| `site_lon` | Island longitude | deg | -3.55 | |
| `grid_name` | Grid connection point | | *Grid substation* | |
| `grid_lat` | Grid point latitude | deg | 47.64 | |
| `grid_lon` | Grid point longitude | deg | -3.16 | |
| `h2_name` | Hydrogen offtake point | | *Hydrogen offtaker* | |
| `h2_lat` | Hydrogen offtake latitude | deg | 47.27 | |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| h2_lon | Hydrogen offtake longitude | deg | -2.2 | |
| route_method | How route lengths are found | | sea route | |
| routing_factor | Routing factor (route / straight line) | | 1.2 | |
| link_distance_override_km | Export cable route length (0 = computed) | km | 0 | |
| pipe_distance_override_km | Pipeline route length (0 = computed) | km | 0 | |
| avg_water_depth_m | Water depth at the island | m | 75 | |
| fixed_floating_depth_m | Fixed/floating threshold depth | m | 60 | v9 |
| decentralized_h2 | Turbine-level electrolysis (no power cable) | | off | |
| existing_pipeline_available | Existing gas pipeline on the route can be repurposed | | off | |
| sea_area_km2 | Usable sea area | $km^2$ | 38 | |
| wind_density_mw_km2 | Wind power density | $MW/km^2$ | 6.3 | site data |
| pv_density_mw_km2 | Floating PV density • | $MW/km^2$ | 60 | assumption |
| deck_area_max_m2 | Island deck area available | $m^2$ | 200 000 | v9 |
| area_onshore_max_m2 | Onshore plot area | $m^2$ | 1 000 000 | v9 |
| **Connections** | | | | |
| elec_connection | Electricity connection | | direct | |
| hub_name | Nearby offshore substation or cable | | *Nearby offshore substation* | |
| hub_lat | Hub latitude | deg | 47.42 | |
| hub_lon | Hub longitude | deg | -3.45 | |
| hub_spare_mw | Spare capacity at the hub | MW | 100 | |
| hub_voltage_kv | Hub voltage | kV | 225 | |
| hub_access_fee_eur_mw_yr | Hub access fee • | EUR/MW/yr | 42 000 | v9 |
| export_voltage | Export link | | auto | |
| h2_connection | Hydrogen connection | | direct | |
| tiein_name | Nearby pipeline tie-in point | | *Nearby pipeline* | |
| tiein_lat | Tie-in latitude | deg | 47.45 | |
| tiein_lon | Tie-in longitude | deg | -3.3 | |
| tiein_spare_kgh | Spare capacity in the pipeline | kg/h | 20 000 | |
| tiein_pressure_bar | Pipeline operating pressure | bar | 70 | |
| tiein_fee_eur | Tie-in cost (hot tap, metering) • | EUR | 5 000 000 | assumption |
| **Technologies** | | | | |
| enable_pv | Floating PV | | on | |
| enable_ez_off | Electrolyser on the island | | on | |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| enable_ez_on | Electrolyser onshore | | on | |
| enable_fc_off | Fuel cell on the island | | on | |
| enable_fc_on | Fuel cell onshore | | on | |
| enable_batt_off | Battery on the island | | on | |
| enable_batt_on | Battery onshore | | on | |
| enable_tank_off | Hydrogen tank on the island | | on | |
| enable_tank_on | Hydrogen tank onshore | | on | |
| enable_cable | Electricity export cable | | on | |
| enable_pipe | Hydrogen export pipeline | | on | |
| **Finance** | | | | |
| discount_rate | Discount rate (real WACC) | fraction | 0.07 | v9 |
| offshore_premium | Offshore cost premium on island equipment * | x | 1.1 | v9 |
| offshore_fom_adder | Extra fixed O&M for batteries and tanks on the island | fraction/yr | 0.005 | v9 |
| **Offshore wind** | | | | |
| wind_capex_eur_mw | CAPEX, bottom-fixed * | EUR/MW | 2 540 000 | v9 |
| wind_capex_floating_eur_mw | CAPEX, floating * | EUR/MW | 3 429 000 | Lanni 2025; NREL 2024 |
| wind_life | Lifetime | yr | 25 | v9 |
| wind_fom | Fixed O&M, bottom-fixed | fraction/yr | 0.025 | v9 |
| wind_fom_floating | Fixed O&M, floating | fraction/yr | 0.03 | v9 |
| turbine_mw | Turbine rating (for the drawing) | MW | 15 | IEA 15 MW |
| **Floating PV** | | | | |
| pv_capex_eur_kw | CAPEX * | EUR/kW | 1500 | v9 |
| pv_life | Lifetime | yr | 25 | v9 |
| pv_fom | Fixed O&M | fraction/yr | 0.02 | v9 |
| **Electrolysis** | | | | |
| elec_capex_eur_kw | CAPEX (stack + balance of plant) * | EUR/kW | 2000 | v9 |
| electrolyzer_efficiency | Efficiency (LHV) | fraction | 0.75 | v9 |
| elec_life | Lifetime (stack replacement) | yr | 15 | v9 |
| elec_fom | Fixed O&M | fraction/yr | 0.015 | v9 |
| area_elec_m2_mw | Deck area | $m^2$/MW | 200 | v9 |
| wt_capex_eur_mw | Water treatment CAPEX | EUR/MW | 20 750 | v9 |
| wt_life | Water treatment lifetime | yr | 20 | v9 |
| wt_fom | Water treatment O&M | fraction/yr | 0.1 | v9 |
| **Compression** | | | | |
| comp_p_in_bar | Inlet pressure | bar | 30 | v9 |
| comp_p_out_bar | Outlet pressure | bar | 70 | v9 |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| comp_isentropic_eff | Isentropic efficiency | fraction | 0.75 | v9 |
| comp_stages | Stages | | 2 | v9 |
| comp_inlet_temp_k | Inlet temperature | K | 313 | v9 |
| comp_z | Compressibility factor | | 1.05 | v9 |
| comp_gamma | Heat capacity ratio of hydrogen | | 1.41 | physics |
| comp_capex_eur_mw | CAPEX | EUR/MW | 5 080 000 | ACER |
| comp_life | Lifetime | yr | 20 | v9 |
| comp_fom | Fixed O&M | fraction/yr | 0.03 | v9 |
| area_comp_m2_mw | Deck area | $m^2$/MW | 100 | v9 |
| **Fuel cell** | | | | |
| fc_efficiency_lhv | Efficiency (LHV) | fraction | 0.5 | v9 |
| fc_capex_eur_kw | CAPEX | EUR/kW | 1170 | v9 |
| fc_life | Lifetime | yr | 15 | v9 |
| fc_fom | Fixed O&M | fraction/yr | 0.03 | v9 |
| area_fc_m2_mw | Deck area | $m^2$/MW | 120 | v9 |
| **Battery** | | | | |
| batt_capex_e_eur_mwh | Energy CAPEX | EUR/MWh | 160 000 | v9 |
| batt_capex_p_eur_mw | Power CAPEX * | EUR/MW | 100 000 | assumption |
| batt_eta | Charge and discharge efficiency (each way) | fraction | 0.95 | v9 |
| batt_life | Lifetime | yr | 15 | v9 |
| batt_fom | Fixed O&M | fraction/yr | 0.02 | v9 |
| batt_vom | Variable O&M | EUR/MWh | 1.19 | v9 |
| area_batt_m2_mwh | Deck area | $m^2$/MWh | 25 | v9 |
| batt_max_mwh | Largest battery allowed, energy (each site) | MWh | 100 000 | v9 |
| batt_max_mw | Largest battery allowed, power (each site) | MW | 20 000 | v9 |
| **Hydrogen tank** | | | | |
| tank_capex_eur_kg | CAPEX | EUR/kg | 650 | v9 |
| tank_life | Lifetime | yr | 20 | v9 |
| tank_fom | Fixed O&M | fraction/yr | 0.02 | v9 |
| tank_vom | Variable O&M | EUR/kg | 0.01 | v9 |
| area_tank_m2_kg | Deck area | $m^2$/kg | 1.5 | v9 |
| tank_max_kg | Largest tank allowed (each site) | kg | 5 000 000 | v9 |
| **Export cable** | | | | |
| cable_cost_model | Cable cost model | | catalogue | Chapter 4 |
| ac_install_eur_km | AC cable installation per circuit-km | EUR/km | 910 000 | ENTSO-E; ACER |
| ac_supply_factor | AC cable supply cost factor | x | 1 | |
| usd_eur | EUR per USD for ORBIT prices | EUR/USD | 0.924 | ECB 2024 |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| grid_power_factor | Power factor at the grid connection | | 0.95 | assumption |
| cable_loss_factor | AC loss factor for screen and armour | x | 1.5 | Pérez-Rúa 2022 |
| cable_tan_delta | XLPE dielectric loss factor tan(delta) | | 0.001 | IEC 60287 |
| reactor_eur_mvar | Shunt reactor cost per MVAr | EUR/MVAr | 20 000 | literature |
| reactor_fixed_eur | Shunt reactor fixed cost per site and circuit | EUR | 1 000 000 | literature |
| allow_midroute | Allow a mid-route compensation platform | | on | |
| midroute_platform_eur | Mid-route compensation platform • | EUR | 50 000 000 | assumption |
| loss_value_eur_mwh | Value of no-load cable losses | EUR/MWh | 50 | literature |
| cable_max_circuits | Largest number of circuits or converter units | | 12 | model limit |
| vh_np_eur_mw | HVDC converter, per MW (each end) | EUR/MW | 112 990 | Vrana–Härtel 2018 |
| vh_n0_eur | HVDC converter, fixed per unit (each end) | EUR | 23 500 000 | Vrana–Härtel 2018 |
| vh_sp_eur_mw | HVDC offshore station, per MW | EUR/MW | 723 420 | Vrana–Härtel 2018 |
| vh_s0_eur | HVDC offshore station, fixed per unit | EUR | 57 320 000 | Vrana–Härtel 2018 |
| vh_blp_eur_mw_km | HVDC cable, per MW and km | EUR/MW/km | 980 | Vrana–Härtel 2018 |
| vh_bl_eur_km | HVDC cable, per km | EUR/km | 270 000 | Vrana–Härtel 2018 |
| vh_b0_eur | HVDC cable, fixed per unit | EUR | 3 630 000 | Vrana–Härtel 2018 |
| hvdc_unit_mw | Largest HVDC unit | MW | 2000 | Vrana–Härtel 2018 |
| hvdc_cost_factor | HVDC cost escalation factor | x | 1 | Table 4.3 |
| hvdc_converter_loss | HVDC converter losses (both stations) | fraction | 0.02 | PyPSA; v9 |
| hvdc_cable_loss_per_100km | HVDC cable loss per 100 km at full load | fraction | 0.0045 | Equation (4.4) |
| hvdc_float_premium | Floating offshore-station premium (floating island) • | x | 1.5 | v9 |
| tso_fee_eur_mw_yr | TSO grid fee (per MW injected into the grid) • | EUR/MW/yr | 42 000 | v9 |
| cable_fom_hvac | Fixed O&M, HVAC link | fraction/yr | 0.005 | v9 |
| cable_fom_hvdc | Fixed O&M, HVDC link | fraction/yr | 0.01 | v9 |
| hvac_hvdc_threshold_km | v9 linear model: HVAC up to | km | 60 | v9 |
| hvac_var_eur_mw_km | v9 linear model: HVAC cost per MW per km | EUR/MW/km | 1360 | v9 |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| hvac_fix_eur_km | v9 linear model: HVAC fixed cost per km | EUR/km | 206 130 | v9 |
| hvac_loss_per_km | v9 linear model: HVAC loss per km | fraction/km | $6.7 \times 10^{-5}$ | v9 |
| hvdc_cable_eur_mw_km | v9 linear model: HVDC cable per MW per km | EUR/MW/km | 1000 | v9 |
| hvdc_cable_loss_per_km | v9 linear model: HVDC cable loss per km | fraction/km | $3.5 \times 10^{-5}$ | v9 |
| v9_converter_pair_eur_mw | v9 linear model: HVDC converter pair | EUR/MW | 300 000 | v9 |
| tso_fee_hvdc_eur_mw_yr | v9 linear model: TSO grid fee, HVDC | EUR/MW/yr | 0 | v9 |
| cable_life | Lifetime | yr | 30 | v9 |
| **Hydrogen pipeline** | | | | |
| pipe_cost_fixed_eur_km | New pipeline cost, fixed part | EUR/km | 1 550 000 | ACER, Equation (4.9) |
| pipe_cost_per_inch_eur_km | New pipeline cost, per inch of diameter | EUR/km/inch | 37 857 | ACER, Equation (4.9) |
| pipe_repurposed_capex_eur_km | Repurposed 36-inch: cost | EUR/km | 1 180 000 | ACER |
| pipe_offshore_factor | Subsea cost factor | x | 1.7 | EHB 2022 |
| pipe_p_out_bar | Delivery pressure at the end of the pipeline | bar | 30 | assumption |
| tiein_dp_bar | Pressure drop allowed in a tie-in spur | bar | 5 | assumption |
| pipe_max_velocity_ms | Design gas velocity | m/s | 20 | DEA 2025 |
| pipe_velocity_fraction | Gas velocity limit, share of erosional velocity | fraction | 0.5 | Singlitico 2021 |
| pipe_erosional_c | Erosional velocity constant (SI) | (kg/m3)^0.5 m/s | 122 | API RP 14E |
| pipe_roughness_mm | Wall roughness | mm | 0.05 | Singlitico 2021 |
| pipe_gas_temp_k | Gas temperature in the pipeline | K | 288 | assumption |
| pipe_id_4in | 4-inch: inner diameter | m | 0.1023 | ASME B36.10 |
| pipe_id_6in | 6-inch: inner diameter | m | 0.1541 | ASME B36.10 |
| pipe_id_8in | 8-inch: inner diameter | m | 0.2027 | ASME B36.10 |
| pipe_id_12in | 12-inch: inner diameter | m | 0.3049 | ASME B36.10 |
| pipe_id_16in | 16-inch: inner diameter | m | 0.3842 | ASME B36.10 |
| pipe_id_20in | 20-inch: inner diameter | m | 0.4826 | ASME B36.10 |
| pipe_id_24in | 24-inch: inner diameter | m | 0.5814 | ASME B36.10 |
| pipe_id_36in | 36-inch: inner diameter | m | 0.8758 | ASME B36.10 |
| pipeline_efficiency | Delivery efficiency | fraction | 0.99 | v9 |
| pipeline_life | Lifetime | yr | 40 | v9 |
| pipeline_fom | Fixed O&M | fraction/yr | 0.009 | v9 |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| **Island platform** | | | | |
| platform_cable_coef | Electrical platform structure, per MW of export capacity (c_E) • | kEUR/MW | 84.651 | DNV GL 2018; Bouma 2023 |
| platform_elec_coef | Process platform structure, per MW of island equipment (c_H) • | kEUR/MW | 139.281 | DNV GL 2018; Bouma 2023 |
| platform_install_factor | Installation, engineering and contingency factor (k_inst) | x | 1.5 | Bouma 2023; Equation (4.13) |
| platform_escalation | Price escalation of the platform coefficients (k_esc) | x | 1 | assumption |
| platform_fixed_eur | Fixed platform cost (C_0) | EUR | 0 | assumption |
| platform_w_comp | Weight of compressor MW on the process platform | | 1 | assumption |
| platform_w_fc | Weight of fuel-cell MW on the process platform | | 1 | assumption |
| platform_w_bat | Weight of battery MW on the process platform | | 1 | assumption |
| platform_ref_depth_m | Reference depth of the coefficients | m | 30 | DNV GL 2018 |
| platform_jacket_share_h | Jacket share of the process platform (s_H) | fraction | 0.28 | DNV GL 2018 |
| platform_jacket_share_e | Jacket share of the electrical platform (s_E) | fraction | 0.22 | DNV GL 2018 |
| platform_depth_exponent | Jacket cost exponent on depth (alpha) | | 1 | Ekhar 2021; BVG 2024 |
| platform_float_topside_uplift | Floating: topside uplift (u) | x | 1.1 | assumption |
| platform_float_hull_ratio | Floating: hull and moorings vs jacket (r) | x | 3 | BVG 2024; Myhr 2014 |
| platform_life | Lifetime | yr | 25 | v9 |
| platform_fom | Fixed O&M, bottom-fixed and turbine-level | fraction/yr | 0.02 | Bouma 2023 |
| platform_fom_floating | Fixed O&M, floating central platform | fraction/yr | 0.03 | assumption |
| dyn_cable_km | Floating: dynamic section of each export circuit | km | 0.5 | assumption |
| dyn_cable_premium | Floating: dynamic cable premium | fraction | 0.75 | Ibrahim 2022 |
| dyn_cable_max_kv | Floating: highest qualified dynamic export voltage | kV | 66 | DNV 2021; BVG 2024 |
| dyn_cable_restrict | Floating: allow only qualified dynamic voltages | | off | |
| turbine_integration_factor | Turbine-level: integration factor (f_C) | | 0.5 | assumption |
| turbine_ez_scale_factor | Turbine-level: electrolyser scale penalty (k_scale) | x | 1.3 | Singlitico 2021 |

All inputs, continued

| Key | Label | Unit | Default | Source |
|---|---|---|---|---|
| infield_pipe_eur_km | Turbine-level: in-field hydrogen pipeline | EUR/km | 550 000 | Travaglini 2025 |
| infield_km_per_turbine | Turbine-level: in-field pipeline per turbine | km | 1 | ERM 2019 |
| infield_pipe_fom | Turbine-level: in-field pipeline O&M | fraction/yr | 0.005 | Travaglini 2025 |
| infield_manifold_eur | Turbine-level: collection manifold | EUR | 50 000 | Lanni 2025 |
| array_cable_credit_eur_mw | Turbine-level: array cables not needed | EUR/MW | 145 000 | Lanni 2025 |
| turbine_ez_eff_gain | Turbine-level: avoided electrical losses | fraction | 0.005 | Lanni 2025 |
| **Solver** | | | | |
| shortfall_penalty_eur_mwh | Penalty for undelivered electricity | EUR/MWh | 3000 | v9 |
| shortfall_penalty_eur_kg | Penalty for undelivered hydrogen | EUR/kg | 100 | v9 |
| lhv_h2_kwh_kg | Hydrogen LHV | kWh/kg | 33.3 | v9 |
| cap_ub_mw | Upper bound on any converter or cable capacity | MW | 5000 | v9 |
| cap_renewable_ub_mw | Upper bound on wind and PV capacity | MW | 10 000 | v9 |

# B

# Output files

## B.1 Results workbook

The **Results workbook (Excel)** button on the **Download** tab writes one file (`report.py`) with the sheets of Table B.1. All quantities come from the full-year solution.

**Table B.1.** Sheets of the results workbook.

| Sheet | Content |
|---|---|
| Summary | LCOE, LCOH (per kg and per MWh LHV), cost of generation, hydrogen share of shared costs; connection, export voltage, rating per circuit, link built, circuits or converter units, link losses, reactive compensation, investment per MW and efficiency; net and total annual cost, revenues, investment; energy delivered, curtailed and undelivered; net unit costs; configuration, export link, pipeline class, route lengths and how they were found; TDA blocks and shape error; net cost on the reduced year; repair factors; run time and solver |
| Capacities | every capacity variable of Table 3.2 with its unit |
| Annual costs | annual cost of each component, the grid fee and variable O&M (EUR/yr) |
| Levelized costs | each component's annual cost, allocation rule, share to electricity and to hydrogen, and its contribution in EUR/MWh and EUR/kg |
| Connection options | when several options were sized: each option and its objective on the reduced year |
| Hourly (full year) | 8 760 rows: wind and PV generation, curtailment, electrolyser input on the island and onshore, cable flows, power to the grid, battery charge and discharge, hydrogen produced, piped, delivered and used in fuel cells, storage levels, shortfalls, price |
| TDA blocks | each block's length in hours and the hour it ends |
| Inputs used | every input with its group, key, label, value, unit, placeholder flag, whether it differs from the default, and the source of the default |

## B.2 Inputs workbook

The inputs workbook of a run (Section 7.5) can be opened in the app to restore the case.

## B.3 Drawings

The route section, the island and the connection atlas are self-contained HTML files with the drawing as SVG. They open in any browser and can be printed to PDF or inserted in reports; the

fonts load from the internet and fall back to system fonts offline.

# C

# Hosting your own copy

The planner is a public GitHub repository plus a Streamlit Community Cloud app that reads it. This appendix lists the steps to publish a copy, update it, and run it elsewhere.

## C.1 Publish from GitHub

1. Create a GitHub account if needed, then a new repository (for example `EVOLVE-Floating-Shore`), public, without a README.
2. On the repository page choose **Add file** → **Upload files**, and drag in the *contents* of the unzipped project folder (select everything inside it, including the `.streamlit` folder). GitHub keeps the subfolders.
3. Write a short message and press **Commit changes**.
4. Sign in at `https://share.streamlit.io` with the GitHub account and choose **Create app** → **Deploy a public app from GitHub**. Select the repository, branch `main`, main file `app.py`, and Python 3.12 under the advanced settings.
5. Press **Deploy**. The first build takes a few minutes; the app's address can then be shared.

## C.2 Update

Upload the changed files to the repository the same way (files with the same name are replaced) and commit. Streamlit Cloud redeploys within a minute or two; if it does not, choose **Reboot** from the app's menu on `https://share.streamlit.io`. Nothing has to be created again.

## C.3 Run elsewhere

**Local.** See Chapter 6. Run the tests with `pip install pytest` and `pytest -q` (about one minute).

**Docker.** The repository contains a `Dockerfile`:

```
docker build -t evolve-planner .
docker run -p 8501:8501 evolve-planner
```

This is the way to run the app on a university or company server with more memory than the free tier.

## C.4 Repository layout

| Path | Content |
|---|---|
| `app.py` | Streamlit app: the six steps |
| `evolve_planner/params.py` | every input with label, unit, default and help; derived quantities (Section 3.7) |
| `evolve_planner/model.py` | the optimisation model (Chapter 3) |
| `evolve_planner/tda.py`, `timestructure.py` | persistence and the block reduction (Chapter 5) |
| `evolve_planner/workflow.py` | reduce, size, check (Section 5.6) |
| `evolve_planner/levelized.py` | LCOE and LCOH (Section 3.9) |
| `evolve_planner/geo.py` | coastline, sea routes, nearest coast |
| `evolve_planner/weather.py`, `workbook.py`, `report.py` | data import and downloads, inputs and results workbooks |
| `ui/` | theme, charts, interactive map and drawings |
| `assets/landmask_120.npz` | 1 km land mask (GLOBE, bit-packed) |
| `examples/` | data guide, blank inputs workbook, Belle-Île series, presets and inputs workbook |
| `tests/` | tests of the reduction, the workflow, the drawings and the routes |

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